ROYAL CANADIAN NAVY
QUALIFICATION STANDARD
AND PLAN (QSP)

Marine Technician (MAR TECH) (00379)

RQ S3 Basic Maintainer (ALJS)

MITE Course ID: 122592

MARPAC RDIMS #983746
Chap 1 approved 10 05 2024
Chap 2, 3 approved 10 05 2024

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
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FOREWORD

2. This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following References.

a. A-P9 - 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification 1 May 19;

c. NAVORD 4500-0 “Structured Delegation”; and

d. NPTORD 4500-22 “Qualification Standard and Plan Production and Management”.

3. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

4. This QSP is effective upon approval and supersedes previous versions.

5. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NTDC(P) MSE div), who will forward to NPTG for approval, as appropriate.

6. The designated TEs for this training are indicated below:

TE	**Phase / Module**	**MITE Course ID**	**NQual**
NFS (A & P)	All	122592	ALJS

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY

COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the MAR TECH Basic Maintainer. This QSP is designed to assist the training system in developing, conducting, evaluating, and validating the training.

COURSE DELIVERY METHODS

2. The Basic Maintainer course has two different course delivery methods.

NOTE: Mod A, B and ODS (1.02) are prerequisites and are completed by use of either method below.

a. Method 1: This method can be done in less time through the Defence Learning Network (DLN) to allow flexibility in delivery. Once all modules on the DLN are completed candidates will be loaded onto the practical portion of the course. Basic Maintainers who complete Method 1 will be granted a separate MITE code for each module as well as the overall MITE and NQual after completion of all modules. Annex O (Modularization Details) includes a detailed breakdown of the different modules including preferred order, pre-requisites, etc.

b. Method 2: Is available for Basic Maintainer candidates who are available for full time enrollment in the course. This method involves a combination of computer-based training (CBT) using the Defence Learning Network (DLN) and practical portions in a shop setting. An instructor is assigned for increased coaching opportunities. Basic Maintainers who complete Method 2 will get only the overall MITE code.

COURSE DURATION

3. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP.

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points	25745
Enabling Checks (ECs)	4345
Performance Checks (PCs)	650
ATT Sub-total	**33,070 (83 days)**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	

Base Administrative Time (BAT)	1020
Training Administrative Time (TAT)	1020
Training Travel Time (TTT)	2250
Physical Fitness (PT)	2700
Divisions	480
Sub-total	**7470** **(19 days)**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days (rounded up)**	
Total Training Days	38210 (96 days)

Ref: NAVRESORD 5031-8 NAVRES Policy Class A Remuneration for Distance Learning and Distributed Learning

4. For Naval Reserve students who are completing the Distributed Learning on part-time service, the authorized class A days for the DL portion of each module are shown at the bottom of the table for each module. Time for logging into the Defence Learning Network; overcoming minor computer or network problems; interacting with the instructors and fellow students; completing the course introduction, reviews and enabling checks; as well as completing the training material are included in the Class A days calculation. The course joining instructions will include milestones for payment with the course schedule.

5. The class A days were calculated with the following formula:

   a. Total ATT minutes divided by 50 to give training hours divided by 7.5 to give training days plus ¼ day (2 hours) for each training day, rounded to the nearest ½ day.

   b. For example: 2460 minutes = 6.56 training days x 1.25 = 8.2 rounded to 8 Class A days. 50 x 7.5

Module A: DL

Allocation of ATT	Minutes
Lesson Objectives (LOs) / Teaching Points 1 001.03 DL 1080 mins – Perform Basic Mathematics	1080
Enabling Checks (ECs)	60

1 EC 001.03 60 mins	
ATT Sub-total	**1140**
No allocation of BAT, TAT, TTT, PT for DL modules	
Course Duration Calculation Total mins/400 = # days (rounded up)	
Total Training Days	**3 days**
Class A Days allocated	**4 days**

Module B: DL

Allocation of ATT	Minutes
Lesson Objectives (LOs) / Teaching Points 1 001.01 DL 410 mins – Maintain Safety Requirements for Engineering Operations 2 001.04 DL 150 mins – Interpret Drawings 3 004.01 DL 440 mins – Inspect Hull Structures 4 002.01 DL 400 mins – Apply Foundational Science 5 002.02 DL 550 mins – Inspect Fluid Systems Components 6 002.03 DL 575 mins – Inspect Components of Pressurized Systems 7 002.04 DL 175 mins – Maintain the Liquid Damage Control Systems 8 002.05 DL 325 mins – Maintain Petroleum, Oils and Lubricant Systems 9 002.06.01 DL 150 mins – Utilize Shop Tools	3175
Enabling Checks (ECs) 1 EC 001.01 60 mins 2 EC 001.04 60 mins 3 EC 004.01 60 mins 4 EC 002.01 60 mins 5 EC 002.02 60 mins 6 EC 002.03 60 mins 7 EC 002.04 60 mins 8 EC 002.05 60 mins	480
ATT Sub-total	**3505**
No allocation of BAT, TAT, TTT, PT for DL modules	

Course Duration Calculation Total mins/400 = # days (rounded up)	
Total Training Days	**10.5 days**
Class A Days allocated	**12.5 days**

Module C: At TE (including CBT)

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points 1 001.05 CBT 90 mins – Maintain DRMIS as a maintainer 2 002.06.02 200 mins – Utilize Shop Tools 3 002.06.03 450 mins - Install Metal Shapes and Components 4 002.07.01 1200 mins - Maintain Valves 5 002.07.02 750 mins - Maintain Pumps	2690
Enabling Checks (ECs) 1 EC 001.05 10 mins 2 EC 002.06 425 mins 3 EC 002.07 150 mins	585
Performance Checks (PCs)	0
ATT Sub-total	**3275**
Allocation of BAT, TAT, TTT, PT (Course < 10 days)	
Base Administrative Time (BAT)	60
Training Administrative Time (TAT)	60
Training Travel Time (TTT)	0
Physical Fitness (PT)	0
Sub-total	**120**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**9 days**

Module D: DL (including HRAI training)

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points 1 001.02 DL 350 mins – Apply ODS regulations via HRAI Canada	800 mins

2 002.11 DL 350 mins –Maintain Refrigeration Systems 3 007.01.01 DL 100 mins – Describe Oxy Acetylene Safe Work Practices	
Enabling Checks (ECs) 1. 001.02 EC 90 mins 2. 002.11 EC 200 mins	290 mins
Performance Checks (PCs)	0
ATT Sub-total	**1090**
Total Training Days	**3 days**
Class A Days Allocated	**3.5 days**

Module D: At TE

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points 1 002.08 F2F 1320 mins - Repair Pipes, Hoses and Insulation 2 002.09 F2F 740 mins - Maintain Miscellaneous Fluid Components 3 002.10 F2F 625 mins - Maintain Air Systems 4 004.02 F2F 1150 mins - Maintain Closures 5 007.01.02 100 mins - Perform Oxy Acetylene Safety 6 007.01.03 100 mins - Perform Oxy Acetylene Equipment Set-up 7 007.02 1200 mins - Weld Metal with Oxy-Acetylene Torch 8 007.03 250 mins - Cut Metal with Oxy-Acetylene Torch 9 007.04 1050 mins - Cut Metal with Exothermic Torch 10 007.05 2300 mins - Perform Basic Shielded Metal Arc Welding	8835
Enabling Checks (ECs) 1 EC 002.08 250 mins 2 EC 002.10 150 mins 3 EC 002.12 200 mins 4 EC 004.02 100 mins	700
Performance Checks (PCs) – PC 007	400
ATT Sub-total	**9935**
Allocation of BAT, TAT, TTT, PT (Course < 31 days)	
Base Administrative Time (BAT)	120
Training Administrative Time (TAT)	120

Training Travel Time (TTT)	600
Physical Fitness (PT)	900
Sub-total	**1740**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**11675** **30 days**

Module E: DL

Allocation of ATT	Minutes
Lesson Objectives (LOs) / Teaching Points 1 003.01 DL 810 mins – Inspect DC Circuits 2 003.02 DL 330 mins – Inspect AC circuits 3 003.03 DL 500 mins – Inspect AC machines 4 003.04 DL 150 mins – Inspect DC machines 5 003.05 DL 200 mins – Maintain basic control devices 6 003.06 DL 120 mins – Maintain ship's service lighting 7 003.07 DL 125 mins – Perform basic maintenance on ship's cabling and connectors 8 003.08 DL 600 mins – Inspect shipboard PG&D	2835
Enabling Checks (ECs) 1 EC 003.01 60 mins 2 EC 003.02 60 mins 3 EC 003.08 60 mins	180
Performance Checks (PCs)	0
ATT Sub-total	**3015**
Total Training Days	**8 days**
Class A Days allocated	**10 days**

Module E: At TE

Allocation of ATT	Minutes
1 Lesson Objectives (LOs) / Teaching Points 2 003.03 F2F 650 mins	4905

3 003.04 F2F 800 mins 4 003.05 F2F 1900 mins 5 003.06 F2F 200 mins 6 003.07 F2F 1355 mins	
1 Enabling Checks (ECs) 2 EC 003.03 60 mins 3 EC 003.04 60 mins 4 EC 003.05 720 mins 5 EC 003.06 140 mins 6 PC 003.07 320 mins	1300
ATT Sub-total	**6205**
Allocation of BAT, TAT, TTT, PT (Course < 31 days)	
Base Administrative Time (BAT)	120
Training Administrative Time (TAT)	120
Training Travel Time (TTT)	450
Physical Fitness (PT)	540
Sub-total	**1230**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**7435** **19 days**

Module F: DL

Allocation of ATT	Minutes
Lesson Objectives (LOs) / Teaching Points 1 005.01 DL 525 mins – Inspect Small Engines 2 006.01 DL 405 mins - Inspect gearing and shaft lines 3 006.02 DL 225 mins – Inspect steering systems	1115
Enabling Checks (ECs) 1 EC 005.01 60 mins 2 EC 006.01 60 mins 3 EC 006.02 60 mins	180
Performance Checks (PCs)	0

ATT Sub-total	**1295**
No allocation of BAT, TAT, TTT, PT for DL modules	
Course Duration Calculation Total mins/400 = # days (rounded up)	
Total Training Days	**3 days**
Class A Days allocated	**4.5 days**

Module F: At TE

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points 1 005.02 F2F 930 mins - Maintain Small Engines	930
Enabling Checks (ECs) 2 EC 005.02 260 mins	260
ATT Sub-total	**1190**
Allocation of BAT, TAT, TTT, PT (Course < 10 days)	
Base Administrative Time (BAT)	60
Training Administrative Time (TAT)	60
Training Travel Time (TTT)	0
Physical Fitness (PT)	0
Sub-total	**120**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**3 days**
Overall Training Day and Class A Totals	
Grand Total Training Days with Additional Class A Days Added (The breakdown is 56 days at the TE and 34.5 days Class A at the NRD)	**90.5 Days**

RESOURCES

6. The following table details key personnel required to conduct the training:

Required Course Staffing					
Position	**Rank**	**Occupation**	**Qty**	**Module**	**PO / EO**
Instructor	MS	Any as content is general	1	B	001.01

Instructor	MS	HRAI Instructor (outside DND)	1	D	001.02
Instructor	MS	Any as content is academic	1	A	001.03
Instructor	MS	Any as content is general	1	B	001.04
Instructor	MS	Any as content is general	1	C	001.05
Instructor	MS	Any as content is academic	1	B	002.01
Instructor	MS	MARTECH - Mech	1	B	002.02
Instructor	MS	MARTECH - Mech	1	B	002.03
Instructor	MS	MARTECH - Mech	1	B	002.04
Instructor	MS	MARTECH - Mech	1	B	002.05
Instructor	MS	MARTECH - Mech	1	C	002.06
Instructor	MS	MARTECH - Mech	1	C	002.07
Instructor	MS	MARTECH - Mech	1	D	002.08
Instructor	MS	MARTECH - Mech	1	D	002.09
Instructor	MS	MARTECH - Mech	1	D	002.10
Instructor	MS	MARTECH - Mech	1	D	002.11
Instructor	MS	MARTECH - Elec	1	E	003.01
Instructor	MS	MARTECH - Elec	1	E	003.02
Instructor	MS	MARTECH - Elec	1	E	003.04
Instructor	MS	MARTECH - Elec	1	E	003.05
Instructor	MS	MARTECH - Elec	1	E	003.06
Instructor	MS	MARTECH - Elec	1	E	003.07
Instructor	MS	MARTECH - Elec	1	E	003.08
Instructor	MS	MARTECH - Mech	1	B	004.01
Instructor	MS	MARTECH - Mech	1	D	004.02
Instructor	MS	MARTECH - Mech	1	F	005.01
Instructor	MS	MARTECH - Mech	1	F	005.02
Instructor	MS	MARTECH - Mech	1	F	006.01
Instructor	MS	MARTECH - Mech	1	F	006.02
Instructor	MS	MARTECH - Mech	1	D	007.01
Instructor	MS	MARTECH - Mech	1	D	007.02
Instructor	MS	MARTECH - Mech	1	D	007.03
Instructor	MS	MARTECH - Mech	1	D	007.04

7. The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP. Details on material included in specific packages are found in Annex H.

Critical Training Resources	Module	EO/PO
Stationary	All	All
Scientific Calculator (on loan)	A, B	001, 002, 003
General Classroom	All	All
Universal Classroom	All	All
Laptop Computer (DWAN) – each trainee	All	All
DWAN Drop – each trainee	All	All
Lab Volt Equipment	E	003
Basic Tools	D, F	002, 005
Power Tools	D, F	002, 005
Testing Equipment	E	003
LOTO Equipment	B	001, 002, 003
Oxy / Acetylene Equipment	D	007
Electrical Tools	E	003
Safety Equipment	All	All

COURSE CAPACITY

8. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification can be found in Annex C.

a. Method 1:

Module A, B, C, D, E, F: DL	Module C, D, E, F: In-house
Maximum – 60	Maximum – 12
Minimum – 1	Minimum – 1
Optimum – 30	Optimum – 12

b. Method 2:

(1) Maximum – 12;

(2) Minimum – 1; and

(3) Optimum – 6.

MITE

9. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

10. Scheduling. It is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends, and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays

PREREQUISITES

11. The loading authority must ensure that trainees have the following prerequisites before being selected for this course: Basic Military Qualification (BMQ)

QUALIFICATION.

12. Successful completion of this course constitutes the primary means of attaining the MAR TECH RQS3 Basic Maintainer (NQUAL: ALJS, Final MITE ID 122592. The following MITE Course IDs will be granted based on training delivery method:

Method	MITE Course ID
Full Time	**Final MITE ID:** 122592 **Full time:** 122591 **Additional MITE Codes Granted:** 1 DRMIS Maintainer via CBT classroom during MOD C - 119096 2 ODS via CBT classroom during MOD D - 109133 **One of the following:** 1 USQ Roundsperson Halifax Class Residential Phase –121178 2 USQ Roundsperson HDW Class Residential Phase –TBD 3 USQ Roundsperson JSS Class Residential Phase –TBD 4 USQ Roundsperson JSS Class Residential Phase –122629
Part Time	**Final MITE ID:** 122592 **Part time:** 122603 1 Mod A DL: 122593 2 Mod B DL: 122594 3 Mod C DL: 122595 (this was moved to Mod B DL in its entirety)

	4 Mod C TE: 122596 & DRMIS Maintainer 11909 5 Mod D DL: 122597 & ODS via DL through HRAI 109133 6 Mod D TE: 122598 7 Mod E DL: 122601 8 Mod E TE: 122602 9 Mod F DL: 122599 10 Mod F TE: 122600 **One of the following to occur year 2 summer:** 1 USQ Roundsperson Halifax Class Residential Phase –121178 2 USQ Roundsperson Kingston Class Residential Phase – 121180 3 USQ Roundsperson Orca Class Residential Phase –121182

COURSE REPORTING

13. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

SPECIAL INSTRUCTIONS

15. All trainees must be in possession of a valid Test Fitting Card for the Half Mask prior to commencing PO 002 Perform Basic Maintenance on Fluid System and PO 007 Perform Basic Naval Welding Operations.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

GENERAL

1. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

2. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

3. Dress and deportment, ethics, conduct, and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

4. Enabling Checks (ECs). ECs are formative assessments that help the trainee, and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

5. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES vol 6 with the unsatisfactory progress documented in their Trainee File.

6. Critical Enabling Checks (CECs). CECs are summative assessments at the end of an EO which are considered critical because trainees cannot progress in the training without first successfully completing them. They are related to elements such as safety and security where harm could result from poor performance. CECs are treated the same as PCs.

7. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

8. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

9. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 “Training Review Board Process”. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

10. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

11. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

PROGRESS MONITORING

12. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained. Formative assessment may include: ECs, Homework, Assignments, Quizzes, Practice periods, and any other method or tool used to develop the trainee.

13. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

14. Continuous monitoring of candidate progress is required to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for any required TRBs; and

d. Information to assist in completion of the CF 377 Course Report.

15. Trainees experiencing difficulty in any area of performance are to be counselled regarding the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

16. Instructional staff shall maintain individual trainee files, including the results of all:

a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);

b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and

c. Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

17. TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process”.

18. In the event of prolonged trainee absence, TE Staff shall determine, on a case-by-case basis, the necessity to convene a TRB.

COURSE REPORTS

19. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications”.

20. Trainees will be graded pass or fail whenever possible. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

21. The Course Report narrative comments could include:

a. Ability to work with others;

b. Ability to work under pressure;

c. Written and verbal communication skills;

d. Organizational skills; and

e. Leadership ability.

22. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

PO 1 Complete Technical Administration

1. Performance. Complete technical administration.
2. Conditions.
   a. Given:
      (1) PPE;
      (2) LOTO tools and devices;
      (3) Hazardous materials spill response kits;
      (4) DRMIS account;
      (5) DLN 3.0 account; and
      (6) References.
   b. Denied: Nil.
   c. Environmental: Any condition, at sea or ashore.
3. Standard The basic maintainer shall independently complete basic technical administration IAW refs by:
   a. Adhering to safety requirements for engineering operations to include:
      (1) Inspecting personal protective equipment;
      (2) Applying safety precautions in industrial environments;
      (3) Applying lock out tag out devices and procedures; and
      (4) Following hazardous materials and spills guidelines and procedures.
   b. Apply ODS regulations (Task#) IAW Halon 1301 Extinguishing System Maintainer QSP MITE 109149.
   c. Interpreting drawings to include the following:
      (1) System schematics; and
      (2) Schematic symbols.
   d. Maintain DRMIS logs IAW Mod 1 of DRMIS Course QSP MITE 119096 to include utilizing DRMIS portal as a maintainer
4. Reference
5. Task, Skill, Knowledge Numbers.
6. Training Limitations. Nil.
7. Remarks.
   a. This PO consolidates various administrative and general components of tasks completed by a S3 MAR TECH.
   b. “Mod 0” is included in the first EO as two lessons within the DLN. The first lesson is a brief introduction to the RCN, and the second is an introduction to the MAR TECH trade.

   c. The subtasks for Applying ODS regulations and DRMIS are as described in the QSP MITE 109149 and QSP MITE 119096 respectively and will need to be updated should that document be revised.

8. Test Details. The assessment for this PO will be conducted as an individual EC at the end of each EO. Test details are in Annex L.

EO 1.01 Adhere to Safety Requirements for Engineering Operations

1. Performance. Adhere to safety requirements for engineering operations.
2. Conditions.
   a. Given:
      (1) PPE;
      (2) Drafting tools (IAW Annex H);
      (3) LOTO devices and tools;
      (4) Hazardous material spill kits
      (5) DRMIS account;
      (6) DLN 3.0 account; and
      (7) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any condition, at sea or ashore;
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall adhere to safety requirements for engineering operations IAW refs by:
   a. Inspecting personal protective equipment;
   b. Applying safety precautions in industrial environments;
   c. Applying lock out tag out devices and procedures; and
   d. Following hazardous materials and spills guidelines and procedures.
4. Time Required.
   a. EO – 590 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A1, A2, A3, A45, A46, A60, A61, A62, A63, D5
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. For method 1, EO 001.01 will be part of Module B and taught by CBT. It is a prerequisite for completion prior to Module C, D, E, and F.
   b. For both methods, EO 001.01 shall be completed prior to any training within workshops, labs, or other environments requiring safety precautions.
   c. Module 0 is delivered in the DLN and should be done at the very beginning of the training. In Module 0, you will learn about the career lifestyle you can expect in the Royal Canadian Navy (RCN) as well as the roles and responsibilities that are undertaken as a MAR TECH S3. In addition, you will be given a brief overview of the ship vessel and rank structure adhered in the RCN as well as how the MAR TECH S3 training program will be delivered.

d. Trainees will be continually assessed on safety practices throughout the entirety of the RQ-S3 course with safety being considered critical success criteria during all practical assessments.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
Module 0		**Method / Media:** DL **Trg Environment:** Classroom / CBT	180
Mod 0 is broken into 2 DLN Lessons which are: 1 Introduction to the RCN 2 Introduction to MAR TECH Training Introduction to the RCN: 3 Explain what will be learned in NETP 4 Describe the tasks done during life at sea 5 Explain the characteristics of life onboard ship 6 Describe a typical work schedule of a sailor 7 Identify every ship vessel available in the RCN 8 Identify the different ship departments available in a ship vessel 9 Identify the rank structure administered in the RCN Introduction to MAR TECH Training • Describe how the MAR TECH S3 training program will be delivered • Identify the seven EOs in the MAR TECH S3 training program • Describe the tasks and responsibilities of a MAR TECH S3 • Explain the career path of a MAR TECH In Module 0, you will learn about the career lifestyle you can expect in the Royal Canadian Navy (RCN) as well as the roles and responsibilities that are undertaken as a MAR TECH S3. In addition, you will be given a brief overview of the ship vessel and rank structure adhered in the RCN as well as how the MAR TECH S3 training program will be delivered.			
LO 001.01.01: Identify appropriate Personal Protective Equipment			
1 Define the purpose of PPE; 2 Describe types and use of generic PPE within a work environment to include: a. Goggles; b. Face Shield; c. Welding Mask;		**Method / Media:** DL **Trg Environment:** Classroom / CBT. **Activity Details:** Trainees will go through the online training module to complete all lessons, knowledge checks and EC's. **Trg Resources:**	75

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
d. Work Gloves; e. Ear Defenders; f. Ear Plugs; g. N95 Mask h. P100 Mask i. Self-Contained Breathing Apparatus (SCBA) j. Hard hats k. Long pants/shirt l. Steel-toe boots; m. Coveralls n. Welding aprons/jacket and pants o. HAZMAT suits		Can be found in the Resources folder for the MAR TECH RQ S3 Group in the DLN.	
LO 001.01.02: Apply Safety Precautions in Industrial Environments			
1 Apply basic shop safety to include: a. Jewellery removal; b. Clothing, hair and loose material hazards; c. Hazards of contact lenses; d. Proper use of a machine/shaft guards; e. Appropriate shop/workplace behaviour and etiquette; and f. Due diligence (never pass a fault).		**Method / Media:** DL **Trg Environment:** Classroom / CBT. **Activity Details:** Trainees will complete the online training module. **Trg Resources:** Video – 38657 6301	50
LO 001.01.03: Apply Lock out / Tag out (LOTO)			
2 Describe the purpose of LOTO 3 Identify LOTO Devices (valve and electrical LOTO devices, LOTO locks and stations). a. Valve b. Turn ball c. Butterfly d. Cable or chain e. Electrical		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will go through the online training module to complete all lessons, knowledge checks and EC's. **Trg Resources:** Lockout Safety Video LOTO Procedures Video	75

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
f. locks and stations g. Tagout devices 4 Describe LOTO Procedure 5 Define: a. Assurance Test; b. Due Diligence; c. Lockout Device; d. Lock out/Tag out; e. Lock out/tag out station; f. Locks; g. Tag out device; and h. LOTO Responsibilities. 6 Describe LOTO SEMS Annexes, to include the following: a. Annex S4A; b. Annex S4B; c. Annex S4C; and d. Annex S4D.			
LO 001.01.04: Hazardous Materials and Spills			
1 Define HAZMAT terms, to include: a. HAZMAT; b. Material Safety Data Sheets (MSDS); c. Spills; and d. Fuelling.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will go through the online training module to complete all lessons, knowledge checks and EC's. **Trg Resources:** Halifax Class SEMS Manual Ship Structure Performance Support Reference Card SEMS Fleet Video	75
LO 001.01.05: Describe Spill Responses			
1 Explain the spill response process from SEMS to include the following:		**Method / Media:** DL **Trg Environment:** CBT/Classroom	135

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
a. Purpose of spill response; b. Personnel responsibilities; c. Initiation of a Spill response (pipe); and d. Action of a discoverer. 2 Explain the spill response process from SEMS to include the following: a. Alongside: (1) Purpose of spill response; (2) Personnel responsibilities; (3) Initiation of a Spill response (pipe); (4) Action of a discoverer; and (5) QHM. b. At Sea: (1) Purpose of spill response; (2) Personnel responsibilities; (3) Initiation of a Spill response (pipe); and (4) Action of a discoverer. 3 Describe RADHAZ safety procedures to include the following: a. Definition of RADHAZ; b. Onboard RADHAZ safety; and c. Relation of RADHAZ to Spill Response. 4 Describe Spill response equipment, to include the following: a. Components and configuration of shipboard small spill response kit. b. Components and configuration of Shipboard large spill response kit. c. Components and configuration of Jetty.		**Activity Details:** Students will complete this lesson via CBT. **Trg Resources:** DLN 3.0 account	

10. Test Details. EO 001.01 will be assessed in an online assessment IAW annex L. There will be no summative assessment for PO 001. Formative assessment will ensure the trainees are prepared to progress on to practical training and each part of tech admin is further assessed during these practical assessments later in the course. PO 001 is summatively assessed as part of PC 003.

EO 1.02 Apply ODS Regulations IAW QSP MITE 109149

EO 1.03 Calculate Basic Mathematics

1. Performance. Perform Basic Mathematics.
2. Conditions.
   a. Given:
      (1) Formulas;
      (2) DLN 3.0 account; and
      (3) References.
   b. Denied: Nil.
   c. Environmental: Any condition, at sea or ashore.
      (1) Any location with access to DLN 3.0; and
      (2) Classroom.
3. Standard. The basic maintainer shall independently calculate basic mathematics IAW refs by:
   a. Calculating positive and negative integers;
   b. Conducting order of operations;
   c. Calculating fractions;
   d. Performing round and significant figures;
   e. Performing scientific notation;
   f. Solving mathematical equations using ratios, integers and fractions;
   g. Calculating decimals;
   h. Performing percentage and estimate calculations;
   i. Calculating powers, exponents, square numbers and roots;
   j. Performing arithmetic operations and evaluating expressions;
   k. Performing algebra;
   l. Calculating area and volume;
   m. Performing arithmetic using the quadratic formula; and
   n. Performing graphing.
4. Time Required.
   a. EO – 1080 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References.
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. In method 2, EO 001.03 must be completed prior to all other POs.
   b. In method 1, EO 001.03 is Module A and is a prerequisite for all other modules.
   c. EC 001.03 is to be invigilated by an instructor or regional instructor.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 001.03.01: Calculate with Positive and Negative Integers			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	50
LO 001.03.02: Order of Operations			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to solve problems with instructor/regional instructor support as required. **Trg Resources:** Videos describing simplifying expressions	50
LO 001.03.03: Perform Calculations with Fractions			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given information, videos, and calculations to complete for performing arithmetic calculations with instructor/regional instructor support as required. **Trg Resources:** Videos describing arithmetic calculations	50
LO 001.03.04: Calculate with Positive and Negative Fractions			
• Description. • Examples.		**Method / Media:** DL **Trg Environment:** CBT/Classroom	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
• Practice.		**Activity Details:** Trainee will be given information, videos, and calculations to complete for performing algebraic calculations with instructor/regional instructor support as required. **Trg Resources:** Videos describing algebraic problems	
LO 001.03.05: Scientific Notations			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given information, videos, and calculations to complete for performing basic geometric calculations with instructor/regional instructor support as required. **Trg Resources:** Videos describing geometry concepts	50
LO 001.03.06: Solving Problems Using Ratios Integers and Fractions			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	50
LO 001.03.07 Decimals			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required.	150

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		Trg Resources: DLN 3.0 account	
LO 001.03.08 Perform Arithmetic Calculations Part 1			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:**	50
LO 001.03.09 Perform Arithmetic Calculations Part 2			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	50
LO 001.03.10 Arithmetic Operations and Evaluating Expressions			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	50
LO 001.03.11 Use Algebra to Solve Problems			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical	150

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		calculations with instructor/regional instructor support as required. **Trg Resources:** Notes: Once this LO has been completed students are allotted 100 min for an assignment. Timing is included in this LO.	
LO 001.03.12 Using Geometry to Determine Area with Volume			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	50
LO 001.03.13 Using Quadratic Formula			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	90
LO 001.03.14 Graphing Mechanics			
• Description. • Examples. • Practice.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information, videos, and calculations to complete for performing basic mathematical calculations with instructor/regional instructor support as required. **Trg Resources:** DLN 3.0 account	190

10. Test Details. EO 001.03 will be tested in EC 001.03 IAW Annex K. Test details can be found in Annex L.

EO 1.04 Interpret Drawings

1. Performance. Interpret drawings.
2. Conditions.
   a. Given:
      (1) Handout of symbol legends;
      (2) NDID table;
      (3) DRMIS account;
      (4) DLN 3.0 account; and
      (5) Assistance;
      (6) Supervision; and
      (7) References.
   b. Denied: Nil.
   c. Environmental: Any condition, at sea or ashore.
      (1) Any location with access to DLN LMS; or
      (2) Classroom.
3. Standard. The basic maintainer shall interpret drawings IAW refs by:
   a. Identifying system schematics; and
   b. Identifying schematic symbols.
4. Time Required.
   a. EO – 150 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A5, A6, A7, A8, A9, A25, A26, A47, A48, A49, A53, A60, C2, D1, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. For Method 1, EO 001.04 is part of Module B and is completed using CBT
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 001.04.01: Identify Drawing Components			
1 Identify Drawings, to include the following: a. National Defense Index of Documentation (NDID). b. NDID Activity. c. Title Block Drawing Numbers. 2 Interpret Drawings, to include the following: a. Dimensions and Tolerance; b. Expressing Tolerance; c. Actual and Nominal size; d. Additional Size Definitions; e. Dimensioning Lines; f. Dimensioning: The Dos; and g. Dimensioning: The Don'ts. 3 Explain projection drawings, to include the following: a. Orthographic projection; b. Isometric projection; and c. Oblique projection. 4 Explain freehand sketching techniques, to include the following: a. Horizontal lines; b. Vertical lines; c. Incline lines and angles; and d. Arcs, circles, and ellipses, 5 Explain Detail and Assembly Drawings, to include the following: a. Samples; and b. Guidelines.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given information from teaching points. **Trg Resources:** Handout of symbol legends; NDID.	100
LO 001.04.02: Identify System Schematics and Symbols			
1 Identify drawing symbology, to include the following: a. Hydraulic symbols; and b. Mechanical symbols.		**Method / Media:** DL **Trg Environment:** CBT/Classroom	50

	c. Electrical symbols		**Activity Details:** Trainees will be given information from teaching points. **Trg Resources:** Nil	
2	Identify the fundamentals of system schematic drawing, to include the following: a. Flow direction; b. Component indicators; c. Machinery indicators; and d. Different layouts.			

10. Test Details. EO 001.04 will be tested in EC 001.04 IAW Annex K. Test details can be found in Annex L.

EO 1.05 Maintain DRMIS logs IAW Mod 1 of DRMIS QSP MITE 119096

PO 2 Maintain Pressurized Systems

1. Performance: Maintain pressurized systems.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) Schematics;
      (3) PPE;
      (4) Basic tools (IAW Annex H);
      (5) Power tools (IAW Annex H);
      (6) Testing equipment (IAW Annex H);
      (7) LOTO equipment (IAW Annex H);
      (8) Oxy / Acetylene equipment;
      (9) DRMIS account;
      (10) DLN 3.0 account;
      (11) Assistance;
      (12) Supervision; and
      (13) References.
   b. Denied: Nil.
   c. Environment: Any conditions on ship at sea or alongside.
3. Standard: The S3 maintainer shall maintain pressurized systems under supervision IAW refs by:
   a. Inspecting fluid system components to include the following:
      (1) Shipboard fluid systems;
      (2) Pressurized systems;
      (3) Piping and tubing;
      (4) Hoses;
      (5) Tanks;
      (6) Lagging and insulation;
      (7) Shipboard valves;
      (8) Pumps;
      (9) Seals;
      (10) Strainers;
      (11) Filters;
      (12) Heat exchangers;
      (13) Gauges and monitoring equipment; and
      (14) Eductors.
   b. Completing fluid testing.
   c. Inspecting pressurized systems to include the following:

(1) Shipboard freshwater system;
(2) Reverse osmosis desalination;
(3) Freshwater systems on RCN ships;
(4) Shipboard sanitary systems;
(5) Treatment system;
(6) Black and grey water systems;
(7) Feedwater system;
(8) Shipboard domestic steam system;
(9) Boiler;
(10) Chilled water/Hydronic system on RCN ships;
(11) Heating ventilation and air conditioning system;
(12) Chemical, biological, radiological, and nuclear (CBRN) defense system; and
(13) Compressed air systems.

d. Inspecting the liquid damage control system to include the following:
(1) Sea water service systems on RCN ships; and
(2) Shipboard fire suppression systems.

e. Inspecting fuel, oil, and lubricants to include the following:
(1) Petroleum, oils and lubricants (POL's);
(2) Dirty oil system;
(3) Bilge stripping system;
(4) Fuel oil transfer system;
(5) Fuel oil service system;
(6) Lube oil; and
(7) Coalescer as a sub-system of a fuel oil service system.

f. Comparing fuel oil transfer systems across RCN platforms.

g. Repairing metal components, to include:
(1) Recommending repair method;
(2) Preparing area for work;
(3) Installing metal shapes and components by:
(a) Installing metal fasteners;
(b) Cutting metal;
(c) Repairing bolt threads;
(d) Drilling/tapping holes; and
(e) Filing metal.

h. Maintaining pumps and valves to include:
(1) Maintaining basic valves by:
(a) Disassembling valves;

(b) Repairing valves;

(c) Assembling valves; and

(d) Testing valves;

(2) Maintaining pumps by:

(a) Disassembling pumps;

(b) Repairing grease nipples;

(c) Inspecting shock mounts;

(d) Lubricating bearings;

(e) Lubricating mechanical linkage;

(f) Removing/replacing seals; and

(g) Assembling pumps.

i. Repairing pipes, hoses, and insulation to include the following:

(1) Hoses;

(2) Tubing;

(3) Piping; and

(4) Lagging and insulation.

j. Maintaining fluid components to include:

(1) Testing feed water system;

(2) Inspecting strainers;

(3) Cleaning strainers;

(4) Changing filters;

(5) Starting centrifuges; and

(6) Repairing marine vacuum toilets by:

(a) Locking/tagging out system;

(b) Disassembling toilet;

(c) Troubleshooting fault;

(d) Repairing (replacing) components; and

(e) Completing a function test.

k. Maintaining air systems to include:

(1) Inspecting air compressor components for leaks and abnormal noises;

(2) Starting air compressor;

(3) Logging pressures;

(4) Checking oil levels;

(5) Purging water;

(6) Replacing air filters;

(7) Isolating/De-isolating valves;

(8) Venting cooling coils;

(9) Replacing air filters

(10) Removing/replacing 3-way valves; and

(11) Removing/replacing motor fan components.

l. Maintaining refrigeration systems by:

(1) Inspecting refrigeration cycle;

(2) Inspecting refrigeration systems;

(3) Completing refrigeration safety;

(4) Inspecting leak detection process on refrigeration systems;

(5) Testing for leaks on refrigeration systems;

(6) Completing refrigeration rounds to include:

(a) Detecting leaks;

(b) Checking oil levels;

(c) Logging running hours;

(d) Checking suction pressure;

(e) Logging sea water temperature;

(f) Logging fridge temperature; and

(g) Checking for moisture.

4. References.

5. Task, Skill, Knowledge Numbers.

6. Training Limitations. Practical portions of PO 002 require appropriate workshops in the TE to carry out practical assignments and assessments.

7. Remarks.

8. Test Details. The assessment for this PO will be conducted as an individual EC at the end of each EO.

EO 2.01 Apply Foundational Science

1. Performance. Apply foundational science.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS account;
      (3) DLN 3.0 account;
      (4) Assistance;
      (5) Supervision; and
      (6) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall apply foundational science IAW refs by:
   a. Performing calculations using units of measure;
   b. Applying basic mechanical concepts;
   c. Applying basic energy concepts;
   d. Inspecting simple machines;
   e. Applying chemistry concepts;
   f. Applying basic thermodynamic concepts; and
   g. Applying basic fluid dynamics concepts.
4. Time Required.
   a. EO – 400 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References.
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. This EO is the grouping of theory and knowledge components of PO 002.
   b. For Method 1, EO 002.01 is part of Module B and can be completed either by CBT (self-study) or in a classroom dependent on resource availability.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.01.01 Describe basic units of measurements			
1 Recognize measurements and types of descriptions, to include the following: a. Qualitative descriptions; and b. Quantitative descriptions. 2 Describe units of measurement, to include the following: a. Systems of units; b. Base units; c. Mechanical base quantities;; d. Standards; e. Measurement calibration; f. SI Prefixes; g. Derived units; and h. Common units in the RCN. 3 Describe unit conversions, to include the following: a. SI prefix conversion process; b. Conversion factors; and c. Unit conversion procedure.		**Method / Media: DL** **Trg Environment**: CBT/classroom **Activity Details**: Trainees will be given images, videos, animations, and interactive tools to aid in the description and identification of different measurements. Trainees will complete a short assignment on measurements. **Trg Resources**: ● Handout displaying tables of measurements with comparisons between metric and imperial ● Images, videos, and or animations for measurements	50
LO 002.01.02: Describe Basic Mechanics Concepts			
1 Describe the mechanics of physics, to include the following: a. Objects in motion; b. Velocity and acceleration; c. Concept of force; d. Newton's laws of motion; e. Types of forces; f. Normal force; g. Gravitational force; h. Friction; i. Friction force; j. Magnitude of friction; and k. Contact area.		**Method / Media:** DL **Trg Environment:** CBT/classroom **Activity Details:** Trainees to be given text with images, videos, and animations to illustrate the terminology and interactive images to show physic concepts. Trainees will complete a short assignment on basic physics concepts. **Trg Resources:** ● Handout breaking down the core concepts of physics ● Images, videos, and/or animations for physics concepts	50

LO 002.01.03: Describe Basic Energy Concepts			
1 Describe the different energy types to include the following: a. Mechanical; b. Chemical; c. Thermal; and d. Electrical. 2 Describe energy, to include the following: a. Mechanical energy; b. Conservation of energy; and c. Useful energy. 3 Describe work, to include the following: a. Transitional energy; b. Work; c. Power; d. Rotation; and e. Torque.		**Method / Media:** DL **Trg Environment:** CBT/classroom **Activity Details:** Trainees to be given text with images, videos, and animations to illustrate the terminology and interactive images to show physic concepts. Trainees will complete a short assignment on basic physics concepts. **Trg Resources:** ● Handout breaking down the core concepts of physics ● Images, videos, and/or animations for physics concepts	50
LO 002.01.04: Describe Basic Simple Machines			
1 Describe simple machines, to include Mechanical advantage. 2 Describe types of machines, to include the following: a. Classes of lever; and b. Simple machine applications. 3 Describe motion, force, and energy.		**Method / Media:** DL **Trg Environment:** CBT/classroom **Activity Details:** Trainees to be given text with images, videos, and animations to illustrate the terminology and interactive images to show physic concepts. Trainees will complete a short assignment on basic physics concepts. **Trg Resources:** ● Handout breaking down the core concepts of physics ● Images, videos, and/or animations for physics concepts	50
LO 002.01.05: Describe Basic Chemistry Concepts			
1 Describe acids and bases, to include the following:		**Method / Media:** DL **Trg Environment:** CBT/classroom	50

a. Fundamentals of acids; b. Fundamentals of bases; c. Potential of hydrogen (pH); and d. Handling acids and bases.		**Activity Details:** Trainees to be given text with images, videos and animations to illustrate the terminology and interactive images to show chemistry concepts. Instructor to link to feed water testing for context. Trainees will complete a short assignment on basic chemistry concepts. **Trg Resources:** ● Handout of the PH Scale ● Images, videos, and/or animations of chemistry concepts	
LO 002.01.06: Describe Basic Thermodynamic concepts			
1 Describe thermodynamics and energy, to include an introduction to thermodynamics and energy. 2 Describe thermal energy, to include the following: a. Comparing mechanical and thermal energy; and b. Stored thermal energy. 3 Describe temperature and temperature scales. 4 Describe heat, to include the following: a. Material phase; b. Phase changes; c. Sensible heat and latent heat; d. Heat transfer; e. Conduction; f. Convection; and g. Radiation.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee to be given text with images, videos and animations to illustrate the terminology and interactive images to show thermodynamic concepts. Trainees will complete a short assignment on basic thermodynamic concepts. **Trg Resources:** ● Handout showing the basics of thermodynamics ● Images, videos, and/or animations of thermodynamic concepts	75
LO 002.01.07: Describe Basic Fluid Dynamic Concepts			
1 Describe fluid mechanics. 2 Describe pressure, to include the following: a. Definition of pressure; b. Generating pressure;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees to be given text with images, videos, and animations to illustrate the terminology and interactive	75

c. Pressure references; d. Variation of pressure with depth and height; e. Open tanks; and f. Pressure measurement. 3 Describe density, to include the following: a. Definition of density; b. Compressibility; and c. Separation. 4 Describe viscosity, to include the following: a. Definition of viscosity; b. Viscosity management; c. SAE viscosity; and d. Flow rate. 5 Describe Gas, and Gas laws. 6 Describe Fluid power.		images to show fluid dynamic concepts. Trainees will complete a short assignment on basic fluid dynamic concepts. **Trg Resources:** ● Handout showing the basics of fluid dynamics ● Images, videos, and or animations of fluid dynamic concepts	

10. Test Details. Test Details: EO 002.01 will be tested in EC 002.01 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.02 Inspect Fluid System Components

1. Performance. Inspect fluid system components.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS account;
      (3) DLN 3.0 account;
      (4) Assistance;
      (5) Supervision; and
      (6) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect fluid system components IAW refs by:
   a. Observing for abnormal vibrations, smells, and sounds;
   b. Checking for cracks;
   c. Checking for sweating, leaks, mould, and bubbles;
   d. Checking for icing;
   e. Looking for obvious damage;
   f. Checking for corrosion;
   g. Investigating general wear and tear;
   h. Checking for abnormal temperatures;
   i. Checking for alarm indications;
   j. Looking for discoloration; and
   k. Performing fluid testing.
4. Time Required.
   a. EO – 550 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A10
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. For Method 1, EO 002.02 is part of Module B and can be completed either by CBT (self-study) or in a classroom dependent on resource availability.

b. Any LO stating, they will compare systems or equipment "on all RCN Classes" will not be part of the EC or PC. It is a link between the RQS3 and Roundsperson OJPR Package.

c. This is the theory EO for pressurized system components and an introduction to troubleshooting skills that will be completed during OJT.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.02.01 Describe a Basic Shipboard Fluid system			
1 Describe a basic fluid system, to include the following: a. Purpose of shipboard fluid systems; b. Hydraulic and pneumatic systems; c. Common components; and d. Fluid system examples.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic shipboard liquid system. **Trg Resources:** ● Animations and/or handouts emphasizing the three core component concepts of any liquid system.	25
LO 002.02.02 Describe pressurized system safety			
1 Describe pressurized fluid storage and transport, to include the following: a. Pipes, tubes, and hoses; and b. Cylinders. 2 Describe pressurized fluid systems, to include the following: a. Pressure limits; b. Depressurizing; c. Isolation (Shut off valves); d. Double valve isolation; e. Hose management; f. Personal protective equipment (PPE); and g. Static electricity production. 3 Describe pressurized gas cylinder safety, to include the following:		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given images showing types and characteristics of fluid safety. **Trg Resources:** ● Pictures of pressurized system components ● Video on safety for pressurized equipment	50

a. Cylinder construction; b. Cylinder built-in safeties; and c. Proper storage. 4 Describe pressurized gases, to include the following: a. Halon; b. Refrigerants; c. Oxygen; d. Helium e. Acetylene; f. Carbon Dioxide; g. Argon; and h. Nitrogen.			
LO 002.0.03 Describe piping and tubing			
1 Describe piping and tubing, to include the following: a. Purposes of piping and tubing; and b. Shipboard markings. 2 Describe material and identification, to include the following: a. Piping materials; b. Tubing materials; and c. Identification criteria. 3 Describe joining methods, to include the following: a. Pipe fittings; b. Tube fittings; c. Other fittings; d. Permanent joints; and e. Tasks and associated equipment.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given images showing types and characteristics of fluid safety. **Trg Resources:** DLN 3.0 account	25
LO 002.02.04 Describe hoses			
1 Describe the purpose of a hose. 2 Describe the construction of a hose, to include the following; a. Commonalities of hose construction; and;		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given images showing types and characteristics of hoses and fittings,	25

b. End connections. 3 Describe hoses used for different types of fluids, to include the following: a. Pneumatic (Air lines); b. Hydraulic; c. Potable water; d. Fire hose; e. Steam; f. Black water; and g. Fuel. 4 Describe joining methods using fittings, to include the following: a. Threaded; b. Flanges; c. Compression; and d. Specialized fittings. 5 Describe fire hose accessories to include the following: a. Double male/female coupling; b. Water Thief (tri-gate); c. Reducers; d. Wye gate; e. Caps; f. In-line AFFF Educators; and g. Hose spanners. 6 Describe the safety checks conducted on a hose, to include the following: a. Cracks; b. Cuts or gouges; c. Frayed wire; d. Bulging or bubbles; e. Missing o-rings, lubrication, or cleanliness; f. Broken, corroded, or pitted wedding bands or connections; and g. Dry, hard plastic-type rubber.		and types of deterioration and defects. They will also be given animated cut-away views identifying components and video(s) showing care and custody of hoses to cover off all the teaching points. **Trg Resources:** ● Pictures of components ● Video on hose care	

7 Describe basic storage and husbandry for portable hoses, to include the following: a. Proper rolling / coiling; b. Separating hoses by medium; c. Appropriate environment ; d. Cleanliness; and e. Chlorinating potable water hoses.			
LO 002.02.05 Describe tanks			
1 Describe tanks, to include the following: a. Purpose of a tank; b. Common types of tanks; and c. Common components of a tank.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given text and images showing the purpose and components of sumps and tanks. **Trg Resources:** ● Images of sumps and tanks	25
LO 002.02.06 Describe lagging and insulation			
1 Describe lagging and insulation, to include the following: a. Purpose of lagging and insulation; b. Properties of insulation; c. Purpose of vapour barriers; d. Types of insulating materials; e. Materials used with lagging and insulation; and f. Hazards and safety precautions.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given text and images showing the purpose, types and uses of insulation and bonding materials. Video showing installation techniques. **Trg Resources:** ● Images of insulation and lagging	50
LO 002.02.07 Describe Shipboard Valves			
1 Describe the purpose of the valves and their characteristics to include the following: a. Types of valves and their components, to include: b. Gate valves; c. Globe valves;		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given images showing types and characteristics of fluid safety. **Trg Resources:** ● Pictures of pressurized system components	50

d. Diaphragm valves; e. Ball and butterfly valves; f. Check valves; g. Screw down non-return (SDNR) valves; h. Orifice plates; i. Steam traps; and j. Relief valves. 2 Describe valve actuation methods, to include the following: a. Actuation configurations; b. Local; c. Remote; and d. Automatic.		● Video on safety for pressurized equipment	
LO 002.02.08 Describe Shipboard Pumps			
1 Describe the purpose of pumps and their role in fluid systems, to include the following: a. Compressors; and b. Fans. 2 Describe the classification of pumps based on operating principle, to include the following: a. Rotary vs. reciprocating; b. Positive Displacement vs. Non-Positive Displacement; c. Reciprocating pump operation; d. Centrifugal pump operation; e. Specific pump types; and f. Energy sources for pumps. 3 Describe pump operational concepts, to include the following: a. Discharge valves; b. Relief valves; c. Air locking; d. Cavitation; and e. Priming.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees to be given images of pumps, compressors and fans, with prime movers identified (e.g. diesel engine driven fire pump, electric motor driven compressor, and electric fan). They will be shown 3D animations and cut-aways showing types of pumps, pump components, pump operation and fluid dynamics of various types of pumps. **Trg Resources:** ● Pictures of pumps ● 3D graphics of pumps	50
LO 002.02.09 Describe Seals			

1 Describe the purpose of seals. 2 Describe types of seals, to include the following: a. Gaskets b. O-rings; c. Lip seals; d. Labyrinth seals; e. Packed seals; and f. Mechanical seals.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees to be given 3D and cut-away images showing types of seals and videos showing replacing seals. **Trg Resources:** ● 3D graphics of seals	50
LO 002.02.10 Describe Strainers			
3 Describe the purpose and characteristics of strainers. 4 Describe types of strainers, to include the following: a. Strainer baskets; b. Simplex; and c. Duplex. 5 Describe Pressure readings, to include checking for strainer conditions.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given graphics, cutaway and animated images to show common types of strainers, their operation and relationship between pressure and strainer condition. **Trg Resources:** ● Images of strainers	25
LO 002.02.11 Describe Filters			
1 Describe the purpose of filters. 2 Describe filter types, to include the following: a. Air; b. Water; c. Oil; and d. Fuel.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given 3D, cutaway, and animated images to show an oil filter and its operation. **Trg Resources:** ● 3D graphics of filters	25
LO 002.02.12 Describe Heat Exchangers			
1 Describe the purpose of heat exchangers. 2 Review the principles of heat transfer, to include a review of basic thermodynamics concepts. 3 Describe types of heat exchangers, to include the following: a. Shell and tube;		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given 3D, cutaway and animated images to show a heat exchanger, its operation, disassembly and assembly for maintenance. **Trg Resources:**	50

b. Plate; c. Finned tube; and d. Heat sink.		● 3D graphics of heat exchangers	
LO 002.02.13 Describe Gauges and Monitoring Equipment			
1 Describe the purpose of gauges and monitoring equipment. 2 Describe the characteristics of gauges and monitoring equipment, to include the following: a. Monitoring; b. Interpretation of reading gauges; Remote monitoring c. Parallax error; d. Units/scale; and e. Communicating information. 3 Describe common types of gauges and monitoring equipment, to include the following: a. Parameters; b. Volume tank level; c. Pressure; d. Flow rate; e. Temperature; and f. Valve position.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given images to show various gauges and monitoring equipment. **Trg Resources:** ● Images / 3D graphics of gauges and monitoring equipment	25
LO 002.02.14 Describe Eductors			
1 Describe eductors, to include the following: a. Purpose of an eductor; b. Venturi effect; and c. Components and theory of operation of an eductor.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given 3D, cutaway, and animated images to show an eductor and its operation. **Trg Resources:** ● 3D graphics of educators and Venturi effect	25
LO 002.02.15 Describe Fluid Testing			
1 Describe the purpose of testing, to include: a. Detecting contaminants;		**Method / Media:** DL **Trg Environment:** CBT / Classroom	50

	b. Determining treatment chemicals required; c. Measuring specific properties; and d. Determining overall quality.		**Activity Details:** Trainees will be given images or a video showing a boiler Feed water test. **Trg Resources:** ● Graphics / video of fluid testing	
2	Describe common test methods, to include the following: a. Titration; b. Refractometer; c. Falling Ball; d. Visual Inspection; and e. Filtration.			

12. Test Details. EO 002.02 will be tested in EC 002.02 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.03 Inspect Components of Pressurized Systems

1. Performance. Inspect components of pressurized systems.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS account;
      (3) DLN 3.0 account;
      (4) Assistance;
      (5) Supervision; and
      (6) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect components of pressurized systems IAW refs by:
   a. Observing for abnormal vibrations, smells, and sounds;
   b. Checking for cracks;
   c. Checking for sweating, leaks, mould, and bubbles;
   d. Checking for icing;
   e. Looking for obvious damage;
   f. Checking for corrosion;
   g. Investigating general wear and tear;
   h. Checking for abnormal temperatures;
   i. Checking for alarm indications;
   j. Looking for discoloration; and
   k. Checking for burn marks.
4. Time Required.
   a. EO – 575 mins to deliver training.
   b. EC – 60 mins (total for all trainees).
5. References. A10
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. This EO is the theory of pressurized system components and comparison of fluid systems.
   b. For Method 1, EO 002.03 is part of Module B and can be completed either by CBT (self-study) or in a classroom dependent on resource availability.

c. Any LO stating, they will compare systems or equipment "on all RCN Classes" will not be part of the EC or PC. It is a link between the RQ-OS and Roundsperson.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.03.01: Describe a Freshwater System			
1 Describe the purpose of a freshwater system, to include the following: a. Importance of water conservation; and b. Common uses of freshwater on ships. 2 Describe freshwater system components, to include the following: a. Primary components; b. Freshwater tanks; c. Freshwater service pumps; d. Hot freshwater sub-system; e. Desalination units; and f. Chemical injection unit. 3 Explain the primary functions of a freshwater system. 4 Explain the bromine chemical injection system, to include the following: a. PPE when handling bromine; and b. Bromine test kits.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic platform non-specific shipboard freshwater system. **Trg Resources:** ● Animations and/or handouts on the freshwater system	50
LO 002.03.02: Describe a Reverse Osmosis Desalination (ROD) plant as a Sub-System of a Shipboard Fresh Water System			
1 Describe the purpose of a Reverse Osmosis Desalination (ROD) plant. 2 Describe the basic theory of osmosis and reverse osmosis. 3 Describe ROD plant components, to include the following: a. Strainers; b. Clean in place tank; c. Filters;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to basic platform on-specific sub-systems of a shipboard freshwater system. **Trg Resources:**	75

d. Membranes; e. Salinity monitor; and f. High Pressure pump. 4 Explain the operation of a ROD Plant.		● Graphics and/or handouts of freshwater sub-systems	
LO 002.03.03: Describe the Fresh Water Systems on all RCN Classes			
1 Compare the cold freshwater system on all RCN Classes, to include the following: a. Upper deck filling valves; b. Upper deck riser isolation valves; c. Tank filling valves; d. Transfer main cross connect to FW main; e. Local Pressure gauges; f. Pumps and local control panel; g. Pump supply & discharge valves; h. Leak off valves; i. Priming tank and valves; j. Shipboard Reverse Osmosis Desalination (SROD) system; k. Chemical injection systems (bromide); and l. Supply and return main isolation valves. 2 Compare the hot freshwater system on all RCN Classes, to include the following: a. Calorifiers; b. Local gauges; c. Calorifier pumps; d. Riser isolation valves to HFW main; e. Cold fresh water supply valves; and f. Location of relief valves.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic freshwater system in the above teaching point, the trainees shall compare each RCN Class's freshwater system configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures and interactive videos of freshwater system configurations / schematics	25
LO 002.03.04: Describe a Shipboard Sanitary System			
1 Describe sanitary systems, to include the following:		**Method / Media:** DL **Trg Environment:** CBT/Classroom	50

a. Purpose of a sanitary system; and b. Pressure, vacuum, and gravity type fluid systems. 2 Describe sanitary system safety, to include chemical cleaners. 3 Describe black water system components. 4 Describe grey water collection system components. 5 Explain the operation of a sanitary system, to include Marine vacuum system advantages.		**Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic platform non-specific shipboard sanitary collection system. **Trg Resources:** ● Animations and/or handouts of freshwater sub-systems	
LO 002.03.05: Describe a Treatment System as a Sub-system of a Sanitary Collection System			
1 Describe the purpose of a sanitary treatment system. 2 Describe sanitary treatment system components. 3 Explain the operation of a sanitary treatment system.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to basic platform non-specific sub-systems for shipboard sanitary collection systems. **Trg Resources:** ● Animations and/or handouts of freshwater sub-systems	25
LO 002.03.06: Describe Black and Grey Water systems on all RCN Classes			
1 Compare the black water system on all RCN Classes, to include the following: a. Local control panel raw sewage discharge; b. Eductor pumps; c. Eductor pumps supply and return for pumps; d. Raw sewage discharge pumps; e. Raw sewage discharge supply, discharge, cross connect, shore and direct overboard; f. Upper deck discharge; g. Flushing connection;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic black and grey water systems in the above teaching point, the trainees shall compare each RCN Class's black and grey water systems configuration in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures and interactive videos of black and grey water systems configuration / schematics	25

h. Local control panel sewage treatment and discharge; i. Electro catalytic Book Cell Plant; j. Media Tanks; k. Ultra-Violet Sterilizer; and l. Treated sewage discharge pumps supply, discharge and direct overboard. 2 Compare the grey water system on all RCN Classes, to include the following: a. Inboard/outboard three way valves; b. Vacuum bellows; c. Grease traps; d. Bulkhead isolations; and e. Tanks.			
LO 002.03.07: Describe a Feed water System			
1 Describe the purpose of a feed water system. 2 Describe the feed water system primary components, to include the following: a. Tanks; b. Transfer pumps; c. Chemical injection unit; and d. Hotwells. 3 Explain the operation of a feed water system.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic platform non-specific shipboard Feed water system. **Trg Resources:** ● Animations and/or handouts of feed water system	25
LO 002.03.08: Describe a Shipboard Domestic Steam System			
1 Describe the purpose of a domestic steam system, to include the following: a. Basic steam principles b. Describe domestic steam system primary components, to include: c. Feed water reservoirs/hotwells; d. Feed water pumps; e. Boilers/steam generators;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic shipboard domestic steam water system. **Trg Resources:**	25

f. Condensers/drain coolers. g. Safety valves; and h. Steam traps. 2 Explain the operation of a domestic steam system.		● Animations and/or handouts of freshwater sub-systems	
LO 002.03.09: Describe a boiler as a sub-system of the domestic steam system			
1 Describe the purpose of a boiler. 2 Describe boiler types, to include the following: a. Fire-tube boiler; b. Water-tube boiler; and c. Packaged boiler. 5 Describe boiler primary components, to include the following: a. Fuel pumps; b. Draft blowers; c. Burners; d. Soot blowers; e. Main stop valves; and f. Safety valves. 6 Explain the operation of a boiler.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a platform-nonspecific sub-system of a shipboard domestic steam water system. **Trg Resources:** ● Animations and/or handouts of shipboard domestic steam system	25
LO 002.03.10: Compare the Domestic Steam system on all RCN Classes			
1 Compare the Domestic Steam System and Steam Generators on all RCN Classes, to include the following: a. Steam Generators; b. Shore steam pressure regulator; c. Low point drains; d. Drains cooler; e. Hot well; f. Shore connections; g. Feed tanks; and h. Feed transfer pumps.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic domestic steam system in the above teaching point, the trainees shall compare each RCN Class's domestic steam system configurations in the order of complexity, from least to greatest. **Trg Resources:** Pictures and interactive videos of domestic steam system configurations /schematics.	15
LO 002.03.11: Describe a Chilled Water/Hydronic System			
1 Describing the purpose of a chilled water/hydronic system, to include the following:		**Method / Media:** DL **Trg Environment:** CBT/Classroom	25

a. Chilled water liquid composition. b. Describing chilled water/hydronic system components, to include the following: c. Circulation Pumps; d. Chillers; e. A/C Plants; and f. Fan units. 2 Explain the operation of a chilled water/hydronic system.		**Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic platform-non-specific shipboard main chilled water/hydronic system. **Trg Resources:** ● Animations and/or handouts of hydronic system	
LO 002.03.12: Compare chilled water systems on all RCN Classes			
1 Compare the chiller and main chilled water systems for all RCN Classes, to include the following: a. Chiller unit and local controllers; b. Aux. seawater circulation (ASC) over boards from condensers; c. Local gauges; d. Evaporator; e. Condenser; f. Circulation pumps; g. Chemical mixing tank; h. Filter locations; i. Expansion tanks; j. Local gauges; k. Main isolation valves; l. Venting locations; and m. Filling locations.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic main chilled water system in the above teaching point, the trainees shall compare each RCN Class's main chilled water system configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures / graphics of main chilled water system configurations / schematics	20
LO 002.03.13: Describe an HVAC system			
1 Describe the purpose of an HVAC system. 2 Describe the basic principles of air conditioning. 3 Describing HVAC system components, to include the following: a. Air Conditioning plants; b. Distributed A/C units, to include: c. Fan coil unit (FCU);		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using images, animations, videos and interactive tools, the trainee will be given information on basic, platform non-specific shipboard HVAC Systems **Trg Resources:**	50

d. Booster cooler (KU); and e. Machinery space ventilation. 4 Explaining the operation of an HVAC system, to include smoke control.		● Animations / handouts describing the basic HVAC system	
LO 002.03.14: Describe HVAC systems on all RCN Classes			
1 Compare the Heating Ventilation and Air Conditioning Systems (HVAC) on all RCN Classes, to include the following: a. PAC plant Locations; b. Filter locations and types; c. Ventilations crash stops; d. Smoke evacuation dampeners; e. Machinery space ventilation; f. Ventilation space rounds; g. KU locations; h. FCU locations; and i. Local ventilation fan controller.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic HVAC systems in the above teaching point, the trainees shall compare each RCN Class's HVAC systems configuration in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures and interactive videos of HVAC systems configuration / schematics	25
LO 002.03.15: Describe CBRN defense systems			
1 Describing the purpose of a CBRN defense system. 2 Describing CBRN defense system components. 3 Describe CBRN Filtration Units to include the following: a. Explosion protection valves; b. Filter banks; and c. Fans. 4 Explaining the operation of a CBRN defense system.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using images, animations, videos and interactive tools, the trainee will be given information on the CBRN System **Trg Resources:** ● Animations / Handouts describing the basic CBRN system	25
LO 002.03.16: Compare CBRN systems on all RCN Classes			
1 Compare the CBRN systems on all RCN Classes, to include the following: a. Location of Filter Units; b. Types of filters; c. Intakes; d. Exhaust route;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic CBRN systems in the above teaching point, the trainees shall compare each RCN Class's CBRN systems configuration in the order of complexity, from least to greatest.	15

e. Heaters; f. Citadel sections; g. Air locks; and h. Cleansing Stations.		**Trg Resources:** ● Pictures and interactive videos of CBRN systems configuration / schematics	
LO 002.03.17: Describe compressed air systems			
1 Describe purpose of compressed air systems. 2 Describe general uses of compressed air. 3 Describe compressed air system components, to include the following: a. Compressors; b. Dryers; c. Double seated valves; d. Air cylinders; e. Reducing stations. f. Describe compressor principles, to include: g. Single or multi-stage compression, and h. Requirements of cooling. 4 Describe dryer principles, to include the following: a. Adsorption; b. Desiccant; and c. Regeneration. 5 Explain the operation of a compressed air system. 6 Describe compressed air safety considerations, to include the following: a. Dieseling b. Leaks; and c. Icing.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given information on generic, platform non-specific compressed air systems **Trg Resources:** ● Animations / cut-away of parts and operation	50
LO 002.03.18: Compare HP/LP compressed air systems on all RCN Classes			
1 Compare the High Pressure (HP) Air systems on all RCN Classes, to include the following:		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic HP/LP compressed air	25

a. Compressors and local controllers; b. Chilled water isolation valves; c. HP air dryers; d. Dryer in/out & bypass valves; e. Alarms and trips; f. Action state; g. Lead / Lag set points; h. Flask inlet valves; i. Flask drain valves; j. Regulators; k. Outside machinery space valves; and l. Upper deck connection valves. 2 Compare the Low Pressure (LP) Air systems on all RCN Classes, to include the following: a. Compressors and local controllers; b. SWC isolation valves; c. LP air dryers; d. Dryer in/out & bypass valves; e. Flask inlet valves; f. Flask drain valves; g. Regulators; h. Outside machinery space valves; i. Upper deck connection valves; j. Action state; k. Alarms and trips; and l. Lead / lag set points. 3 Compare the Shipboard Breathing Air systems on all RCN Classes, to include the following: a. Dräger filling station; b. Bauer portable compressor; c. Dive air compressors; and d. Emergency Breathing System (EBS) rail.		systems in the above teaching point, the trainees shall compare each RCN Class's HP/LP compressed air systems configuration in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures and interactive videos of HP/LP compressed air systems configuration / schematics	

13. Test Details. EO 002.03 will be tested in EC 002.03 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.04 Inspect the Liquid Damage Control System

1. Performance. Maintain pressurized systems.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) Schematics;
      (3) DRMIS account;
      (4) DLN 3.0 account;
      (5) Assistance;
      (6) Supervision; and
      (7) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect the liquid damage control systems IAW refs by:
   a. Observing for abnormal vibrations, smells, and sounds;
   b. Checking for cracks;
   c. Checking for sweating, leaks, mould, and bubbles;
   d. Checking for icing;
   e. Looking for obvious damage;
   f. Checking for corrosion;
   g. Investigating general wear and tear;
   h. Checking for abnormal temperatures;
   i. Checking for alarm indications; and
   j. Looking for salt buildup. .
4. Time Required.
   a. EO – 175 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A10
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. This EO is theory-based information for liquid damage control systems.
   b. For Method 1, EO 002.04 is part of Module B and can be completed either by CBT (self-study) or in a classroom dependent on resource availability.

c. Any LO stating, they will compare systems or equipment "on all RCN Classes" will not be part of the EC or PC. It is a link between the RQ-OS and Roundsperson.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.04.01: Describe Seawater Service Systems			
1 Describe the purpose of a seawater service (SWS) system. 2 Describe seawater service system components, to include the following: a. Sea bays; b. Pumps; and c. Overboard discharges. 3 Explain the primary functions of a SWS system, to include the following: a. DC systems, including: (1) Fire main/Fire Hoses; (2) AFFF Proportioning Unit; (3) Quartzoid Bulb Sprinklers; (4) Magazine Flood and Spray; (5) Bilge Salvage Eductors; and (6) Pre-wet. b. Cooling; c. Ballast; and d. Sanitary.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic (platform non-specific) shipboard seawater services system. **Trg Resources:** ● Schematic of basic seawater services system	50
LO 002.04.02: Describe Shipboard Seawater Services on all RCN Classes			
1 Compare the Sea Water Service (Fire main) System on all RCN Classes, to include: a. Fire main working pressure (min/max); b. Pump Management; c. Location of all fire pumps; d. Location of main isolations;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic Shipboard Seawater Service system in the above teaching point, the trainees shall compare each RCN Class's seawater service system configurations in the order of complexity, from least to greatest.	35

e. Arrangement of piping; f. Location of all hydrants; g. Direction of flow; and h. Cold weather precautions. 2 Compare the bilge eductor systems on all RCN Classes, to include the following: a. Eductor flow rates; b. Valve activation sequence; c. Direction of flow; d. Location of mode of water; e. Location of Suction; f. Location Overboard discharge; g. Location of Lazy rods/deck fitting; h. Location of Emergency overboard discharge connections; and i. Location of submersible pumps.		**Trg Resources:** ● Pictures and interactive videos of freshwater system configurations / schematics	
LO 002.04.03: Describe Shipboard Fire Suppression Systems			
1 Describe the purpose of a fire suppression system. 2 Describe the following types of fitted fire suppression systems to include the following: a. Aqueous Film Forming Foam (AFFF); b. Quartzoid Bulb Sprinklers; c. Magazine Flood and Spray; d. Range guard (Karbaloy); e. Novex enclosure/fine water spray; f. CO2; g. Halon; and h. Twin agent unit.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic (platform non-specific) shipboard fire prevention system. **Trg Resources:** ● Schematic of basic liquid fire prevention systems	50
LO 002.04.04: Describe Shipboard liquid Fire Prevention Systems on all RCN Classes			

1 Compare the Aqueous Film Forming Foam (AFFF) fitted fire prevention systems on all RCN Classes, to include the following: a. Tank capacities; b. Tank Locations; and c. Activation methods. 2 Compare the Flood and Spray fire prevention systems on all RCN Classes, to include the following: a. Magazine Spray; b. Quartzoid System; and c. Floodable Lockers. 3 Compare the range guard fire suppression systems on all RCN classes. 4 Compare the CO2 Fire Suppression Systems on all RCN classes. 5 Halon fire suppression system on HFX class. 6 TAU fire suppression system on HFX class. 7 Compare Novex/Fine water spray fire suppression systems onboard all RCN classes.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic shipboard liquid fire prevention systems in the above teaching point, the trainees shall compare each RCN Class's liquid fire prevention system configurations in the order of complexity, from least to greatest. **Trg Resources:** Computer, DLN access, DWAN/internet access.	40

10. Test Details. EO 002.04 will be tested in EC 002.04 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.05 Inspect Fuel, Oil and Lubricant Systems

1. Performance. Inspect fuel, oil, and lubricant systems.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) Schematics;
      (3) DRMIS account;
      (4) DLN 3.0 account;
      (5) Assistance;
      (6) Supervision; and
      (7) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside;
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall Inspect fuel, oil, and lubricant systems IAW refs by:
   a. Observing for abnormal vibrations, smells, and sounds;
   b. Checking for cracks;
   c. Checking for sweating, leaks, mould, and bubbles;
   d. Checking for icing;
   e. Looking for obvious damage;
   f. Checking for corrosion;
   g. Investigating general wear and tear;
   h. Checking for abnormal temperatures;
   i. Checking for alarm indications;
   j. Looking for discoloration; and
   k. Comparing fuel oil transfer systems across RCN platforms.
4. Time Required.
   a. EO – 325 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A2, A10, A11, A12
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. This EO is the theory for fuel, oil, and lubricant systems, sub-systems, and components.
   b. For Method 1, EO 002.05 is part of Module B and can be completed either by CBT (self-study) or in a classroom dependent on resource availability.
   c. Any LO stating, they will compare systems or equipment “on all RCN Classes” will not be part of the EC or PC. It is a link between the RQ-OS and Roundsperson.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.05.01: Describe Petroleum (fuel) Oils and Lubricants			
1 Explain POL definitions and properties to include the following: a. Flashpoint; b. Cloud point; c. Auto-ignition temperature; d. Pour point; and e. Viscosity. 2 Explain POL handling to include the following: a. Safety; b. POL Storage; and; 3 POL condition analysis to include the following: a. OCCAP; and b. Onboard testing.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information from the teaching points. **Trg Resources:** ● Interactive videos showing appropriate POL procedures ● POL Handbook ● SEMS	50
LO 002.05.02: Describe a Dirty Oil System			
1 Describe the purpose of a Dirty Oil System. 2 Describe arrangements for collecting dirty oil. 3 Describe equipment that produces Dirty Oil.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given images, videos, and interactive activities to aid in instruction of teaching points. **Trg Resources:** ● Video demonstrating sounding of a tank with a dip tape	25
LO 002.05.03: Describe a Bilge Stripping System			
1 Describe the purpose of a bilge stripping system. 2 Describe environmental concerns related to bilge stripping systems including the following: a. Regulations (SEMS); and		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic (platform non-specific) shipboard bilge system.	25

b. Requirements to pump bilge water overboard 3 Describe bilge system components, to include the following: a. Collection tanks; b. Bilge pumps; and c. Oily water separators. 4 Describe oily water separator components, to include the following: a. Solid contaminant removal; b. Free oil removal; c. Emulsified oil removal; and d. Oil Content Monitors. 5 Explain the operation of a bilge system. 6 Describe Bilge Management to include the following: a. Effects of debris in the bilge; b. Reasons to limit oils and fuels in bilge; c. Effects of Detergents in Bilge water; and d. Consolidation of Bilge water.		**Trg Resources:**	
LO 002.05.04: Describe Bilge Stripping Systems on all RCN Classes			
1 Compare the Oil pollution and abatement systems on all RCN Classes to include the following: a. Bilge pumps; b. Bilge suction valve locations (inside and outside machinery spaces); c. Bulkhead isolation valves; d. Hydromen location; e. Batch tank; f. Oil content monitor;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic bilge system in the above teaching point, the trainees shall compare each RCN Class's bilge system configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Interactive video/animation comparing generic components of all RCN bilge systems	25

g. Normal System parameters; and h. Automatic functions.			
LO 002.05.05: Describe a Fuel Oil Transfer System			
1 Describe the purpose of a FOT System. 2 Describe FOT System components, to include the following: a. Fuel Tanks; b. Transfer Pumps; and c. Centrifuges. 3 Explain the primary functions of a FOT System, to include the following: a. Receiving fuel; b. Transferring fuel; and c. Supplying fuel.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic platform non-specific shipboard fuel oil and diesel oil transfer system. **Trg Resources:** ● Animations of a FOT system	25
LO 002.05.06: Describe a centrifuge as Sub-system of a Fuel Oil Transfer System;			
1 Describe the purpose of a centrifuge 2 Describe centrifuge components, to include the following: a. Bowl assembly; b. Disk Stack; and c. Gravity disk. 3 Explain the operation of a centrifuge.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given 3D, cutaway, and animated images to show a centrifuge, its operation, disassembly, and assembly for maintenance. **Trg Resources:** ● 3D graphics of centrifuges ● Video showing disassembly and assembly of a centrifuge	25
LO 002.05.07: Describe Fuel Oil Transfer Systems on all RCN Classes			
1 Compare the FOT system on all RCN Classes to include the following: a. Tank filling valve locations; b. Bulkhead isolations; c. Settling tank valve locations; d. Transfer pump;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic FOT system in the above teaching point, the trainees shall compare each RCN Class's FOT system configurations in the order of complexity, from least to greatest.	25

e. Centrifuge; f. Normal System parameters; and g. Automatic functions.		**Trg Resources:** Interactive video/animation comparing generic components of all RCN FOT systems	
LO 002.05.08: Describe a Fuel Oil Service System			
1 Describe the purpose of a Fuel Oil Service System. 2 Describe FOS system components, to include the following: a. Service Tanks; b. Pumps; c. Filters/Water Separators; d. Pressure regulators; e. Supply and return piping; and f. Emergency Supply. 3 Explain the operation of a Fuel Oil Service System.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic (platform non-specific) shipboard FOS system. **Trg Resources:** ● Animations of a FOS system	25
LO 002.05.9: Describe a coalescer as sub-system of a fuel oil service system;			
1 Describe the purpose of a coalescer to include the following: a. Components; b. Housing; c. Coalescer element; and d. Sump. 2 Explain the operation of a coalescer.		**Method / Media:** DL **Trg Environment:** CBT / Classroom **Activity Details:** Trainees will be given cutaway and animated images to show a coalescer and its operation. **Trg Resources:** ● 3D graphics of coalescers	25
LO 002.05.10: Compare Fuel Oil Service Systems on all RCN Classes			
1 Compare the FOS system on all RCN Classes to include the following: a. Boost pumps and local controllers; b. J-main / ring main; c. Emergency fuel tank; d. RIB filling;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic FOS system in the above teaching point, the trainees shall compare each RCN Class's FOS system configurations in the order of complexity, from least to greatest. **Trg Resources:**	25

e. Oil / water separator; f. Supply / return; g. Pneumatic supply valves; h. Bulkhead isolations; i. Service tanks; j. Emergency FOT/FOS cross connect valves; k. Local pressure gauges; l. DDFP day tanks; m. Pressure regulators; n. Normal System parameters; and o. Automatic functions.		● Interactive video/animation comparing generic components of all RCN FOS systems	
LO 002.05.11: Describe a Lube Oil System			
1 Describe the purpose of Lube Oil Systems, to include the following: a. Lube oil supply systems; and b. Lube oil transfer systems. 2 Describe LO system components, to include the following: a. Storage Tanks; b. Sumps; c. Pumps; d. Filters; e. Pressure Regulators; f. Temperature Regulators; and g. Heat Exchangers. 3 Explaining the operation of a lube oil system, to include the following: a. Lube oil supply systems; and b. Lube oil transfer systems, to include: (1) receiving oil;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to a basic platform non-specific shipboard lube oil system. **Trg Resources:** ● Animations of a lube oil system	25

(2) transferring oil; and (3) cleaning oil.			
LO 002.05.12: Compare Lube Oil Systems on all RCN Classes			
1 Compare the Main Lube Oil (MLO) system on all RCN Classes, to include the following: a. MLO coolers; b. Pressure regulators; c. Combined thrust and plumber block; d. Flow sight glasses; e. Manifold; f. MLO filters; g. Sump; h. Temperature regulator; i. Air turbine; j. Local gauge panels; k. Signal Conditioning Unit (SCU); l. Normal System parameters; and m. Automatic functions. 2 Compare the Diesel lube oil fill and transfer (DLO) system on all RCN Classes to include the following: a. Upper deck connection; b. Upper deck isolation valve; c. Bulkhead isolation valves; d. Transfer pump; e. Flow meters; f. Normal System parameters; and g. Automatic functions.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic MLO system in the above teaching point, the trainees shall compare each RCN Class's MLO system configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Interactive video/animation comparing generic components of all RCN MLO systems	25

10. Test Details. EO 002.05 will be tested in EC 002.05 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.06 Repair Metal Components Without Heat

1. Performance. Repair metal components without heat.
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Basic tools (IAW Annex H);
        (5) Power tools (IAW Annex H);
        (6) Testing equipment (IAW Annex H);
        (7) DRMIS account;
        (8) DLN 3.0 account;
        (9) Assistance;
        (10) Supervision; and
        (11) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions on ship, at sea or alongside;
        (2) Workshop;
        (3) Any location with access to DLN 3.0; or
        (4) Classroom.
3. Standard. The basic maintainer shall repair metal components without heat IAW refs by:
    a. Inspecting basic shop tools;
    b. Utilizing shop tools;
    c. Applying safety considerations;
    d. Preparing area for work;
    e. Recommending repair method (without heat);
    f. Installing metal shapes and components by:
        (1) Installing metal fasteners;
        (2) Cutting metal;
        (3) Repairing bolt threads;
        (4) Drilling/tapping holes;
        (5) Filing metal; and
        (6) Securing metal components together.
4. Time Required.
    a. EO – 800 mins to deliver training.
    b. PC – 425 mins (total for all trainees).
5. References. A13, A14, A15, A16
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.

8. Remarks. For Method 2, EO 002.06 is part of Module C and will be completed F2F at NFS for tool access.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.06.01: Describe basic shop tools			
1 Describe the purpose and how to utilize the following hand tools properly and safely: a. Wrenches/sockets; b. Pliers; c. Pincers; d. Screwdrivers; e. Files; f. Hack saw; g. Hammers; h. Pry bars; i. Vise-grips; j. Drill bits; k. Taps & dies; l. Punch; and m. Chisels. 2 Describe the purpose, how to utilize the following measuring equipment properly and safely: a. Various types of scales; b. Calipers; c. Protractors; d. Rulers/tape measure; e. Square and straight edge; f. Vernier calipers; g. Height gauges; h. Depth Gauges; i. Micrometers; j. Dial indicators; k. Thickness gauges;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will use images, videos, and interactive activities in order to gain the fundamental knowledge to utilize hand and measuring tools. They will complete small confirmation assignments throughout to assess their understanding. **Trg Resources:** ● Pictures of all hand tools and measuring equipment ● Physical hand tools and measuring equipment as available	150

l. Radius gauges; and m. Screw pitch gauges.			
LO 002.06.02: Utilize common shop tools			
1 Demonstrate the purpose and how to properly and safely utilize the following power tools to include: a. Drill; b. Grinder: c. Impact wrench; d. Band Saw; and e. Pneumatic tools. 2 Complete basic maintenance on the power tools listed above. 3 Test the power tools listed above. 4 Care and custody of tools		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Trainee will be given a shop tour and shown the proper use and basic maintenance of power tools. After demonstration, trainee will practice on all tools in a round robin. **Trg Resources:** ● Properly equipped workshop with all required tools	200
LO 002.06.03: Installing metal shapes and components			
1 Tool Familiarization to include: a. Identifying local hazards prior to work (e.g. electrical components, depth of water tight bulkheads, etc). b. Prepping area for work. 2 Installing metal components by: c. Repairing a bolt thread. d. Cutting a metal component. e. Drilling and tapping a metal component. f. Filing a metal component. g. Securing two metal components together.		**Method / Media:** Group Assignment **Trg Environment:** Workshop **Activity Details:** Instructor will demonstrate how to perform the skills. Trainee will complete the provided project IAW the provided marking guide, which involves the listed basic tool skills. **Trg Resources:** ● Properly Equipped Workshop with required tools and materials	450

11. Test Details. EO 002.06 will be tested in PC 002.06 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.07 Maintain Pumps and Valves

1. Performance. Maintain pumps and valves.
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Basic tools (IAW Annex H);
        (5) Power tools (IAW Annex H);
        (6) Testing equipment (IAW Annex H);
        (7) LOTO equipment (IAW Annex H);
        (8) DRMIS account;
        (9) DLN 3.0 account;
        (10) Assistance;
        (11) Supervision; and
        (12) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions on ship, at sea or alongside
        (2) Any location with access to DLN 3.0; or
        (3) Classroom.
3. Standard. The basic maintainer shall maintain pumps and valves IAW refs by:
    a. Maintaining basic valves to include:
        (1) Disassembling valves;
        (2) Repairing valves;
        (3) Assembling valves; and
        (4) Testing valves.
    b. Maintaining pumps to include:
        (1) Disassembling pumps;
        (2) Repairing grease nipples;
        (3) Inspecting shock mounts;
        (4) Lubricating bearings;
        (5) Lubricating mechanical linkage;
        (6) Removing/replacing seals; and
        (7) Assembling pumps.
4. Time Required.
    a. EO – 1950 mins to deliver training.
    b. EC – 150 mins (total for all trainees).
5. References. A2, A17, A18, A19, A20, A21, D7
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.

8. Remarks.

a. For Method 1, EO 002.07 is part of Module D and will be completed at an NFS for workshop access.

b. Training facilities will require a test bench to pressurize basic valves.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.07.01: Maintain basic valves			
1 Assembling/Disassembling the following basic valves: a. Gate Valve; b. Globe Valve; c. Diaphragm Valve; d. Ball Valve; e. Butterfly Valve; f. Check Valve; g. Non-return Valve; h. Orifice plates; i. Steam trap; j. Pressure Relief Valve; and k. Pressure Regulating valve. 2 Describe what gaskets are, and their purpose. 3 Perform basic valve maintenance on the above-mentioned valves, to include: a. Lubricating valve spindles. b. Replacing the packing. c. Replacing the gaskets. d. Testing the above-mentioned valves under pressure, to include: (1) Ensuring the gaskets do not leak; and		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Instructor to explain and demonstrate how to perform all the teaching points outlining main components, materials, and functions. One type of each valve is to be laid out on workbenches. Trainee will rotate by each valve, carrying out some basic maintenance on each, incorporating all the teaching points simultaneously. **Trg Resources:** ● Workshop equipped for valve maintenance ● Test bench ● Multiple examples of each valve ● Materials for maintenance (ex. gaskets)	1200

(2) Ensuring the valve can perform its designated task.			
LO 002.07.02: Maintain pumps			
1 Disassembling the following pumps: a. Positive Displacement; b. Centrifugal; c. Rotary; and d. Reciprocating. 2 Perform routine maintenance by: a. Repairing grease nipples; b. Inspecting shock mounts; c. Lubricating bearings; and d. Lubricating mechanical linkage. 3 Repair / replace the following seals: a. O-rings; b. Packed seal; and c. Mechanical seal. d. Assemble the following pumps: e. Positive Displacement; f. Centrifugal; g. Rotary; and h. Reciprocating.		**Method / Media:** EDI **Trg Environment:** Workshop appropriate for pump repair **Activity Details:** Demonstration and practice of disassembly/assembly of maintenance on various pumps. **Trg Resources:** Tools, PPE, multiple examples of various pumps	750

12. Test Details. EO 002.07 will be tested in EC 002.07 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.08 Repair Pipes, Hoses, and Insulation

1. Performance. Repair pipes, hoses, and insulation.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) Schematics;
      (3) PPE;
      (4) Basic tools (IAW Annex H);
      (5) Power tools (IAW Annex H);
      (6) Testing equipment (IAW Annex H);
      (7) LOTO equipment (IAW Annex H);
      (8) Oxy / Acetylene equipment;
      (9) DRMIS account;
      (10) DLN 3.0 account;
      (11) Assistance;
      (12) Supervision; and
      (13) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside;
      (2) Ventilated workshop equipped with pipes and hoses;
      (3) Any location with access to DLN 3.0; or
      (4) Classroom.
3. Standard. The basic maintainer shall repair pipes, hoses, and insulation IAW refs by:
   a. Finding fault or break;
   b. Isolating system;
   c. Preparing area for work;
   d. Recommending repair method;
   e. Repairing break or fault; and
   f. Completing a function test.
4. Time Required.
   a. EO – 1320 mins to deliver training.
   b. EC – 250 mins (total for all trainees).
5. References. A2, A22, A23, D3, D4, D5
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. EO 007.01 must be completed prior to EO 002.08 as 007.01 contains safety and equipment set up procedures relating to oxy acetylene usage.
   b. For method 1, EO 002.08 is part of module D and shall be completed within an NFS that's properly equipped with ventilation, tools and equipment related to pipes, hoses, and insulation.

c. The Lesson Objective (LO): complete basic maintenance on piping, will need to be scheduled prior to the LO: Repair insulation and lagging. This is to allow the trainees to utilize the project created in piping for the lagging demo.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.08.01: Complete basic maintenance on hoses			
1 Demonstrate the proper use of hose repairing tools, to include the following: a. Wrenches; b. Spanners; c. Cutting tools; and d. Patching kits. 2 Repair fluid hoses, flexible couplings and quick disconnect fittings to include the following: a. Hoses; b. Clamps; c. Patches; and d. Fittings.		**Method / Media:** EDI **Trg Environment:** Workshop equipped for plumbing and fluid system maintenance. **Activity Details:** Instructor to explain and demonstrate use of tools and repair of equipment. Trainees to complete a project replacing the coupling/fitting. **Trg Resources:** ● Hose maintenance tools (including Teflon tape) ● Hoses and fittings ● Gauges ● PPE	150
LO 002.08.02: Perform basic maintenance on tubing			
1 Demonstrate the proper use of tube repair tools, to include the following: a. Spanners and wrenches; b. Flaring tool; c. Tube bender; and d. Tube cutters. 2 Fabricate tubing sections with fittings to include: a. Cutting the tubing to length; b. Bending the tubing; and c. Installing the fittings using the flaring tool. 3 Demonstrate the proper use of the portable propane torch.		**Method / Media:** EDI **Trg Environment:** Workshop equipped for plumbing and fluid system maintenance **Activity Details:** Instructor to explain and demonstrate use of tools and repair of equipment. Trainees to complete a project to fabricate tubing section. Once complete, they will drill a small hole in the project to repair. They will also sweat off one of the fittings. Instructor will give feedback as required. **Trg Resources:** ● Tubing maintenance tools (including propane for the torch, Teflon tape) ● Piping and fittings	300

4 Utilizing appropriate PPE by: a. Following safety precautions; b. Turning on the gas; and c. Lighting the torch. 5 Complete small repairs on damaged tubing using a portable propane torch to include: a. Filling in a small hole; b. Adding a fitting; and c. Sweating off a fitting.		● Solder and flux ● Gauges ● PPE	
LO 002.08.03: Perform basic maintenance on piping			
1 Utilize the oxy-acetylene torch by: a. Utilizing appropriate PPE; b. Demonstrating proper cylinder transportation techniques; c. Demonstrating proper method of securing cylinders; d. Cleaning cylinder valves; e. Inspecting regulators; f. Checking for visible signs of damage (broken glass, damaged threads etc.) to include: (1) Checking for ease of operation of regulator adjusting screw; and (2) Checking for evidence of oil/grease. g. NOTE: Do not use Teflon when connecting any fittings Teflon; h. Connecting regulators to appropriate cylinders; i. Backing off the adjusting screw;		**Method / Media:** EDI **Trg Environment:** Workshop equipped for plumbing and fluid system maintenance **Activity Details:** Instructor to explain and demonstrate use of various piping types and fittings for repairing and assembling. Trainees will fabricate and thread piping IAW a project that incorporates all key teaching points and is in line with the EC checklist Trainees are to notify their instructor of any unsafe condition. Instructors will give feedback throughout the project. **Trg Resources:** ● Plumbing tools (including torch and fuel, Teflon tape) ● Piping and fittings ● Solder and flux ● Gauges ● PPE	570

	j. Opening cylinder valves – close cylinder valves, check for pressure drop on first stage of regulators. (open oxygen Vv slowly to prevent damage to regulator diaphragm); k. Inspecting and connecting flashback arrestors to regulators; l. Inspecting and connecting hoses; m. Inspecting and connecting check Vv's to torch body (2 check Vv's on each hose or 1 flashback arrestor connected to regulator); n. Inspecting and connecting torch body assembly to hose; o. Inspecting and connecting torch tip; p. Ensuring all Vv's are closed and regulator adjusting screws are backed off; q. Opening slowly oxygen cylinder valve fully; r. Opening acetylene cylinder valve ¼ to ½ turn; s. Adjusting regulators to "working pressure"; t. Checking for leaks at all connections/fittings by – sound, smell, and sight (snoop); and u. Purging the system (blow through).			
3	Demonstrate the proper techniques to fabricate a small piping section utilizing an oxy-			

acetylene torch to include the following: a. Torch safety; b. Soldering techniques; c. Filling in a hole; and d. Adding, or replacing a fitting. 4 Replace threaded fittings to include the following: a. Pipe and hose threads; b. Use of Teflon; and c. Use of o-rings. 5 Fabricate a small section of copper pipe, to include: d. Adding a fitting (both threaded and soldered); and e. Filling a hole.			
LO 002.08.04: Repair insulation and lagging			
1 Repair lagging and insulation to include: a. Utilizing appropriate PPE; b. Following Safety precautions; c. Inspecting lagging and insulation for defects to include the following: (1) Tears; (2) Worn components; and (3) Missing components. d. Removing lagging and insulation; e. Assembling/disassembling lagging and insulation; and f. Replacing and repairing the lagging and insulation using proper materials.		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Instructor to explain and demonstrate the installation techniques of various types of insulation and lagging. The trainees will complete a project where they lag the piping project that was created in the previous LO. This LO shall be scheduled after the piping LO. **Trg Resources:** ● Insulation maintenance tools ● Piping ● Adhesives ● Insulation ● PPE	300

13. Test Details. EO 002.08 will be tested in EC 002.08 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.09 Maintain Miscellaneous Fluid Components

1. Performance. Maintain miscellaneous fluid components.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) Schematics;
      (3) PPE;
      (4) Basic tools (IAW Annex H);
      (5) Power tools (IAW Annex H);
      (6) Testing equipment (IAW Annex H);
      (7) LOTO equipment (IAW Annex H);
      (8) DRMIS account;
      (9) DLN 3.0 account;
      (10) Assistance;
      (11) Supervision; and
      (12) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside;
      (2) Workshop
      (3) Any location with access to DLN 3.0; or
      (4) Classroom.
3. Standard. The basic maintainer shall maintain miscellaneous fluid components IAW refs by:
   a. Testing boiler feed water;
   b. Inspecting strainers;
   c. Cleaning strainers;
   d. Changing filters;
   e. Starting centrifuge;
   f. Repairing toilets to include:
      (1) Locking/tagging out system;
      (2) Disassembling toilet;
      (3) Troubleshooting fault;
      (4) Repairing (replacing) components; and
      (5) Completing a function test.
4. Time Required.
   a. EO – 740 mins to deliver training
   b. EC – 90 mins (total for all trainees)
5. References. A2, A24, A25, A26, A27, A28, A29, A30, A31, A32, A33, A34, A35
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. For method 1, EO 002.09 is part of module D and shall be completed within a NFS.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.09.01: Complete Boiler Feed Water Testing			
1 Describe the method to test Feed water, to include the following: a. Purpose; b. Quality tests; c. Treatment chemicals; d. Typical Feed system arrangements; and e. Salinity cells. 2 Calibrate the test kit; and 3 Perform a Feed water test, to include: a. Ensuring proper PPE is utilized; b. Drawing a feed water sample; c. Analyzing the sample; and d. Ensuring proper disposal of the test sample.		**Method / Media:** EDI **Trg Environment:** Classroom / Workshop **Activity Details:** Instructor to explain and demonstrate how to perform boiler feed water testing. Trainees will work individually or in pairs to conduct testing with instructor feedback. **Training Resources:** ● Feed water samples, ● Test kits.	150
LO 002.09.02: Complete basic maintenance on strainers.			
1 Disassemble a strainer, to include: a. Removing the basket cover; b. Removing the strainer/basket; c. Removing the drain plug; and d. Removing the gaskets. 2 Inspect the components of a strainers, to include the following: a. Outer body; b. Entry and exit routes; c. Screen/basket; d. Basket assembly; e. Cover flanges; f. Gaskets; and g. Drain plug. 3 Clean the screen/basket of all debris.		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Instructor to explain and demonstrate basic maintenance on strainers. Trainees will work individually or in pairs to conduct the listed maintenance tasks with instructor supervision and feedback. **Training Resources:** ● Strainers ● Tools and equipment	45

4 Ensure entry and exits routes are clear of any debris or obstructions. 5 Assemble a strainer, to include: a. Replacing the gaskets; b. Replacing the drain plug; c. Replacing the strainer/basket; and d. Replacing the basket cover.			
LO 002.09.03: Complete basic maintenance on heat exchangers.			
1 Disassembling. 2 Cleaning. 3 Assembling.		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Instructor to explain and demonstrate basic maintenance on heat exchangers. Trainees will work individually or in pairs to conduct the listed maintenance tasks with instructor supervision and feedback. **Training Resources:** ● Heat exchanger ● Tools and equipment	45
LO 002.09.04: Complete basic maintenance on a fuel Coalescer			
1 Replacing the filter elements; and 2 Cleaning the bowl.		**Method / Media:** EDI **Trg Environment:** Appropriate Workshops **Activity Details:** Instructor to explain and demonstrate basic maintenance on coalesces. Trainees will work individually or in pairs to conduct the listed maintenance tasks with instructor supervision and feedback. **Training Resources:** ● Coalescers ● Tools and equipment	80
LO 002.09.05: Complete basic maintenance on in-line AFFF educators.			
1 Disassemble an in-line AFFF eductor, to include: a. Removing the wand; b. Removing the hose; c. Removing the O-rings; and		**Method / Media:** EDI **Trg Environment:** Appropriate Workshops **Activity Details:** Instructor to explain and demonstrate basic maintenance on educators. Trainees will work	100

d. Removing the positive setting meter. 2 Inspect the in-line AFF eductor for the following: a. Corrosion; b. Damage; and c. Foreign debris. 3 Ensure the positive setting meter can have the value easily changed to 0, 0.5, 1, 3 or 6%; 4 Clean all the components with soapy water; and 5 Assemble an in-line AFFF eductor, to include: a. Replacing the positive setting meter; b. Replacing the O-rings; c. Replacing the hose; and d. Replacing the wand.		individually or in pairs to conduct the listed maintenance tasks with instructor supervision and feedback. **Training Resources:** ● Eductors ● Tools and equipment	
LO 002.09.06: Complete basic maintenance on centrifuges.			
1 Disassembling. 2 Cleaning. 3 Assembling.		**Method / Media:** EDI **Trg Environment:** Appropriate Workshops **Activity Details:** Instructor to explain and demonstrate basic maintenance on centrifuges. Trainees will work individually or in pairs to conduct the listed maintenance tasks with instructor supervision and feedback**.** **Training Resources:** ● Centrifuges ● Tools and equipment	200
LO 002.09.07: Complete basic maintenance on marine vacuum toilets			
1 Disassemble a marine vacuum toilet common component, to include: a. Vacuum Interface valve; b. Discharge valve housing; c. VT Control mechanism; d. Discharge valve assembly; and		**Method / Media:** EDI **Trg Environment:** Appropriate Workshops **Activity Details:** Instructor to explain and demonstrate basic maintenance on toilets. Trainees will work individually or in pairs to conduct the listed maintenance tasks with	350

e. EVAC toilet assembly. 2 Inspect the marine vacuum toilet components for the following: a. Corrosion; b. Tears and rips; and c. Cracks and holes. 3 Repair by replace any damaged parts. 4 Assemble a marine vacuum toilet common component, to include the following: a. Vacuum Interface valve; b. Discharge valve housing; c. VT Control mechanism; d. Discharge valve assembly; and e. EVAC toilet assembly. 5 Perform an auto evacuation routine by connecting the vacuum line to the discharge valve.		instructor supervision and feedback **Training Resources:** ● Toilets ● Tools and equipment	

14. Test Details. EO 002.09 will be tested in EC 002.09 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.10 Maintain Air Systems

1. Performance. Maintain Air Systems
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Fan coil unit Mock-up;
        (5) Basic tools (IAW Annex H);
        (6) Power tools (IAW Annex H);
        (7) Testing equipment (IAW Annex H);
        (8) LOTO equipment (IAW Annex H);
        (9) DRMIS account;
        (10) DLN 3.0 account;
        (11) Assistance;
        (12) Supervision; and
        (13) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions on ship, at sea or alongside;
        (2) Workshop;
        (3) Any location with access to DLN 3.0; or
        (4) Classroom.
3. Standard. The basic maintainer shall maintain air systems IAW refs by:
    a. Inspecting air compressor components (for leaks and abnormal noise);
    b. Starting air compressor;
    c. Logging pressures;
    d. Checking oil level;
    e. Purging water;
    f. Replacing air filters;
    g. Isolating/de-isolating valves;
    h. Venting cooling coil;
    i. Replacing air filters;
    j. Removing/replacing 3-way valves; and
    k. Removing/replacing motor fan components.
4. Time Required.
    a. EO – 625 mins to deliver training.
    b. EC – 150 mins (total for all trainees).
5. References. A2, A36, A37
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.

8. Remarks.

a. For method 1, EO 002.10 is part of module D and shall be completed within an NFS that's properly equipped with air system components.

b. LO 002.10.01 "Perform Basic Maintenance on an FCU" will likely be taught prior to the Electrical phase therefore trainees will not yet have learned to electrically disconnect/connect any wiring. Instructor to emphasise that by the end of the Basic Maintainer course, the trainee will be able to achieve what is being shown when they disconnect/ connect the motor/fan combo.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.10.01: Complete Basic Maintenance on Air Compressors			
1 Inspect the following components: a. Hoses; b. Piping; c. Discharge valves; and d. Pressure gauge panel. 2 Replace the oil filter. 3 Replace the air filter; 4 Drain and refill Oil. 5 Inspect for leaks and abnormal noises.		**Method / Media:** EDI **Trg Environment:** Classroom/Workshop **Activity Details:** Trainee will learn how to perform maintenance on compressor through instructor demonstration and then trainees conduct practical. Oil change and filter change get done at the same time. **Trg Resources:** ● PPE, ● Test equipment, ● LOTO equipment, ● compressor, ● compressor oil, ● filters.	250
LO 002.10.02: Complete basic Maintenance on a Booster cooler (KU)			
1 Vent the booster cooler cooling coils. 2 Change filter media on booster cooler units. 3 Lubricate pillow block bearings. 4 Describe proper alignment of drive pulley.		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Trainee will perform the listed maintenance on a physical KU training aid. Instructor to ensure all safety and LOTO procedures are implemented. **Trg Resources:** ● KU Mock-up	125

5 Remove and replace drive pulley. 6 Adjust motor alignment. 7 Adjust belt tension.			
LO 002.10.03: Perform basic Maintenance on a Fan Coil Unit (FCU)			
1 Vent a FCU's cooling coil. 2 Replace air filter on fan coil unit. 3 Remove and replace three-way valve. 4 Remove and replace the motor/fan combo.		**Method / Media:** EDI **Trg Environment:** Workshop **Activity Details:** Instructor to demo electrically disconnecting/reconnecting the motor/fan combo. Trainee to perform the listed maintenance on a physical FCU training aid. If trainees have not learned to electrically disconnect/ connect any wiring, instructor to emphasise they will be able to disconnect/ connect the motor/fan combo after the electrical phase. Instructor to ensure all safety and LOTO procedures are implemented. **Trg Resources:** ● FCU Mock-up	250

15. Test Details. EO 002.10 will be tested in EC 002.10 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based EC's.

EO 2.11 Maintain Refrigeration Systems

1. Performance. Maintain refrigeration systems.
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Refrigeration leak detectors;
        (5) Basic tools (IAW Annex H);
        (6) Power tools (IAW Annex H);
        (7) Testing equipment (IAW Annex H);
        (8) LOTO equipment (IAW Annex H);
        (9) DRMIS account;
        (10) DLN 3.0 account;
        (11) Assistance;
        (12) Supervision; and
        (13) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions on ship, at sea or alongside;
        (2) Workshop;
        (3) Any location with access to DLN 3.0; or
        (4) Classroom.
3. Standard. The basic maintainer shall maintain refrigeration systems IAW refs by:
    a. Inspecting refrigeration cycle;
    b. Inspecting refrigeration systems;
    c. Performing refrigeration safety;
    d. Inspecting main refrigeration systems on RCN ships;
    e. Inspecting leak detection process on refrigeration systems;
    f. Testing for leaks on refrigeration systems; and
    g. Completing refrigeration rounds to include:
        (1) Detecting leaks;
        (2) Checking oil levels;
        (3) Logging running hours;
        (4) Checking suction pressure;
        (5) Logging sea water temperature;
        (6) Logging fridge temperature; and
        (7) Checking for moisture.
4. Time Required.
    a. EO – 350 mins to deliver training.
    b. EC – 200 mins (total for all trainees).

5. References. A2, A19, A42, A43, A44, A45
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
   a. For method 1, EO 002.11 is part of module D and shall be completed within an NFS that's properly equipped with refrigeration system components.
   b. EO 002.11 will be scheduled in conjunction with EO 001.02 (ODS) whenever possible as a refrigeration qualified instructor is required for both.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 002.11.01: Describe the Refrigeration Cycle			
1 Describe the purpose of the refrigeration cycle. 2 Review the principles of heat transfer as they relate to heat flow in the refrigeration cycle. 3 Describe the basic refrigeration cycle to include the flow pattern using a schematic drawing template.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to refrigeration systems via DLN. **Trg Resources:** Nil.	25
LO 002.11.02: Describe Refrigeration Systems			
1 Describe the purpose of a refrigeration system. 2 Describe the properties of refrigerants and commonly used refrigerants. 3 Describe common components found within refrigeration systems, to include the following: a. Compressors: (1) Types; (2) Reciprocating; (3) Screw; (4) Rotary Vane; (5) Configurations; (6) Open;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to principles of refrigeration cycle. **Trg Resources:** Nil	50

(7) Hermetic; and (8) Semi-Hermetic. b. Condensers: (1) Finned tube radiators; (2) Shell and tube heat exchangers; (3) Throttles; (4) Capillary tube; (5) TXV; (6) Evaporators; (7) Finned tube radiators; and (8) Shell and tube heat exchangers.			
LO 002.11.03: Describe Refrigeration Safety			
1 Describe the Hazardous Properties of Refrigerant to include the following: a. Odourless; b. Displaces oxygen; c. Irritant; d. Ozone depleting substance; e. Global warming potential; 4 Describe refrigeration safety practices, to include the following: f. First aid; g. Skin contact; h. Refrigerant in eyes; i. Overcome by expelled refrigerant; j. Recording; k. Due diligence; and l. Procedure for loss of refrigerant.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to safety concepts related to working with refrigerant. **Trg Resources:** Nil	50
LO 002.11.04: Describe Main Refrigeration systems on all RCN Classes			

1 Compare the Main Refrigeration system on all RCN Classes, to include the following: a. Compressor; b. Condenser; c. Evaporator; d. Thermal control valve; e. Local Gauges; f. Para sense location; g. Control panel; and h. Oil sump.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic main refrigeration system in the above teaching point, the trainees shall compare each RCN Class's main refrigeration system configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures and interactive images of main refrigeration system configurations / schematics	25
LO 002.11.05: Describe Leak Detection Testing on Refrigeration System			
1 Describe purpose of leak detection in refrigeration systems. 2 Describe types of leak detector systems, to include the following: a. Portable leak detectors; b. Fitted leak detectors; and c. Describe Shipboard fitted sensors for leak detection. 3 Describe leak detection methods using a portable leak detection device to include: a. Describing the leak detector and its capabilities; b. Describing the parts of the leak detector; c. Describing how the leak detector is used; and d. Describing common areas where leaks are found on refrigeration system.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Using animations, images, videos and interactive tools, the trainee shall be introduced to refrigeration leak detection methods. **Trg Resources:** Nil	50
LO 002.11.06: Testing for Leaks on Refrigeration Systems			

1 Describe use of a respiratory protection device in case of a major refrigeration leak. 2 Demonstrate the procedures in case of skin/eye contact with the leaked refrigerant gas. 3 Utilize a leak detector on a refrigeration system.		**Method / Media:** EDI **Trg Environment:** Refrigeration workshop **Activity Details:** Using a leak detection device, the instructor will demonstrate how to perform leak detection procedures, followed by the trainee practicing operation of leak detector and leak detection procedures. Instructor to ensure trainees follow all safety procedures. **Trg Resources:** ● Leak detection device, ● Refrigeration system	150

16. Test Details. EO 002.11 will be tested in PC 002 IAW Annex K. Test Details found in Annex L. Application of this knowledge will be practically assessed throughout the skill-based PC.

PO 3 Maintain Electrical Systems

1. Performance. Maintain electrical systems.
2. Conditions
    a. Given:
        (1) PPE;
        (2) Electrical tools (IAW Annex H);
        (3) Testing equipment (IAW Annex H);
        (4) LOTO equipment (IAW Annex H);
        (5) Lab Volt Equipment;
        (6) DRMIS account;
        (7) DLN 3.0 account;
        (8) Assistance;
        (9) Supervision; and
        (10) References.
    b. Denied: Nil.
    c. Environmental: Any conditions on ship, at sea or alongside.
3. Standard. The basic maintainer shall maintain electrical systems IAW refs by:
    a. Inspecting DC Circuits to include:
        (1) Inspecting basic circuits;
        (2) Interpreting DC electrical drawings and symbols;
        (3) Applying electrical safety considerations to include identifying arc flash hazards;
        (4) Locking/tagging out systems as required; and
        (5) Analyzing DC circuits.
    b. Inspect AC Circuits to include:
        (1) Inspecting magnetism and electromagnetic induction;
        (2) Analyzing AC circuits;
        (3) Interpreting AC electrical drawing and symbols;
        (4) Applying electrical safety considerations to include identifying arc flash hazards;
        (5) Locking/tagging out systems as required;
        (6) Inspecting the different types of AC loads; and
        (7) Analyzing AC power.
    c. Inspecting DC machines to include:
        (1) Checking grounding to include confirming integrity of straps and insulation;
        (2) Cleaning commutators;
        (3) Testing insulation of commutators/wires;
        (4) Checking brushes and filters;
        (5) Greasing bearings;
        (6) Investigating bearings;
        (7) Inspecting basic operation of DC motors by:
            (a) Inspecting DC motor components seals (AUT1750);
            (b) Inspecting DC control components;

(c) Conducting voltage test (AT0170);
(d) Conducting resistance checks (AT0070); and
(e) Conducting temperature test (AT0140);

(8) Analyzing speed control and regulation to include completing rotational speed test (AT0080);
(9) Operating a DC motor; and
(10) Operating a DC generator.

d. Inspecting AC machines to include:
(1) Checking grounding to include confirming integrity of straps and insulation;
(2) Testing insulation of wires;
(3) Checking filters; and
(4) Inspecting AC Motors.

e. Troubleshooting basic control devices to include:
(1) Responding to electrical issue;
(2) Applying safety considerations including arc flash hazards;
(3) Investigating electrical faults;
(4) Proposing possible reason for fault; and
(5) Proposing possible fix of fault.

f. Maintaining ships service lighting to include:
(1) Completing voltage tests (AT0170);
(2) Completing resistance checks (AT0070);
(3) Inspecting lighting circuits;
(4) Testing light circuits;
(5) Replacing light bulbs;
(6) Inspecting lighting sockets;
(7) Testing lighting housing;
(8) Replacing incandescent lighting housing;
(9) Verifying appropriate lighting ballasts;
(10) Replacing light fixtures; and
(11) Repairing lighting ballasts.

g. Troubleshooting ships cabling and connectors to include:
(1) Conducting voltage test (AT0170);
(2) Conducting resistance test;
(3) Inspecting grounding straps;
(4) Inspecting cable insulation;
(5) Testing cable insulation;
(6) Repairing cable insulation;
(7) Inspecting internal wiring; and
(8) Inspecting electrical connections to include:
(a) Cleaning electrical connections; and
(b) Repairing electrical connections.

h. Reporting status of PG&D systems to include:

(1) Investigating PG&D;
(2) Confirming state of components;
(3) Manipulating change over switches;
(4) Manipulating Static Frequency Converter (SFC);
(5) Manipulating UPS to include cleaning filters;
(6) Manipulating breakers; and
(7) Applying lock out/tag out procedures.

4. References.

5. Task, Skill, Knowledge Numbers.

6. Training Limitations. PO 003 requires appropriate workshops in the TE to carry out practical assignments and assessments.

7. Remarks.

8. Test Details. The assessment for this PO will be conducted as an individual EC at the end of each EO.

EO 3.01 Inspect DC Circuits

1. Performance. Inspect DC circuits.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0;
      (3) Classroom.
3. Standard. The basic maintainer shall inspect DC circuits IAW refs by:
   a. Interpreting DC electrical drawings and symbols;
   b. Applying electrical safety considerations to include identifying arc flash hazards;
   c. Lock-out and Tag-out systems as required; and
   d. Analyzing DC circuits using algebraic formulas.
4. Time Required.
   a. EO – 810 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. C1, C2, D7, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.01.01: Describe Basic Atomic Theory			
1 Describe Electro-Chemical reactions to include the following: a. Matter. b. Elements. c. Compounds. d. Molecules. e. Atoms. f. Protons. g. Neutrons.	Floyd 7th edition, Principles of Electric Circuits	**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** Video "Basic Electricity Part 1" Video "Electricity Fundamentals"	90

h. Electrons. i. Free electrons. j. Electron flow. k. Ionic flow.			
LO 003.01.02: Describe Electrical Fundamentals			
1 Identify the sources of electricity to include the following: a. Friction. b. Magnetism. c. Chemical. d. Light. e. Heat. f. Mechanical Pressure. 2 Describe fundamental electrical concepts to include the following: a. Charge. b. Electromotive force. c. Current. d. Electron flow. e. Current direction. f. Resistance. g. Electric power. h. Power dissipation.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** Nil	120
LO 003.01.03: Describe Basic Circuits			
1 Describe basic circuits to include the following: a. Switches. b. DC Power supplies. c. Load. 2 Describe types of basic circuits to include the following:		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** Resistor colour code handout	180

a. Open circuit. b. Closed circuit. c. Short circuit. d. Series circuit. e. Parallel circuit. f. Series/parallel circuit. 3 Describe resistors to include the following: a. Types of resistors. b. Determining resistor values. c. Resistor color code. d. Resistor values. e. Power ratings. f. Flashcard activity. 4 Solve basic electrical problems to include the following: a. Ohm's law; and b. Power formula. 5 Measure electrical values to include the following: a. Multimeters. b. Voltmeter. c. Ammeter. d. Ohmmeter. e. Continuity. f. Decade box.			
LO 003.01.04: Analyze Series Circuits			
1 Describe basic series circuits and analysis to include the following: a. Definition; b. Total resistance in series; c. Constant current;		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. They will be required to complete an assignment on Series, Parallel, Series/parallel circuit analysis	90

d. Series voltage drop; e. Kirchhoff's Voltage Law; f. Series circuit analogy; g. series power h. Analyzing series circuits; i. Practice analysing series circuits; j. Aiding and opposing voltages; k. Short circuits; and l. Open circuits. 2 Describe basic parallel circuits and analysis to include the following: a. Definition; b. Current in parallel; c. Resistance in parallel; and d. Parallel circuit rules. 3 Describing basic series/parallel circuits and analysis to include: a. Finding resistance total; b. Resistance string in parallel; c. Series/parallel voltages; and d. Series/parallel current.		and calculations with instructor/regional instructor support and feedback. **Trg Resources:** Video "Basic Electricity Part 4" Video "Basic Electricity Part 5" Video "Basic Electricity Part 6" Video "Basic Electricity Part 7" Video "Basic Electricity Part 13"	
LO 003.01.05: Analyze Parallel Circuits			
1 Describe basic parallel circuits and analysis to include the following: a. Definition;		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. They will be required to	90

b. Current in parallel; c. Resistance in parallel; and d. Parallel circuit rules. 2 Describing basic series/parallel circuits and analysis to include: a. Finding resistance total; b. Resistance string in parallel; c. Series/parallel voltages; and b. Series/parallel current.		complete an assignment on Series, Parallel, Series/parallel circuit analysis and calculations with instructor/regional instructor support and feedback. **Trg Resources:** Video "Basic Electricity Part 4" Video "Basic Electricity Part 5" Video "Basic Electricity Part 6" Video "Basic Electricity Part 7" Video "Basic Electricity Part 13"	
LO 003.01.06: Analyze Series/Parallel Circuits			
1 Describing basic series/parallel circuits and analysis to include the following: a. Introduction; b. Finding resistance total; c. Resistance string in parallel; d. Series/parallel voltages; and e. Series/parallel current.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. They will be required to complete an assignment on Series, Parallel, Series/parallel circuit analysis and calculations with instructor/regional instructor support and feedback. **Trg Resources:** Video "Basic Electricity Part 4" Video "Basic Electricity Part 5" Video "Basic Electricity Part 6" Video "Basic Electricity Part 7" Video "Basic Electricity Part 13"	90
LO 003.01.07: Describe Electro Chemical Elements			
1 Describe basic electro-chemical battery circuit to include the following: a. Battery definition.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** Nil	120

b. Battery electro-chemical action to include: (1) Anodes; (2) Cathodes; (3) Dissimilar metals; and (4) Potential difference. c. Voltaic cell. d. Internal resistance. e. Electromotive series of elements. f. Series. g. Parallel. h. Series/Parallel. i. Gel cell battery. j. Battery maintenance.			
LO 003.01.08: Describe Electrical Safety			
1 Describe the workplace electrical safety bulletin. 2 Describe the purpose of electrical PPE to include the following: a. Determining the correct PPE; b. Types of PPE; and c. Electrical PPE. 3 Define the following electrical safety terms to include the following: a. Qualified Person; b. Supervisor; c. Responsibilities; d. Due diligence; and e. General Hazards.	Safety bulletin	**Method / Media:** DL **Trg Environment:** Classroom/ CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. They will be shown all electrical safety equipment either in pictures or physically **Trg Resources:** Video – 02807 “Electrical Safety” Pictures of / physical PPE	60

4 Describe Electrical shock first-aid to include the following: a. Definition of electrical shock; b. Effects of electric shock; c. Individual resistance to electric shock; and d. Pathways through the body.			
LO 003.01.09: Describe Arc Flash and Arc Hazard			
1 Describe arc flash hazard terminology to include the following: a. Arc flash; b. Arc hazard; c. Arc flash hazard analysis; d. Arc flash boundary; e. Arc flash suit; f. Hazard risk category (HRC); and g. Incident Energy. 2 Describe arc flash hazard to include the following: a. Arc flash event; b. Arc flash event causes; c. Arc flash magnitude factors; and d. Severity of energy factors.		**Method / Media:** DL **Trg Environment:** Classroom/ CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. They will be shown an Arc Flash suit. **Trg Resources:** Arc Flash Suit	60

10. Test Details. EO 003.01 will be tested in EC 003.01 IAW Annex K.

EO 3.02 Inspect AC Circuits

1. Performance. Inspect AC circuits.
2. Conditions.
    (1)
    b. Given:
        (1) Computer;
        (2) DRMIS;
        (3) Assistance; and
        (4) References.
    c. Denied: Nil.
    d. Environmental:
        (1) Any conditions on ship, at sea or alongside
        (2) Any location with access to DLN 3.0; or
        (3) Classroom.
3. Standard. The basic maintainer shall inspect AC circuits IAW refs by:
    a. Interpreting AC electrical drawings and symbols;
    b. Applying electrical safety considerations to include identifying arc flash hazards;
    c. Lock-out and Tag-out systems as required; and
    d. Analyzing AC circuits using algebraic formulas.
4. Time Required.
    a. EO – 330 mins to deliver training
    b. EC – 60 mins (total for all trainees)
5. References. C1, C2, D7, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.02.01 Describe Magnetism and electromagnetic induction			
1 Describe magnetism to include: a. Magnet characteristics; and b. Principles of a magnetic field. 2 Describe magnet types and materials to include the following:		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points using videos and animations. **Trg Resources:** ● Video 38599; ● Video 38613; and	90

a. Permanent; and b. Rare earth; c. Describe electromagnetism and electromagnetic induction to include: d. Basics of electromagnetism; e. Coils and core materials; f. Demagnetization; g. Saturation and flux density; h. Principles and laws for electromagnetic induction; and i. Relative motion. 3 Describe electromagnetism at work to include: a. Solenoid; b. Relay; c. Magnetic vibrators; and d. Bells.		● Video 38754	
LO 003.02.02 Analyze AC Fundamentals			
1 Describe sine waves to include: a. Basics of AC; b. AC waveform – sine waves; and c. Sine wave terms. 2 Analyze sine waves to include: a. Instantaneous value; b. Effective values in sine waves; and c. Average value.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points using videos and animations. They will complete an assignment with the calculations with feedback given by the instructor. **Trg Resources:** ● Videos/animations of SINE wave theory	90
LO 003.02.03 Describe the Different Types of AC Loads			
1 Describe resistors to include: a. AC circuits; and		**Method / Media:** DL **Trg Environment:** Classroom/CBT	50

b. Resistive circuits. 2 Describe inductors to include: a. Inductive circuits; b. Inductors in series and parallel; and c. Inductive reactance. 3 Describe capacitors to include: a. Capacitive circuits; b. Operation of capacitors; c. Capacitor energy storage and capacitance factors; d. Capacitors in series and parallel; e. Capacitive reactance and reactive power; and f. Phase shift mnemonics.		**Activity Details:** Trainees will be given the information to cover off the teaching points using videos and animations. **Trg Resources:** ● Videos/animations of different types of loads	
LO 003.02.04 Describe Power in AC			
1 Describe power in AC circuits to include: a. True power; b. Reactive power; and c. Apparent power. 2 Describe 3 phase power to include: a. Phase relationships; b. Phase separation; and c. Identifying the phases.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points using video animations. **Trg Resources:** ● Videos/animations of power phases	50
LO 003.02.05 Describe Transformer Theory			
1 Describe transformers to include: a. Basics of transformer; b. Mutual induction; and c. Turns ratio; d. Isolation transformer. 2 Describe rectification to include:		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points using video animations. **Trg Resources:** ● Videos/animations of transformer theory	50

a. Rectification overview; and b. Half-ware and full-wave bridge rectifier.			

10. Test Details. EO 003.02 will be tested in EC 003.03 IAW Annex K. Test details can be found in Annex L.

EO 3.03 Inspect DC Machines

1. Performance. Inspect DC machines.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect DC machines IAW refs by:
   a. Checking grounding to include confirming integrity of straps and insulation;
   b. Cleaning commutators;
   c. Testing insulation of commutators/wires;
   d. Checking brushes and filters; and
   e. Inspecting DC motor components.
4. Time Required.
   a. EO – 1250 mins to deliver training.
   b. EC – 60 mins (total for all trainees).
5. References. C1, C2, D7, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.03.01 Describe Principle and Construction of DC Generators			
Describe principles and construction of DC generators to include the following: 1 Generation of power; 2 DC generator action; 3 Construction of DC generators; 4 Commutators; 5 Commutation;	Direct Current Fundamentals 4th edition	**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** ● Video – 38599 “Basic Electricity Part 3”; and	150

6 Generator output; 7 Galvanometers; 8 Converting AC to DC power; and 9 Power losses.		● Video – 38613 “Electricity-Magnetism”.	
LO 003.03.02 Describe Armature Reaction			
1 Explain armature reaction to include: a. Demagnetizing effect; and b. Cross-magnetizing effect. 2 Describe methods in correcting; armature reaction to include: a. Shift the brushes; b. Compensating windings; and c. Interpoles.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** DLN 3.0 Account	70
LO 003.03.03 Describe Basic Operations of DC Generators			
1 Voltage regulation to include: a. Classification of DC generators: (1) Permanent magnet generators; (2) Separately excited generators; and (3) Self-excited generators. b. Field connections: (1) Series; (2) Shunt; and (3) Compound.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** DLN 3.0 Account	70
LO 003.03.04 Describe Mechanical Motion from Electrical Energy			
1 Principles of DC motors. 2 Motor action. 3 Producing a rotating field. 4 Elementary DC motor requirements.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points.	100

5 Torque and rotary motion. 6 Motor power losses. 7 Need for added armature coils. 8 Armature reaction.		**Trg Resources:** DLN 3.0 Account	
LO 003.03.05 Describe Basic Operations of DC Motors			
1 Classification of DC motors. 2 Self excited motors to include: a. Series filed motors; b. Shunt field motors; and c. Compound field motors. 3 Comparison of DC motors.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** DLN 3.0 Account	70
LO 003.03.06 Describe Speed Control and Regulation			
1 Speed control. 2 Speed regulation. 3 Reversal rotation.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. **Trg Resources:** DLN 3.0 Account	40
LO 003.03.07 Carry out Lab-Volt safety practices			
1 Describe Lab-volt equipment and station. 2 Perform lab volt safe practices.		**Method / Media:** EDI **Trg Environment:** Lab Volt workshop **Activity Details:** Trainees will be shown the Lab Volt equipment and will complete DC Theory Lab-Volt exercises 4 and 5 on the safe handling of lab volt equipment. **Trg Resources:** ● Lab Volt Equipment ● Electrical toolbox	100
LO 003.03.08 Inspect DC Motor components			
1 **Identify DC components to include:** a. Armature winding b. Stator poles c. Shunt Field Winding d. Series Field Winding	Lab-Volt Project Manual	**Method / Media:** EDI **Trg Environment:** Lab-Volt workshop **Activity Details:** Trainees will be applying the information introduced in the teaching points. They will be required to perform DC Machine Lab-Volt exercises 11, 12, 24-30.	650

e. Commutator f. Brushes g. Neutral brush position h. Rheostat. 2 Complete LabVolt exercises to include: a. The Direct Current Motor Part 1 b. The Direct Current Motor Part 2 c. The DC Shunt Motor d. The DC Series Motor e. The DC Compound Motor f. The DC Separately Excited Shunt Generator g. The DC Self Excited Shunt Generator. 3 Inspecting motor impeller (AUT1740). 4 Inspecting fan impeller (AKT0890). 5 Inspecting motor seals (AUT1750).		Debrief trainees after run-throughs with instructor. **Trg Resources:** ● Lab-Volt Equipment, ● Electrical tool kit	

10. Test Details. EO 003.03 will be tested in EC 003.03 IAW Annex K. Test details can be found in Annex L.

EO 3.04 Inspect AC Machines

1. Performance. Inspect AC machines.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect AC machines IAW refs to include:
   a. Checking grounding to include confirming integrity of straps and insulation;
   b. Testing insulation of wires;
   c. Checking filters; and
   d. Inspecting AC Motor components.
4. Time Required.
   a. EO – 850 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. C1, C2, D7, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.04.01 Describe AC generators			
1 Describe AC generators to include: a. Basic of AC generators; b. Rotating armature type and rotating field type; c. Three phase alternators; d. Output frequency; e. Output voltage; f. Voltage regulation; g. Synchronization;		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainee will be given the information to cover off the teaching points using videos and animations of AC machines. **Trg Resources:** ● Video 38608 ● Video 38070	75

h. Checking phase rotation; i. Synchro scope; j. Synchronous speed; and k. Generator cooling.			
LO 003.04.02 Describe AC Motor Types			
1 Describe three phase motor basics. 2 Describe synchronous speed. 3 Describe construction, operation and application of motor types to include the following: a. Squirrel-cage induction; b. Synchronous motor; c. Single phase AC motors; d. Resistance-start, Induction run motor; e. Capacitor-start, induction run motor; f. Capacitor-start, capacitor run; g. Shaded-Pole motors theory of operation; and h. Universal motors.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainee will be given the information to cover off the teaching points using videos and animations of motors. **Trg Resources:** Animations/videos of different motors. AC Motor 3D Model animation AC Motor HAL class locations animation	75
LO 003.04.03 Test AC Machines			
1 Analyze AC Motors to include: a. Split phase induction motor; b. Capacitive start motor; and c. Universal motor. 2 Analyze AC generators to include: a. Squirrel cage induction; and b. Synchronous motor.	Lab volt textbook	**Method / Media:** EDI **Trg Environment:** Lab-Volt workshop **Activity Details:** Instructor will explain and demonstrate the operation of AC machines. Trainees will individually operate AC machines with instructor supervision and feedback. Trainees will be required to complete Lab Volt assignments 31-34, 36, 52-54, 56-57. **Trg Resources:** ● Lab-Volt Equipment ● Electrical tool kit	700

10. Test Details. EO 003.04 will be formatively assessed during the labs therefore there will be no formal EC for this EO.

EO 3.05 Troubleshoot Basic Control Devices

1. Performance. Troubleshoot basic control devices.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall troubleshoot basic control devices IAW refs by:
   a. Responding to electrical issue;
   b. Applying safety considerations including arc flash hazards;
   c. Investigating electrical fault;
   d. Proposing possible reason for fault; and
   e. Proposing possible fix of fault.
4. Time Required.
   a. EO – 2100 mins to deliver training
   b. EC – 720 mins (total for all trainees)
5. References. A7, A8, A47, A49, D6, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.05.01 Explain Principles of Electric Motor Controllers			
1 Explain the principles of motor control to include: a. Types of Motor Control b. Basic Control Theory c. Types of Control Systems	American Electrician's Handbook	**Method / Media:** DL **Trg Environment:** Classroom/ CBT **Activity Details:** Trainees will be given the information to cover off the teaching points. They will be shown all electrical components **Trg Resources:**	200

		Pictures of, or physical controller components	
LO 003.05.02 Explain Basic Control Devices			
1 Purpose of control devices. 2 Transformers. 3 Control devices to include: a. Push button; b. Selector switches; c. Automatic switches; d. Pressure switches; e. Thermal switches; f. Variable resistors; g. Contactors; and h. Indicating lights.		**Method / Media:** DL **Trg Environment:** Classroom/ CBT **Activity Details:** Trainees will be introduced to basic control devices. **Trg Resources:** Pictures, models, cut-away.	100
LO 003.05.03 Examine a Simple Electric Motor Controller			
1 Investigate circuit continuity, opens and shorts. 2 Compare normally open to normally closed contacts of relays, toggle switches, and push buttons. 3 Compare make before break and break before make contacts. 4 Maintain overload protection devices to include the following: a. Control circuit functions; b. Component and terminal identification; c. Toggle switches for local and remote motor reversal and speed control; d. Pull-in and pull-out properties of electromagnetic contactors; e. Operating speed of the electromagnetic contactor;		**Method / Media:** EDI **Trg Environment:** Lab-Volt workshop **Activity Details:** Trainees will apply the information introduced in the teaching points. Trains will be required to complete Lab Volt assignments 3-5, 6, Lab-Volt 9 fig (9-1, 9-2 & 9-3) procedure 6, 11. **Trg Resources:** ● Lab-Volt Equipment, ● Electrical tool kit	1000

f. Pull-in and drop-out properties of time delay relays; g. Measurements of the time delay; h. Across the line magnetic starters; i. Tripping time vs overload for the thermal overload relay; and j. Two station control of a motor. 5 Maintain basic timing relays to include: a. Timing operations using the time delay relay; b. The delayed start magnetic starter; and c. High inertia starting loads; 6 Maintain interlocks to include: a. Reversing starter with a mechanical interlock; and b. Reversing starter with an electrical interlock; 7 Maintain reversing controllers to include: a. Reversing started with mechanical and double-electrical interlocks; and b. Instant reversing starters; 8 Maintain a reduced voltage starting controller to include: a. Reduced voltage starter using autotransformers; b. Torque per ampere characteristics of an autotransformer starter; and c. Autotransformer starter and primary resistor starter comparison.			
LO 003.05.04 Troubleshoot a Motor Controller			

1 Troubleshoot motor controllers, to include the following: a. Voltage; b. Resistance; c. Speed Test; and d. Temperature test. 2 Confirm power at equipment using DVM and as fitted indication to include: a. Checking feed breaker; b. Checking input power; and c. Checking secondary power of control transformer. 3 Confirm Status of Protective Devices to include: a. Control fuse(s); and b. Overloads.		**Method / Media:** EDI **Trg Environment:** Lab-Volt workshop **Activity Details:** Trainees will apply the information introduced in the teaching points. Trainees will be required to complete Lab Volt assignments **Trg Resources:** Lab-Volt Equipment, Electrical tool kit	100
LO 003.05.05 Complete Fault finding on control circuits			
1 Operate control circuit to include: a. Safety precautions when working with live circuits; b. open/short circuits c. Open windings; d. Single phasing of AC motors; e. Faulty switches f. Contactors; and g. Tripped or faulty overloads.	Lab-Volt project manual	**Method / Media:** EDI **Trg Environment:** Lab-Volt workshop **Activity Details:** Trainees will apply the information introduced in the teaching points. Instructor will introduce faults into a control circuit (or use faulty lab-volt components to provide realistic training scenarios). Trainees will complete lab volt assignments. **Trg Resources:** Lab-Volt Equipment, Electrical tool kit	700

10. Test Details. EO 003.05 will be tested in EC 003.05 IAW Annex K. Test details can be found in Annex L.

EO 3.06 Troubleshoot Ship's Service Lighting

1. Performance. Troubleshoot ship's service lighting.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall troubleshoot ship's service lighting IAW refs by:
   a. Conducting voltage test (AT0170);
   b. Conducting resistance checks (AT0070);
   c. Inspecting lighting circuits;
   d. Testing light circuits;
   e. Replacing light bulbs;
   f. Inspecting lighting sockets;
   g. Testing lighting sockets;
   h. Testing lighting housing;
   i. Replacing incandescent lighting housing;
   j. Verify appropriate lighting ballasts;
   k. Replacing light fixtures; and
   l. Repairing lighting ballasts.
4. Time Required.
   a. EO – 320 mins to deliver training
   b. EC – 140 mins (total for all trainees)
5. References. A7, A8, A47, A49, D6, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.06.01 Describe Shipboard lighting			
1 Navigation lighting components.		**Method / Media:** DL **Trg Environment:** CBT/Classroom	40

2 Navigation light locations. 3 Ships service lighting: a. Red lighting; b. Surgical lighting; c. Emergency lighting; d. Ceremonial lighting; e. Explosion proof lighting; f. Weather deck lighting; g. Yardarm blinker lights; h. Signal and search lights; i. Replenishment at sea lighting; and j. Vertical replenishment lighting.		**Activity Details:** Trainees will be given the information to cover the TPts, including pictures or physical training aids and lighting schematics. **Trg Resources:** Pictures or physical training aids, Lighting schematics	
LO 003.06.02 Describe Types of Bulbs			
1 Halogen 2 Incandescent. 3 Fluorescent. 4 Mercury vapor. 5 Light-emitting diode.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given the information to cover the TPts. **Trg Resources:** DLN 3.0 Account	25
LO 003.06.03 Describe Types of Fixtures			
1 Screw/bayonet. 2 Halogen. 3 Fluorescent. 4 Ballast. 5 Batteries. 6 Explosion Proof. 7 Starters.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given the information to cover the TPts. **Trg Resources:** DLN 3.0 Account	25
LO 003.06.04 Describe Electrical Drawings			
1 Reading a drawing.		**Method / Media:** DL	30

2 Types of drawings to include: a. Electrical schematics; b. Wiring diagram; and c. Block diagram.		**Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given the information to cover the TPts. **Trg Resources:** DLN 3.0 Account	
LO 003.06.05 Test Lighting Circuits			
3 Observe electrical safety considerations. 4 Confirming the power supplies including the feeder breaker. 5 Operating the switch. 6 Inspect lighting to include: a. Fixtures and bulbs; b. Cabling; and c. Ballast. 7 Repair lighting to include: a. Replacing Bulb; b. Replacing Fixture; and c. Replacing Ballast. 8 Replace Starter 11 Test lighting to include: a. Restoring power; and b. Confirming operation.		**Method / Media:** EDI **Trg Environment:** Electrical Workshop **Activity Details:** Instructor will explain and demonstrate. Trainee will individually repair lighting IAW TPts, using the EC checklist to self-assess their work with instructor support as required. Trainees will be required to apply LOTO procedures where applicable. **Trg Resources:** ● Physical training aids ● Schematics ● Electrical toolbox and multi-meters for each trainee and instructor	200

10. Test Details. EO 003.06 will be tested as part of EC 003.06 IAW Annex K. Test details can be found in Annex L.

EO 3.07 Troubleshoot Ship's Cabling and Connectors

1. Performance. Troubleshoot ship's cabling and connectors.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall troubleshoot ship's cabling and connectors IAW refs to by:
   a. Conducting voltage test (AT0170);
   b. Conducting resistance test;
   c. Inspecting grounding straps;
   d. Inspecting cable insulation;
   e. Testing cable insulation;
   f. Repairing cable insulation;
   g. Inspecting internal wiring;
   h. Inspecting electrical connections to include:
      (1) Cleaning electrical connections; and
      (2) Repairing electrical connections.
4. Time Required.
   a. EO – 1080 mins to deliver training
   b. PC – 320 mins (total for all trainees)
5. References. A7, A8, A47, A49, D6, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.07.01 Describe Cabling Onboard HMC Ships			
1 Explain the purpose of cabling to include: a. Cable types;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:**	75

b. Transit blocks and upper deck cabling; c. Cable identification; d. System identification designations; e. Grounding straps; f. Bonding straps; and g. Preventative maintenance. 2 Describe cabling termination accessories to include: a. Sleeving; b. Non-shrinkable sleeving; c. Aperseal; d. Nylon watertight connector; and e. Metal non-watertight connector.		Trainees will be given the information from the TPts including images or physical training aids, and drawings. **Trg Resources:** ● Images/physical training aids ● Cabling drawing	
LO 003.07.02 Describe Connecting and Splicing			
1 Bare metal connectors. 2 Nylon insulated ring connector. 3 Standard round with various hole sizes. 4 Standard fork. 5 Male and female plugs. 6 Inline. 7 Describe cable splicing to include: a. Western union/staggered splice; b. Pigtail splice; c. T-splice; and d. Mariette's.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given the information from the TPts including images or physical training aids, and drawings. **Trg Resources:** DLN 3.0 Account	50
LO 003.07.03 Repair Ships Cabling			
8 Repair ships cabling by:		**Method / Media:** EDI	775

a. Demonstrate the purpose and how to properly and safely utilize the following tools to include: (1) Wire Strippers (2) Side Cutters (3) Soldering to include; (a) Iron (b) Resin/Flux (c) Solder Extractor (d) Wet Sponge (e) Heat sinks (f) Ventilation b. Identifying faulty section of cable to include: (1) Completing a visual check; and (2) Utilizing multi-meters to conduct cable insulation and continuity test. c. Removing faulty section of cable. d. Inserting new section of cable by splicing cable together using: (1) Western Union/staggered splice; (2) Pigtail splice; and (3) Tee splice. e. Soldering spliced cable.		**Trg Environment:** Electrical Workshop **Activity Details:** Instructor will give safety brief for entry to electrical workshop and will explain and demonstrate maintenance of cabling. Trainees will individually repair cabling IAW TPts using the EC checklist to self-assess their work with instructor support as required. **Trg Resources:** ● Cabling ● schematics, ● electrical toolbox, ● multi-meters for each trainee and instructor ● Soldering equipment shared by the group ● heat guns will be shared by the group.	
LO 003.07.04 Repair Electrical Connectors			
1 Identify connector types to include: (1) Glands; and		**Method / Media:** EDI **Trg Environment:** Electrical Workshop	100

(2) Stakons. 2 Repair Cable Connectors by: a. Preparing the cable; and b. Attaching the appropriate connector. 3 Install cable glands by: a. Identifying cable gland types; and b. Securing a cable gland.		**Activity Details:** Instructor will explain and demonstrate. Trainees will individually repair connectors IAW TPts using the EC checklist to self-assess their work with instructor support as required. **Trg Resources:** ● Connectors ● Schematics, ● Electrical toolbox, ● Multimeters for each trainee and instructor	
LO 003.07.05 Construct a Lighting Circuit			
1 Construct a lighting circuit to include: a. Connection box; b. Duplex outlet; c. Rotary switch; d. Light fixture; e. Male receptacle; and f. Appropriate wiring and glands. 2 Confirm operation of the lighting circuit by: a. Confirming power is present using a digital voltmeter in: (1) junction boxes; (2) receptacles; and (3) lighting sockets using a digital voltmeter. b. Confirm circuit operation power and continuity.		**Method / Media:** EDI **Trg Environment:** Electrical Workshop **Activity Details:** Instructor will explain and demonstrate. Trainees will individually build a lighting circuit board IAW TPts using the EC checklist to self-assess their work with instructor feedback as required. All trainees must produce a functioning light circuit with the light turning on. **Trg Resources:** ● Cabling and connectors ● schematics ● electrical tool kit ● multimeters for each trainee and instructor ● Soldering equipment shared by the group ● heat guns shared by the group	480

10. Test Details. EO 003.07 will be tested in PC 003 IAW Annex K. Test details can be found in Annex L.

EO 3.08 Report Status of Shipboard Power Generation and Distribution

1. Performance. Report status of shipboard power generation and distribution.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall report status of shipboard power generation and distribution IAW refs by:
   a. Investigating PG&D;
   b. Confirming state of components;
   c. Manipulating Change over switches;
   d. Manipulating SFCs (static frequency converter);
   e. Manipulating UPS to include cleaning filters;
   f. Manipulating breakers; and
   g. Applying lock-out and tag-out procedures.
4. Time Required.
   a. EO – 600 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A7, A8, A47, A49, D6, D14, D15
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 003.08.01 Describe PG&D			
1 Describe the purpose of the PG&D system. 2 Define primary power to include: a. Normal power; and b. Alternative power.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given the information from the TPts including pictures or	50

3 Define secondary power. 4 Describe the common components that make up the PG&D system to include: a. Generator; b. Prime mover; c. Local control panel; and d. Primary switchboard. 5 Describe distribution systems to include: a. Ring main; b. Radial power; and c. Tree structure.		physical training aids, and PG&D schematics. **Trg Resources:** ● Pictures or physical training aids ● PG&D schematics	
LO 003.08.02 Describe Power Distribution			
1 Explain Cable Markings to include: a. The purpose; b. Cable prefixes; c. Cable marking and secondary power systems; and d. Location of cable tags. 2 Explain power distribution panels to include: a. Location numbering; b. Tally plate numbering; and c. Diagram numbering. 3 Explain an Automatic Change Over Switch (ACOS) to include: a. The purpose; b. Description; c. Normal source failure; and d. Test Switch.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given the information from the TPts including pictures or physical training aids, and PG&D schematics. **Trg Resources:** ● Pictures or physical training aids ● PG&D schematics	50

4 Explain a Manual Change Over Switch (MCOS) to include: a. The purpose; b. Description; and c. The operation. 5 Explain step-down transformers to including: a. 440 Vac / 115 Vac transformers; b. 440 VAC / 220 Vac transformers; and c. Explain purpose of frequency conversion devices. 6 Explain uninterruptible power supply. 7 Explain the purpose and safety considerations of Casualty Power runs. 8 Explain PG&D Ground fault detection to include: a. The purpose; b. Earth ground; c. Switchboard control panel; d. Ground fault detection example to include: (1) No fault; (2) Test; (3) A phase fault; (4) Troubleshooting; and (5) Breaker open.			
LO 003.08.03 Describe PG&D Systems on all RCN Classes			
1 Describe common components that make up PG&D system to include: a. IPMS; b. Switchboard;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic PG&D system in the above teaching point, the trainees shall compare each RCN Class's PG&D	50

c. ACOS; d. MCOS; e. UPS; f. Shore power; and g. Casualty power.		system configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures and interactive videos of PG&D system configurations / schematics	
LO 003.08.04 Describe Ship's Structural Cathodic Protection System			
1 Describe cathodic protection to include: a. Purpose; b. Corrosion; c. Oxidation; d. Galvanic corrosion to include: (1) Anode; (2) Cathode; (3) Galvanic series; and (4) Factors affecting corrosion. 2 Describe cathodic protection systems to include: a. Galvanic or sacrificial system; b. Impressed current systems; c. Passive shaft grounding; d. Slip ring and brush assembly; e. Complete systems; and f. Anti-fouling systems. 3 Describe basic maintenance of cathodic protection systems to include: a. Over the side readings;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given the information from the TPts including pictures or physical training aids, and Cathodic protection system schematics. **Trg Resources:** Pictures or physical training aids, system schematics	50

b. Abnormal conditions; and c. Inspection and safety.			
LO 003.08.05 Describe a Degaussing System			
1 Describe the concept of degaussing to include: a. The definition of degaussing; b. The purpose of degaussing; c. Earth's magnetic field; d. Inducted and permanent magnetism and a ship's magnetic signature; e. Basic of degaussing; f. Ship's degaussing coils to include: (1) M coil; (2) F coil; (3) Q coil; and (4) A coil; b. Degaussing range; c. Degaussing system; and d. Deperming.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given the information from the TPts including pictures or physical training aids, and Degaussing system schematics. **Trg Resources:** ● Pictures or physical training aids ● PG&D schematics	50
LO 003.08.06: Operate PG&D System Virtual Trainer			
Operate PG&D Virtual Trainer to include: 1. ACOS a. Manual Operation b. Transfer Test c. Pin Out d. Unpin 2. MCOS a. Manual Mode b. Electrical Mode c. Remote Operation 3. Switchboard		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will complete the "Show me", "guide me", "let me" for each section. **Trg Resources:** Access to the PG&D Virtual Trainer.	350

a. Parallel a Generator b. Sync a generator to shore c. Restore Load Shed Condition 4. UPS a. Flash Up b. Shut Down 5. SFC a. Flash Up b. Shut Down 6. Navigation Lights a. Pre-sail Checks b. Shut Down 7. Ship's Whistles a. Pre-sail Checks			

10. Test Details. EO 003.08 will be tested in EC 003.08 IAW Annex K. Test details can be found in Annex L.

PO 4 Maintain Shipboard Structures

1. Performance. Maintain Shipboard Structures.
2. Conditions.
   a. Given:
      (1) Appropriate Personal Protective Equipment (PPE);
      (2) Appropriate Tools;
      (3) DRMIS account;
      (4) DLN 3.0 account; and
      (5) Assistance;
      (6) Supervision; and
      (7) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The S3 basic maintainer shall maintain shipboard structures under supervision IAW refs while adhering to all safety requirements by:
   a. Inspecting hull structures to include:
      (1) Identifying ship's structure components;
      (2) Identifying structural fitted equipment;
      (3) Interpreting a ship's survey;
      (4) Reporting hull markings;
      (5) Identifying non-watertight closures and watertight closures;
      (6) Reporting ship's fluid/liquid levels; and
      (7) Supporting confined space entry team as safety attendant.
   b. Maintaining closures by:
      (1) Adjusting structures and closures;
      (2) Replacing defective components;
      (3) Disassembling doors and hatches;
      (4) Calibrating doors and hatches;
      (5) Greasing doors and hatches;
      (6) Testing doors and hatches;
      (7) Confirming watertight integrity; and
      (8) Identifying Shipboard and Jetty safety equipment.
4. References.
5. Task, Skill, Knowledge Numbers.
6. Training Limitations. PO 004 requires appropriate workshops in the TE to carry out practical assignments and assessments.
7. Remarks. Nil.
8. Test Details. The assessment for this PO will be conducted as an individual EC at the end of each EO.

EO 4.01 Inspect Hull Structures

1. Performance. Inspect hull structures.
2. Conditions.
   a. Given:
      (1) PPE;
      (2) Drafting tools (IAW Annex H);
      (3) DRMIS account;
      (4) DLN 3.0 account; and
      (5) Assistance;
      (6) Supervision; and
      (7) References.
   b. Denied: Nil.
   c. Environmental: Any condition, at sea or ashore.
3. Standard. The basic maintainer shall independently identify hull structures IAW refs by:
   a. Identifying ship's structure components;
   b. Identifying structural fitted equipment;
   c. Interpreting a ship's survey;
   d. Reporting hull markings;
   e. Identifying non-watertight closures and watertight closures;
   f. Identifying ship's fluid tanks; and
   g. Support confined space entry team as safety attendant.
4. Time Required.
   a. EO – 440 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References. A51, A53, A54 A62, A66, A67, A68, A69, A70, A71, A72, A73, A74, A75, A76, D17
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. For method 1, EO 004.01 is part of Module B and will be given by CBT either as a self-study or within a classroom depending on availability of resources and trainees.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 004.01.01: Identify Ship's Structure Components			
1 Keel 2 Frames 3 Longitudinal members 4 Shell Plating 5 Pillars 6 Deck 7 Deck heads 8 Bulkheads 9 Deck grating 10 Ladders 11 Superstructure 12 Hangar 13 Masts 14 Shaft line 15 Bilge keels 16 A brackets 17 Stern flap		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given concepts of ship's hull structure using 3D visual models and interactive training. **Trg Resources:** ● 3D models of hull structure	50
LO 004.01.02: Identify Structural Fitted Equipment			
1 Guard rails 2 Upper deck drains 3 Port lights 4 Reversible eye pads 5 Flush mounted deck fittings 6 Teleflex cable 7 Seatings 8 Mushroom/Ventilation flaps 9 Windows 10 Nose guide winch covers 11 Fuel fittings 12 Fairleads		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given basic concepts of structural fitted equipment through interactive visual aids. **Trg Resources:** DLN 3.0 Account	50
LO 004.01.03: Interpreting a Ship's Survey			

1 Describe the purpose of ship's survey to include why they are completed, and locations involved. 2 Describe the method to visually inspect ship's structure, minor structure, ship's appendage, or equipment to include: a. Deformities; b. Cracks; and c. Excessive rust.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given concepts of basic ship's surveys, including visual depictions of items in good condition compared to items in poor condition. **Trg Resources:** Images of ship with areas to inspect, including good condition and parts with damage to compare	110
LO 004.01.04: Reporting Hull Markings			
1 Describe draught marks to include: a. Purpose; b. Differences between imperial and metric draught marks; c. Common draught mark locations; and d. How to take draught mark readings. 2 Describe the purpose of datum plates located on the upper decks to include taking ullages.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given concepts of ship's hull markings using 3D visual models and interactive training. **Trg Resources:** ● 3D models/images of hull structure with markings	50
LO 004.01.05: Identifying Watertight & Non-Watertight Closures			
1 Describe the purpose of non-watertight closures. 2 Describe the different styles of non-watertight doors to include: a. Corrugated door; b. Mediterranean hatch; c. Dumbwaiter; d. Sliding door; and e. Hangar. 3 Define the common components of non-watertight doors to include: a. Door; b. Locks; c. Hinges; d. Handles; e. Door frame; f. Kick out panel;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainee will be given concepts of ship's non-watertight closures using 3D visual models and interactive training. **Trg Resources:** ● 3D models/images of ship with closures	50

g. Door; and h. Hydraulic door closers.			
LO 004.01.06: Reporting Ship's Fluid Tanks on all RCN Classes			
1 Compare the hull-fitted fluid tanks onboard each class of ship to include: a. Main system components; b. Parameters; and c. Locations.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details: Trainees** will be introduced to different tanks onboard each class of ship and get familiarized with their layouts. **Trg Resources:** Pictures and interactive lesson.	30
LO 004.01.07: Support confined space entry team as safety attendant			
1 Define a confined space 2 Describe regulations for confined space entry to include: 3 SMEs preparation 4 Roles within confined space entry team 5 Responsibilities of safety attendant 6 Identify hazards of confined spaces 7 Describe safety precautions for confined spaces; and 8 React to confined space entry communications and emergencies.		**Method / Media:** DL **Trg Environment:** Classroom and tour (if possible) **Activity Details:** Trainee will be given concepts of confined space entry by pictures and video describing confined space entry procedure **Trg Resources:** Images/3D models of confined space entry	100

10. Test Details. EO 004.01 will be tested in EC 004.01 IAW Annex K. Test Details found in Annex L.

EO 4.02 Maintain Basic Closures

1. Performance. Maintain closures.
2. Conditions.
    a. Given:
        (1) PPE;
        (2) Appropriate tools;
        (3) Functioning mock-ups for watertight doors and hatches;
        (4) References;
        (5) Assistance; and
        (6) Supervision.
    b. Denied: Nil.
    c. Environmental:
        (1) Alongside RCN vessel; and
        (2) Workshop with closures/mock-ups.
3. Standard. The basic maintainer shall maintain closures IAW refs by:
    a. Adjusting structures and closures;
    b. Replacing defective components;
    c. Disassembling doors and hatches;
    d. Calibrating doors and hatches;
    e. Greasing doors and hatches;
    f. Testing doors and hatches;
    g. Confirming watertight integrity; and
    h. Identifying Shipboard and Jetty safety equipment.
4. Time Required.
    a. EO – 1150 mins to deliver training
    b. EC – 620 mins (total for all trainees)
5. References. A51, A53, A54 A62, A66, A67, A68, A69, A70, A71, A72, A73, A74, A75, A76, D17
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
    a. For method 1, EO 004.02 is part of Module D and shall be completed within a coastal NFS.
    b. An RCN vessel with watertight closures shall be scheduled with transport as required. Tour can be completed in conjunction with other ship's tours at any point during the course.
    c. Trainees will require access to functioning mock-ups for watertight doors and hatches for the practical phase of training.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 004.02.01: Identify Structures and Closures on all RCN Classes			
1 Identify Common Ships structures on an RCN Class, to include: a. Keel; b. Frames; c. Longitudinal; d. Shell Plating; e. Pillars; f. Deck heads; g. Decks; and h. Bulkheads. 2 Identify Draught marks on a RCN Class. 3 Identify a Datum plate on a RCN Class. 4 Identify common structural fitted equipment on a RCN Class, to include: a. Guard Rails; b. Upper deck drains; c. Port lights and windows; d. Reversible eye pads; e. Flush mounted deck fittings f. (Teleflex cable); etc.); g. Seating's (winches, capstan, weapons, etc.); h. mushroom/ventilation flaps; i. windows; j. Tail guide winch covers; k. fuel fittings; l. Fairleads; and m. Any other associate fittings. 5 Identify the different styles of non-watertight doors on RCN Classes. 6 Identify the different styles of watertight closures on RCN Classes.		**Method / Media:** Tour **Trg Environment:** RCN Vessel **Activity Details:** Instructor will guide the trainees through a tour of an RCN vessel, identifying all the items listed in this LO. If possible, to be scheduled together with the below LO. **Trg Resources:** Nil	400

LO 004.02.02: Identify Shipboard and Jetty safety equipment			
1 Participating in a tour of the dockyard facilities to visualize the following: a. Small spill response kit (ship); b. Large spill response kit and content (Ship); c. Large spill response kit and content (Jetty); d. Booms; and e. Role of QHM. 7 Participating in a Shipboard Safety walkthrough for Halifax, Kingston, Victoria, or Orca class vessels, to include: a. Engineering spaces; b. Mess Decks; c. Cafeterias; d. Workshops; e. Upper Decks; and f. Damage control spaces.		**Method / Media:** Tour **Trg Environment:** RCN Class and Dockyard Jetty location **Activity Details:** Trainees will participate in guided tour describing ship/vessel/jetty safety. If possible, to be scheduled together with the above LO. **Trg Resources:** ● spill kits, ● jetty safety equipment	200
LO 004.02.03: Maintain Watertight (WT) Closures			
1 Disassemble WT door. 2 Reassemble WT door. 3 Obtain a chalk test by adjusting WT door. 4 Disassemble WT hatches. 5 Reassemble WT hatches. 6 Adjust WT hatch, to include: a. Obtaining a seal; b. Confirm correct spring tension (7 degrees); and c. Obtain a chalk test. 7 Carry out removal / renewal of a WT door and hatch's rubber gasket.		**Method / Media:** EDI **Trg Environment:** Closure workshop **Activity Details:** Instructor will explain and demonstrate basic maintenance procedures for watertight closures. Trainees will work in pairs to perform basic maintenance on watertight closures with instructor supervision **Trg Resources:** ● WT door ● WT Hatch ● Rubber Gasket ● Chalk ● Hand tools ● PPE	550

10. Test Details. EO 004.02 will be tested in the practical PC 004 IAW Annex K. Test Details found in Annex L.

PO 5 Maintain Internal Combustion Engines

1. Performance. Maintain internal combustion engines
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Basic tools (IAW Annex H);
        (5) Power tools (IAW Annex H);
        (6) Testing equipment (IAW Annex H);
        (7) DRMIS account;
        (8) DLN 3.0 account;
        (9) Small Gasoline Engine;
        (10) Diesel Engine;
        (11) Assistance;
        (12) Supervision; and
        (13) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions on ship, at sea or alongside;
        (2) Any location with access to DLN 3.0; or
        (3) Classroom.
3. Standard. The basic maintainer shall maintain internal combustion engines under supervision IAW refs while adhering to all safety requirements by:
    a. Inspecting small (diesel and gas) engines to include:
        (1) Completing Pre-start checks;
        (2) Completing rounds on engines; and
        (3) Starting engine to confirm function.
    b. Maintaining small engines by:
        (1) Completing function test on small boat diesel engine;
        (2) Changing the oil and oil filter in a diesel engine;
        (3) Changing fuel/water separator and filter;
        (4) Inspecting belts on engines;
        (5) Replacing belts on engines;
        (6) Inspecting zinc anodes on engines;
        (7) Replacing zinc anodes on engines;
        (8) Completing function test on small gasoline engine; and
        (9) Troubleshooting small boat engines.

4. References.
5. Task, Skill, Knowledge Numbers.

6. Training Limitations. PO 005 requires an appropriate workshop in the TE or at the TS to carry out practical assignments and assessments.

7. Remarks. The breakdown of this PO are the tasks involved at the S3 level. However, a significant portion of this PO will be a theory driven, to familiarize S3's with theory and operation of engines prior to practical phases during Roundsperson packages, and to incorporate scaffolding of learning for OJT and S2 training.

8. Test Details. The assessment for this PO will be conducted as an individual EC at the end of each EO.

EO 5.01 Inspect Small Engine

1. Performance. Inspect small engine.
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Basic tools (IAW Annex H);
        (5) Power tools (IAW Annex H);
        (6) Testing equipment (IAW Annex H);
        (7) DRMIS account;
        (8) DLN 3.0 account;
        (9) Small Gasoline Engine;
        (10) Diesel Engine;
        (11) Assistance;
        (12) Supervision; and
        (13) References.
    b. Denied: Nil.
    c. Environmental: Any location with access to a computer using either DWAN or internet.
3. Standard. The basic maintainer shall complete small (diesel and gas) engine familiarization IAW refs by:
    a. Inspecting small (diesel and gas) engines to include:
        (1) Completing Pre-start checks; and
        (2) Starting engine to confirm function.
    b. Identify components.
    c. Identify differences between various engines.
    d. Complete rounds on diesel engines.
    e. Complete rounds on pre-sail checks on small boats engines.
4. Time Required.
    a. EO – 525 mins to deliver training
    b. EC – 60 mins (total for all trainees)
5. References. A25, A55, A56, A57
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. For method 1, EO 005.01 will be part of Module F. Training will be by CBT.
9. NOTE – This is small engine familiarization in preparation for the Roundsperson package and OJPR. This EO is all theory based, with the next EO providing basic hands-on skills in preparation for OJT and Rounds.
10. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 005.01.01: Describe basic engine theory			
1 Describe the purpose of an engine. 2 Explain the basic functions of an internal combustion engine to include: a. Heat engines; and b. Internal combustion process. 3 Explain the purpose of enclosure assemblies to include: a. Engine enclosure assembly; b. Drainage systems; and c. Fire suppression system. 4 Describe Human Machinery Interface (HMI) on marine diesel engines and gasoline engines to include: a. Engine performance indicators; b. Local operating panel; and c. Remote control panel.		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given concepts of engines using visual models and images. **Trg Resources:** DLN 3.0 Account	50
LO 005.01.02: Describe Marine Diesel and Gasoline Engines			
1 Describe the purpose of marine diesel engines to include: a. Advantages; and b. Disadvantages. 2 Describe the purpose of gasoline engines to include: a. Advantages; and b. Disadvantages. 3 Explain reciprocating engine cycle principles to include: a. Basic reciprocating engine components; b. Reciprocating engine cylinder arrangement; c. Diesel and gasoline engine operating cycle; d. Two stroke engine cycle;		**Method / Media:** DL **Trg Environment:** Classroom/CBT **Activity Details:** Trainees will be given concepts of marine diesel engines using 3D visual models and interactive training. **Trg Resources:** ● Pictures / 3D model of an engine ● machinery specific CFTO/ OEM manual	250

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
e. Four stroke engine cycle; and f. Diesel vs. Gasoline engines. 4 Describe Diesel and Gasoline Engine Components to include: a. Bedplate; b. Engine block; c. Oil sump; d. Bearings; e. Cylinder head; f. Camshaft; g. Timing gears, Belt or Chain; h. Pushrods; i. Rockers; j. Air intake manifold; k. Air filters; l. Exhaust manifold; m. Superchargers; n. Fuel injectors; and o. Spark Plugs. 5 Describe diesel and gasoline engine subsystems to include: a. Lubrication system; b. Intake and exhaust system; c. Fuel system; d. Cooling system; e. Electrical system; and f. Starting system.			
LO 005.01.03: Describe Marine Diesel Engines on all RCN Classes			
1 Compare propulsion diesel engines on all RCN Classes, to include: a. Main Specs; b. Locations; c. Drive train assembly; and d. Power/speed output.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic marine diesel engine in the above teaching point, the trainees shall compare each RCN Class's marine diesel engine configurations in	25

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
2 Compare diesel engine generators on all RCN Classes, to include: a. Main specs; b. Locations; c. Generator sizes; and d. Outputs. 3 Compare small marine diesel engines on all RCN Classes, to include: a. Small boats; b. Portable generators; c. Portable compressors; and d. Portable/stationary pumps.		the order of complexity, from least to greatest. **Trg Resources:** ● Pictures ● 3D models of engines	
LO 005.01.04: Describe Gasoline Engines on all RCN Classes			
1 Compare small gasoline engines on all RCN Classes, to include: a. Small boats; b. Portable generators; c. Portable compressors; and d. Portable/stationary pumps.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic small gasoline engine in the above teaching point, the trainees shall compare each RCN Class's small gasoline engine configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures ● 3D model of an engine, ● interactive videos of the small gasoline engine configurations / schematics	25
LO 005.01.05: Describe a Gas Turbine Engine (Theory for Roundsperson package and OJPR)			
1 Describe the purpose of gas turbine engines. 2 Explain gas turbine cycle principles to include: a. Intake; b. Compression; c. Ignition;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given concepts of gas turbine engines using 3D visual models and interactive training. Instructor to make comparisons to other engine types.	150

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
d. Expansion; and e. Exhaust. 3 Describe gas turbine components to include: a. Compressor; b. Combustor; c. Turbines; and d. Accessory gearbox. 4 Describe gas turbine engine sub-systems to include: a. Lubrication system; b. Intake air system; c. Exhaust system; d. Fuel system; e. Starting system; and f. Water wash system.		**Trg Resources:** ● Pictures ● 3D model of an engine,	
LO 005.01.06: Describe Gas Turbine Engines on all RCN Classes			
1 Compare Propulsion Gas Turbine Engines on all RCN Classes, to include: a. Main Specs; b. Locations; c. Drive train assembly; and d. Power/speed output.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic Gas Turbine engine in the above teaching point, the trainees shall compare each RCN Class's Gas turbine engine configurations in the order of complexity, from least to greatest. **Trg Resources:** ● Pictures ● Interactive videos of the gas turbine engine configurations / schematics	25

11. Test Details. EO 005.01 will be tested in EC 005.01 IAW Annex K. This EO is not summatively assessed as further information and practical training will be on the Roundsperson training.

EO 5.02 Maintain Small Engines

1. Performance. Maintain of small engines.
2. Conditions.
    a. Given:
        (1) Computer;
        (2) Schematics;
        (3) PPE;
        (4) Basic tools (IAW Annex H);
        (5) Power tools (IAW Annex H);
        (6) Testing equipment (IAW Annex H);
        (7) DRMIS account;
        (8) DLN 3.0 account;
        (9) Small Gasoline Engine;
        (10) Diesel Engine;
        (11) Assistance;
        (12) Supervision; and
        (13) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions on ship, at sea or alongside;
        (2) Workshop;
        (3) Any location with access to DLN 3.0; or
        (4) Classroom.
3. Standard. The basic maintainer shall maintain small engines IAW refs by:
    a. Completing function test on small boat diesel engine.
    b. Changing the oil and oil filter in a diesel engine.
    c. Changing fuel/water separator and filter.
    d. Inspecting belts on engines.
    e. Replacing belts on engines.
    f. Inspecting zinc anodes on engines.
    g. Replacing zinc anodes on engines.
    h. Completing function test on small gasoline engine.
    i. Troubleshooting small boat engines.
4. Time Required.
    a. EO – 930 mins to deliver training
    b. EC – 260 mins (total for all trainees)
5. References. A25, A55, A56, A57
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 005.02.01: Complete function test small boat diesel engines			
1 Conduct pre-start checks on diesel engine. 2 Start a diesel engine. 3 Shut off a diesel engine.		**Method / Media:** EDI **Trg Environment:** Workshop/Maintenance Area with proper ventilation **Activity Details:** Instructor will explain and demonstrate how to operate a diesel engine. Three trainees will carry-out the procedure from start to finish with the rest of the trainees watching and giving feedback (fishbowl activity). **Trg Resources:** ● Engine	200
LO 005.02.02: Change the oil and oil filter in a marine diesel engine			
1 Drain oil from engine. 2 Change oil filter/carbon filter. 3 Refill engine oil.		**Method / Media:** EDI **Trg Environment:** Workshop/Maintenance Area with proper ventilation. **Activity Details:** Instructor/regional instructor will explain and demonstrate how to change the engine oil and filter. Trainees will change the engine oil and filter in small groups with the rest of the trainees watching and giving feedback (fishbowl activity). Each small group will be required to do a minimum of one complete oil change. **Trg Resources:** ● Engine ● Oil ● Filter ● Dirty oil pump (as required) ● HAZMAT disposal items	175
LO 005.02.03: Change fuel/water separator and filter			
1 Change fuel/water separator elements. 2 Change fuel filter. 3 Vent the fuel system following maintenance routines.		**Method / Media:** EDI **Trg Environment:** Workshop/Maintenance Area with proper ventilation. **Activity Details:** Instructor will explain and demonstrate how to change the fuel/water separator and	150

		filter. Trainees will change the fuel/water separator and filter in small groups with the rest of the trainees watching and giving feedback. Each small group will be required to do a minimum of one complete fuel filter change. **Trg Resources:** ● Engine ● Fuel ● Filter ● HAZMAT disposal items	
LO 005.02.04: Inspect and Replace Belts on a Marine Engine			
1 Inspect belt for: a. Wear; b. Cracking; c. Proper tension. 2 Inspect belt for wear and cracks, replace if necessary.		**Method / Media:** EDI **Trg Environment:** Workshop/Maintenance Area. **Activity Details:** Instructor will explain and demonstrate how to replace belts. Trainees will replace belts in small groups of two, with the rest of the trainees watching and giving feedback. Each small group will be required to replace a minimum of one belt. **Trg Resources:** ● Engine ● Required Tools ● Belts	120
LO 005.02.05 Inspect and replace Zinc Anodes on Marine Engines			
1 Inspect zinc anodes; and 2 Replace zinc anodes.		**Method / Media:** EDI **Trg Environment:** Workshop/Maintenance Area. **Activity Details:** Instructor will explain and demonstrate how to inspect and replace zinc anodes. Trainees will inspect and replace zinc anodes in small groups of two, with the rest of the trainees watching and giving feedback. Each small group will be required to replace a minimum of one zinc anode. **Trg Resources:**	75

		● Engine ● Required Tools ● Anodes	
LO 005.02.06: Complete function test on a Small Gasoline engine			
1 Start a small gas engine, to include: a. Starting the small gas engine; b. Adjusting speed of the small gas engine; and c. Stopping the small gas engine.		**Method / Media:** EDI **Trg Environment:** Workshop/maintenance area/small boat. **Activity Details:** Instructor to explain and demonstrate operating a small engine. Trainees will independently practice a minimum of one time and operate the small gasoline engine once. **Trg Resources:** Small gasoline engine, fuel, and oil.	140
LO 005.02.07: Troubleshoot small boat marine engine			
1 Troubleshoot the following difficulties on small boat marine engines and systems: a. Cooling system; b. Fuel system; c. Starting system; d. Air Induction; e. Emissions system; f. Electrical system; and g. Lube Oil system. h. Fuel blockages; i. Restricted inlet; j. Low lube oil pressure; k. High lube oil pressure; l. High water temperature alarms; m. Faulty start relay; and n. No power.		**Method / Media:** EDI **Trg Environment:** Workshop/Maintenance Area. **Activity Details:** Instructor will explain and demonstrate how to troubleshoot small boat marine engines. Trainees will troubleshoot small boat marine engines in pairs. Each pair will be required to troubleshoot 2 items. **Trg Resources:** ● Engine ● Required Tools	70

10. Test Details. EO 005.02 will be tested in EC 005.02 IAW Annex K (basic maintenance on internal combustion engine). This EO is not summatively assessed as further information and practical training will be on the Roundsperson.

PO 6 Inspect Propulsion Systems

1. Performance. Inspect propulsion systems
2. Conditions.
   a. Given:
      (1) DRMIS account;
      (2) DLN 3.0 account; and
      (3) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The S3 maintainer shall Inspect propulsion systems IAW refs by:
   a. Inspecting gearing and shaft lines to include:
      (1) Checking plumber blocks (CPF);
      (2) Looking into enclosures;
      (3) Checking seals;
      (4) Checking for abnormal conditions, sounds, vibrations, smells, etc; and
      (5) Checking for obvious leaks.
   b. Inspecting the steering system to include:
      (1) Checking for hydraulic leaks;
      (2) Checking oil levels;
      (3) Checking temperatures; and
      (4) Checking pressures.
4. References.
5. Task, Skill, Knowledge Numbers.
6. Training Limitations. Nil.
7. Remarks. This Performance Objective is not a true performance-oriented objective. It is designed to be familiarization and preparation of the S3 trainee for the Roundsperson package. Because that QSP is separate from this one, scaffolding of knowledge and theory must occur as opposed to building the Performance fully, and using OJT as a teaching environment or instructional strategy. When RQS3 is reviewed again, it is recommended OJT, Roundsperson packages, and RQ QSPs be analyzed together, and divided for training purposes in the design.
8. Test Details. The assessment for this PO will be conducted as an individual EC at the end of each EO.

EO 6.01 Inspect Gearing and Shaft lines

1. Performance. Inspect gearing and shaft lines.
2. Conditions.
   a. Given:
      (1) Computer;
      (2) DRMIS;
      (3) Assistance; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect gearing and shaft lines IAW refs by:
   a. Checking plumber blocks (CPF);
   b. Looking into enclosures;
   c. Checking seals;
   d. Checking for abnormal conditions, sounds, vibrations, smells, etc; and
   e. Checking for obvious leaks.
4. Time Required.
   a. EO – 405 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References.
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. Theory based EO to prepare for class specific Roundsperson training package.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 006.01.01: Describe Shipboard Propulsion System			
1 Describe the purpose of a Shipboard Propulsion System. 2 Define principles of propulsion to include: a. Thrust; b. Pitch; 3 Identify basic propulsion components and positions within a system layout to include: a. Control system; b. Prime mover; c. Gear Box/Transmission; d. Shaft line; e. Thrust block f. bearings; g. Propeller; and h. control system.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given basic principles of marine propulsion and parts of propulsion system through interactive visual aids. **Trg Resources:** DLN 3.0 Account	50
LO 006.01.02: Describe Gearing			
1 Describe the purpose of gears. 2 Explain gear types to include: a. spur gears: (1) external; (2) internal; (3) rack and pinion. b. helical gears: (1) axial loading; (2) gear pairs. c. bevel gears. d. worm and wheel gears: (1) advantages; (2) disadvantages. 3 Explain fundamental gearing principles, to include: a. driving gear.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given basic principles of marine propulsion and parts of propulsion system through interactive visual aids. **Trg Resources:** DLN 3.0 Account	100

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
b. driven gear. c. Shaft power. d. Power conservation. e. Speed reduction. f. Mechanical Advantage/Gear Ratio. 4 Describe gear arrangements to include: a. Simple train. b. Idler gears. c. Compound train. d. Epicyclic/Planetary gear set. e. Gearbox. f. Marine gear box.			
LO 006.01.03: Describe Components of a Shaft line			
1 Describe the purpose of a shaft line. 2 Describe components of a shaft line to include: a. Shafting. b. Bearings to include: (1) inboard bearings; (2) thrust block; (3) outboard bearings. c. Couplings to include: (1) Rigid; (2) Flexible; (3) Elastomer; (4) Gear. d. Clutches to include: (1) Power transfer methods; (2) Clutch engagement methods; (3) Automatic actuation. e. Seals to include: (1) Bulkhead seals;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given the basic principles of marine propulsion and shown parts of the propulsion system. **Trg Resources:** ● visual aids to describe system schematic	150

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
(2) Pressure activated bulkhead seals; (3) Stern seal. f. Stern tube to include: (1) Fairings; (2) Rope guard. g. Propellers to include: (1) Fixed pitch propeller; (2) Controllable pitch propeller.			
LO 006.01.04: Describe Bearings			
1 Describe the purpose of bearings to include: a. Purpose of bearings; b. Alignment of parts; c. Support loads; d. Reduce friction and wear. 2 Describe types of shaft loads to include: a. Axial loads; b. Radial Loads. 3 Describe bearing care and Lubrication. 4 Describe types of bearings to include: a. Journal (sliding contact) type bearings; b. Anti-friction (rolling contact) bearings.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees will be given basic principles of marine propulsion and parts of propulsion system through interactive visual aids. **Trg Resources:** DLN 3.0 Account	25
LO 006.01.05: Describe Shaft line Operation			
1 Describe how the components of the shaft line operate together with the ship's propulsion system to include: a. Shaft line arrangement; b. Bearing support; c. Coupling connections; d. Propeller shaft;		**Method / Media:** DL **Trg Environment: CBT/**Classroom **Activity Details:** Students learn basic principles of marine propulsion and parts of propulsion system through interactive visual aids and video on DLN **Trg Resources:**	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
e. Hollow shaft lines; f. Propulsion system. 2 Describe various generic marine propulsion systems to include: a. Single Shaft; b. Dual Shaft; c. Z-drive/Azipod; d. Bow thrusters; e. Outboard motors.		• Computer, DLN access, DWAN/Internet, propulsion and steering systems schematics	
LO 006.01.06: Describe Shipboard Propulsion Systems on all RCN Classes			
1 Identify and compare classes of propulsion systems onboard RCN Classes to include: a. HFX Propulsion System. b. KINGSTON Propulsion systems. c. ORCA Propulsion systems.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic shipboard propulsion system in the above teaching point, the trainees shall compare each RCN Class's shipboard propulsion system configurations in the order of complexity, from least to greatest. **Trg Resources:** Images of shipboard propulsion system configurations / schematics	30

10. Test Details. EO 006.01 will be tested in EC 006.01 IAW Annex K. Test details can be found in Annex L. At this time candidates will be tasked and tested to PBM on propulsion systems as a part of their Roundsperson package.

EO 6.02 Inspect Steering Systems

1. Performance. Inspect steering systems.
2. Conditions.
   a. Given:
      (1) PPE;
      (2) Drafting tools (IAW Annex H);
      (3) DRMIS; and
      (4) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any conditions on ship, at sea or alongside
      (2) Any location with access to DLN 3.0; or
      (3) Classroom.
3. Standard. The basic maintainer shall inspect steering systems IAW refs to include:
   a. Checking for hydraulic leaks;
   b. Checking oil levels;
   c. Checking temperatures; and
   d. Checking pressures.
4. Time Required.
   a. EO – 225 mins to deliver training
   b. EC – 60 mins (total for all trainees)
5. References.
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. Theory based EO to prepare for class specific Roundsperson package.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 006.02.01: Describe Shipboard Steering Systems			
1 Describe the purpose of a steering system. 2 Explain different methods/types of steering arrangements to include: a. Rudder; b. Z-drive/azipod; c. Bow thruster; d. Planes. 3 Explain basic hydraulic system principles to include: a. Pumps; b. Reservoirs; c. Control valves; d. Actuators. 4 Describe the components of a rudder based steering system to include: a. Hydraulic components; b. Rotary vane; c. Helm units; d. Rudder locking assembly. 5 Explain the operation of steering systems. 6 Explain emergency steering arrangements to include: a. Centering arrangement; b. Accumulators; c. Manual tiller arm; d. Emergency centering arrangement.		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Trainees learn basic principles of marine steering system through interactive visual aids and video on DLN **Trg Resources:** Computer, DLN access, DWAN/Internet	200
LO 006.02.02: Compare Steering Systems on all RCN Classes			
1 Compare the steering systems on all RCN Classes to include: a. Types; b. Steering positions;		**Method / Media:** DL **Trg Environment:** CBT/Classroom **Activity Details:** Starting from the generic shipboard steering system in the above teaching point, the trainees shall compare each RCN Class's	25

c. Hydro mechanical configuration; and d. Rudder.		shipboard steering system configurations in the order of complexity, from least to greatest. **Trg Resources:** Pictures and interactive videos of shipboard steering system configurations / schematics	

10. Test Details. EO 006.02 will be tested in EC 006.02 IAW Annex K. Test details can be found in Annex L. At this time candidates will be tasked and tested to PBM on propulsion systems as a part of their Roundsperson package.

PO 7 Repair Metal Using Heat

1. Performance. Repair metal using heat
2. Conditions.
    a. Given:
        (1) Safety Equipment / PPE;
        (2) Tools;
        (3) Oxy / Acetylene equipment;
        (4) Emergency burning and welding kit;
        (5) Exothermic torch kit;
        (6) Training aids;
        (7) Assistance;
        (8) Supervision; and
        (9) References.
    b. Denied: Nil.
    c. Environmental:
        (1) Any conditions, day, or night, at sea or alongside; or
        (2) Welding shop.
3. Standard. The S3 maintainer shall repair metal using heat IAW refs by:
    a. Identifying repair method according to leak or fault.
    b. Applying safety considerations.
    c. Completing applicable hot work chits.
    d. Assembling oxy-acetylene torch IAW ref to include:
        (1) Opening gas bottles;
        (2) Inspecting equipment to include the following:
            (a) Bottles;
            (b) Regulators;
            (c) Hoses;
            (d) Flash back arrestor; and
            (e) Torch.
        (3) Testing for equipment leaks to include the following:
            (a) Bottles;
            (b) Regulators;
            (c) Hoses;
            (d) Flash back arrestor; and
            (e) Torch.
    e. Cutting metal with oxy-acetylene torch to include:
        (1) Configuring torch;
        (2) Cutting metal; and
        (3) Filing metal.
    f. Welding Metal with oxy-acetylene torch to include:
        (1) Preparing joint;

(2) Selecting filler metal;
(3) Configuring torch; and
(4) Welding a joint.

g. Cutting metal using exothermic torch IAW ref to include:
(1) Configuring plasma torch by:
(a) Checking battery power;
(b) Plugging-in striker and torch lead;
(c) Connecting oxygen regulator;
(d) Opening cylinder valve;
(e) Placing rod in torch;
(f) Pulling trigger on torch; and
(g) Igniting rod on striker.
(2) Cutting obstruction.

4. References. Specified in applicable EOs.
5. Task, Skill, Knowledge Numbers. AMT0140, AMT0960
6. Training Limitations. Nil.
7. Remarks. Nil.
8. Test Details: PC 007 will have a 300 min practical assessment IAW Annex L.

EO 7.01 Assemble Oxy-Acetylene Torch

1. Performance. Assemble Oxy-Acetylene torch.
2. Conditions.
   a. Given:
      (1) Safety Equipment / PPE;
      (2) Tools;
      (3) Oxy / Acetylene equipment;
      (4) Emergency burning and welding kit;
      (5) Exothermic torch kit;
      (6) Training aids;
      (7) Assistance;
      (8) Supervision; and
      (9) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any condition, at sea or ashore;
      (2) Welding shop.
3. Standard. The basic maintainer shall assemble oxy-acetylene torch IAW refs by:
   a. Opening gas bottles;
   b. Inspecting equipment to include the following:
      (1) Bottles;
      (2) Regulators;
      (3) Hoses;
      (4) Flash back arrestor; and
      (5) Torch.
   c. Testing for equipment leaks to include the following:
      (1) Bottles;
      (2) Regulators;
      (3) Hoses;
      (4) Flash back arrestor; and
      (5) Torch.
   d. Changing tanks.
   e. Labeling pressures on bottles.
   f. Identifying critical low pressure (80 psi min).
   g. Inspecting new bottles.
   h. Moving bottles safely.
   i. Ensuring adequate ventilation.
   j. Preparing space for work.
4. Time Required.
   a. EO – 300 mins to deliver training
   b. EC – confirmation through Instructor feedback

5. References. A59, A61, A62, A63, A64, A65, C3, C4, C5
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. EC will be completed through Instructor Feedback.
9. Teaching Points.

Teaching Points	Refs	Method / Media / Environment / Training Activity Details	Time (mins)
LO 007.01.01 Apply Oxy / Acetylene Safe Work Practices			
1 Describe oxy / acetylene theory to include: a. Oxy / acetylene theory. b. Joining theory to include: (1) Welding; (2) Brazing; (3) Soldering. c. Cutting theory. d. Characteristics of oxygen. e. Oxygen cylinders: (1) Characteristics; (2) Safety precautions. f. Characteristics of acetylene. g. Acetylene cylinders: (1) Characteristics; (2) Safety precautions. 2 Describe steel metallurgy to include: a. Types of steel: (1) Martensite; (2) Bainite; (3) Pearlite. b. Properties of dimensional steel:	A1, A2, A3, A4, A8, A10, A11, A13, C1, C2, C5	**Method:** DL **Media:** Instructor, CBT, PPT, print, video, audio-conference, real equipment **Trg Environment:** Classroom* / designated workshop location **Trg Resources:** ● DWAN Computer ● DLN Account ● PPTs ● A/V equipment ● Videos ● Visual Trg Aids ● Physical Trg Aids ● References	100

(1) Ferrous metals; (2) Non-ferrous metals; (3) Impacts of processing metals. 3 Describe Welding components, hot work and equipment set-up to include: a. Describe Oxy / acetylene components to include: (1) Regulators; (2) Regulator gauges; (3) Flashback arrestors, check valves and shut off valves; (4) Torch body; (5) Tips and Heads; (6) Hoses; (7) Hose precautions; and (8) Miscellaneous tools. b. Describe hot work to include: (1) Hot work requirements; and (2) Fire sentry. c. Describe oxy/acetylene equipment set-up. 4 Describe Flame anatomy to include: a. Three types of flame to include: (1) Neutral; (2) Oxidation; and			

(3) Carbon. b. Parts of flame to include: (4) Inner Cone; (5) Intermediate Cone; and (6) Outer Envelope. c. Flame temperature map.			
LO 007.01.02 Apply Oxy / Acetylene Safety			
1 Describe Welding shop safety to include: a. Fires extinguisher locations. b. Ventilation requirements. c. Eye wash stations. d. First aid kit location. e. PPE location. 2 Identify oxy/acetylene components to include: a. Cylinders; b. Regulating valves; c. Hoses; d. Torch bodies; e. Flashback arresors; f. Torch tips; g. Striker; h. Cylinder pressure gauge; i. Working pressure gauge; j. Flint striker; k. Tip cleaners; l. Chipping hammer; m. Wire brush; n. Brass wrenches.	A1, A2, A3, A4, A8, A10, A11, A13, C1, C2, C5	**Method:** D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location **Trg Resources:** ● Handout ● Visual Trg Aids ● Physical Trg Aids ● References	100

3 Perform safe handling of compressed gas cylinders to include: a. Safety precautions; b. safe handling of compressed gas cylinders.			
LO 007.01.03 Set-up Oxy / Acetylene Equipment			
1 Demonstrate equipment setup IAW oxy / acetylene equipment setup checklist by explaining equipment setup procedure. 2 Demonstrate safe operation of an oxy/acetylene torch to include: a. PPE; b. Flame control techniques; c. Types of flame: (1) pure acetylene; (2) Carburizing flame; (3) Neutral flame; and (4) Oxidizing flame. d. Basic troubleshooting (1) cleaning the tip; (2) backfire.	A1, A2, A3, A4, A8, A10, A11, A13, C1, C2, C5	**Method:** D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location **Trg Resources:** ● Handout ● Visual Trg Aids ● Physical Trg Aids ● References	100

10. Test Details. EO 007.01 will be tested in PC 007 IAW Annex K. Test details can be found in Annex L.

EO 7.02 Weld Metal with Oxy-Acetylene Torch

1. Performance. Weld metal with oxy-acetylene torch.
2. Conditions.
   a. Given:
      (1) Safety Equipment / PPE;

(2) Tools;
(3) Oxy / Acetylene equipment;
(4) Emergency burning and welding kit;
(5) Exothermic torch kit;
(6) Training aids;
(7) Assistance;
(8) Supervision; and
(9) References.

b. Denied: Nil.

c. Environmental:
(1) Any condition, at sea or ashore;
(2) Welding shop.

3. Standard. The basic maintainer shall weld metal with oxy-acetylene torch IAW refs by:
a. Preparing joint;
b. Selecting filler metal;
c. Applying PPE;
d. Completing safety checks;
e. Completing hot work chit;
f. Ensuring proper ventilation;
g. Lighting flame;
h. Completing leak tests;
i. Configuring torch; and
j. Braising a joint.

4. Time Required.
a. EO – 1200 mins to deliver training
b. EC – through instructor feedback.

5. References. A59, A61, A62, A63, A64, A65, C3, C4, C5

6. Task, Skill, Knowledge Numbers.

7. Training Limitations. Nil.

8. Remarks. Nil.

9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LO 007.02.01: Perform Oxy/Acetylene Welding			
1 Demonstrate basic metal grinding.		**Method:** E-D, D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location	1200

2 Demonstrate basic oxy/acetylene torch and filler rod hand skills. 3 Demonstrate basic weld joints to include: a. Lap; b. Tee; c. Butt; d. Gap. 4 Demonstrate basic flat weld. 5 Repair metal to include: a. Cutting a metal shape; b. Welding a flat butt joint. 6 Weld by: a. Preparing joint; b. Selecting filler metal; c. Applying PPE; d. Completing safety checks; e. Completing hot work chit; f. Ensuring proper ventilation; g. Completing leak tests; h. Lighting flame; i. Configuring torch; and j. Braising a joint.		**Activity Details:** **Trg Resources:** ● Visual Trg Aids ● Physical Trg Aids ● References	

10. Test Details. EO 007.02 will be tested in PC 007 IAW Annex K. Test details can be found in Annex L.

EO 7.03 Cut Metal with Oxy-Acetylene Torch

1. Performance. Cut metal with an oxy-acetylene torch.
2. Conditions.
   a. Given:
      (1) Safety Equipment / PPE;
      (2) Tools;
      (3) Oxy / Acetylene equipment;
      (4) Emergency burning and welding kit;
      (5) Exothermic torch kit;
      (6) Training aids;
      (7) Assistance;
      (8) Supervision; and
      (9) References.
   b. Denied: Nil.
   c. Environmental:
      (1) Any condition, at sea or ashore;
      (2) Welding shop.
3. Standard. The basic maintainer shall cut metal with oxy-acetylene torch IAW refs by:
   a. Applying PPE;
   b. Completing safety checks;
   c. Completing hot work chit;
   d. Ensuring proper ventilation;
   e. Configuring torch;
   f. Lighting flame;
   g. Completing leak tests;
   h. Cutting metal; and
   i. Securing area.
4. Time Required.
   a. EO – 250 mins to deliver training
   b. EC – through instructor feedback.
5. References. A59, A61, A62, A63, A64, A65, C3, C4, C5
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. Nil.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)

LO 007.03.01: Cut Metal with Oxy-Acetylene Torch			
1 Explain cutting station equipment to include: a. A welding outfit; b. lighting; c. ventilation; and d. cutting table. 2 Cut metal using an oxy/acetylene torch to include: a. cutting process; b. straight free hand cut; c. straight cut with guide; d. 45 degree cut with guide; e. cut 6.5” radius arc free hand; and f. pierce four ¾” diameter round holes.	A1, A2, A3, A4, A8, A10, A11, A13, C1, C2, C5	**Method:** E-D, D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location **Activity Details:** **Trg Resources:** ● Visual Trg Aids ● Physical Trg Aids ● References	250

10. Test Details. EO 007.03 will be tested in PC 007 IAW Annex K. Test details can be found in Annex L.

EO 7.04 Cut Metal with Exothermic Torch

1. Performance: Cut metal with exothermic torch
2. Conditions:
   a. Given:
      (1) Safety Equipment / PPE;
      (2) Tools;
      (3) Exothermic torch;
      (4) Training Aids;
      (5) References;
      (6) Assistance; and
      (7) Supervision.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) Designated Workshop Location.
3. Standard: The trainee shall cut metal using an exothermic torch IAW refs by:
   a. Applying exothermic safe work practices.
   b. Donning exothermic PPE.
   c. Setting up exothermic equipment.
   d. Applying a sawing motion through obstruction.
4. Time Required.
   a. EO – 1050 mins
   b. EC – Through Instructor Feedback
5. References. A59, A61, A62, A63, A64, A65, C3, C4, C5
6. Task, Skill, Knowledge Numbers.
7. Training Limitations. Nil.
8. Remarks. Nil.
9. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
007.04.01: Apply Exothermic Safe Work Practices			
1 Describe direct current electrical theory to include a short circuit. 2 Explain combustion theory to include the creation of excess heat (combustion)	A1, A2, A3, A4, A8, A9, A13, C1, C3	**Method:** IL, D-P **Media:** Instructor, PPT, real equipment **Trg Environment:** Classroom / Designated Workshop Location **AAP Details:** **Trg Resources:**	150

3 Explain equipment performance theory to include: a. O2 bottle; b. Pre-set regulator; c. O2 hose; d. O2 valve; e. Striker; f. Electrical leads; g. battery pack; h. brass fittings; i. Tool limitations. 4 Explain metallurgy to include: a. Vulnerabilities of certain alloys; b. Self-sustained reaction. 5 Don exothermic PPE to include: a. Welding helmet; b. Welding gloves.		● Visual Trg Aids ● Physical Trg Aids ● References	
007.04.02: Setup Exothermic Equipment			
1 Demonstrate exothermic equipment setup by explaining exothermic pack assembly / disassembly.	A1, A2, A3, A4, A8, A9, A13, C1, C3	**Method:** D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location **AAP Details:** **Trg Resources:** ● Visual Trg Aids ● Physical Trg Aids References	100
007.04.03: Perform Basic Exothermic Cutting			
1 Demonstrate basic / safe exothermic lance handing techniques. 2 Demonstrate operating techniques to include:	A1, A2, A3, A4, A8, A9, A13, C1, C3	**Method:** D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location **AAP Details:**	800

a. Cutting without power; b. Travel speed; c. Electrode to work angle; d. Sawing motion.		**1** Learning Activity 29: D-P **2** Learning Activity 30: D-P **Trg Resources:** ● Visual Trg Aids ● Physical Trg Aids ● References ● Commercial Exo Cutting machine (that can handle unlimited rod starts) ● Or, one "arcair slice" pack per. Student. ● Minimum 4 EXO rods / student.	

10. Test Details. EO 007.04 will be tested in PC 007 IAW Annex K. Test details found in Annex L.

EO 7.05 Perform Basic Shielded Metal Arc Welding

1. Performance: Perform Basic Shielded Metal Arc Welding
2. Conditions:
   a. Given:
      (1) Safety Equipment / PPE;
      (2) Tools;
      (3) Arc Equipment;
      (4) Augmented Arc Training System (Augmented Reality)
      (5) Training Aids;
      (6) References;
      (7) Assistance; and
      (8) Supervision.
   b. Denied: Nil.
   c. Environmental:
      (1) Classroom; and
      (2) Designated Workshop Location.
3. Standard: The trainee shall perform basic shielded metal arc welding (SMAW) IAW refs by:
   a. Performing SMAW safe work practices by:
      (1) Understanding AC / DC electrical theory and safety;
      (2) Understanding SMAW accessories;
      (3) Understanding benefits of SMAW;
      (4) Understanding classifications of SMAW electrodes;
      (5) Understand basic SMAW maintenance;
      (6) Donning SMAW, grinding, and gas cutting PPE by understanding PPE characteristics.
   b. Performing SMAW equipment setup by demonstrating assembly / disassembly of arc welding equipment; and
   c. Performing Basic arc weld by:
      (1) Operating Augmented Welder (Augmented Reality)
      (2) Demonstrating Basic SMAW Techniques

4. Time Required.
   a. EO – 2300 mins
   b. PC – 400 mins (total for all trainees)

References. A1, A2, A3, A4, A8, A9, A11, A12, A13, C1, C4, A59, A61, A62, A63, A64, A65, C3, C4, C5

5. Task, Skill, Knowledge Numbers.

6. Training Limitations. Nil.

7. Remarks. During 7.05.03 Perform Basic Arc Weld, Instructor is to split time; AR: 800 minutes and Real SMAW: 1200 minutes

8. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
007.05.01: Perform SMAW Safe Work Practices			
1. Understand AC / DC electrical theory and safety to include arc current; 2. Understand SMAW accessories to include: a. Ground; b. Electrode; c. Electrode holder work clamp; d. Electrode Oven 3. Understand benefits of SMAW to include: a. Steel metallurgy; b. Basic weld anatomy; c. Welding defects; d. Weld positions; e. Types of weld joints; f. Arc control; g. Arc variables; and h. Troubleshooting welding machine. 4. Understand classifications of SMAW electrodes to include:	A1, A2, A3, A4, A8, A9, A11, A12, A13, C1, C4	**Method:** IL, AIT, D-P **Media:** Instructor, PPT, video, real Equipment **Trg Environment:** Classroom / Designated NFS Workshop Location **AAP Details:** *1. Learning Activity 14: Best Summary* *2. Learning Activity 15: Table Topics* *3. Learning Activity 16: D-P* *4. Learning Activity 17: D-P* **Trg Resources:** ● PPTs ● Visual Trg Aids ● Physical Trg Aids ● References	200

a. Types of electrodes; and b. Types of coatings. 5. Perform basic maintenance on SMAW accessories by: a. Explaining ground clamp and cable, torch and cable, and power supply; and b. Inspecting the ground clamp cable, torch and cable, and power supply. 6. Don SMAW, grinding, and gas cutting PPE by explaining PPE characteristics.			
007.05.02: Perform SMAW Equipment Set-up			
2 Perform SMAW equipment setup by explaining assembly / disassembly of arc welding equipment;	A1, A2, A3, A4, A8, A9, A11, A12, A13, C1, C4	**Method:** D-P **Media:** Instructor, real equipment **Trg Environment:** Designated Workshop Location **AAP Details:** *1. Learning Activity 18: D-P* **Trg Resources:** ● Visual Trg Aids ● Physical Trg Aids ● References	100
007.05.03: Perform Basic Arc Weld			
1. Operate augmented arc welder (augmented reality) to perform	A1, A2, A3, A4, A8, A9,	**Method:** D-P	2000

SMAW to include: a. Flat bead on plate b. Flat butt weld c. Horizontal butt weld d. Vertical butt weld e. Flt T-joint weld f. Vertical t-joint weld 2. Demonstrating SMAW techniques.	A11, A12, A13, C1, C4	**Media:** Instructor, augmented reality, real equipment **Trg Environment:** Designated Workshop Location **AAP Details:** *1. Learning Activity 19: D-P* *2. Learning Activity 20: D-P* *3. Learning Activity 21: D-P* *4. Learning Activity 22: D-P* *5. Learning Activity 23: D-P* *6. Learning Activity 24: D-P* *7. Learning Activity 25: D-P* **Trg Resources:** ● Augmented arc training system (augmented reality) ● Visual Trg Aids ● Physical Trg Aids ● References	800 AR Weld 1200 SMAW

10. Test Details. EO 007.05 will be tested in PC 007 IAW Annex K. Test details found in Annex L.

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication (published textbooks)

D = Other publications (data sheets, URLs, etc.)

Refs. Code	Publication Number	Publication Title
CF or other official Canadian government publication		
A1	C-02-040-009/AG-001	General Safety Program, Volume 2, General Safety Standards
A2	http://halifax.mil.ca/MarlantSafetyEnvironment/pages/sems_sems.html	Safety and Environmental Management System (SEMS)
A3		Captain's Standing Orders (CSO)
A4		Ship's Standing Orders (SSO)
A5	HFX-D28-590-000-01	Guardrails And Stanchions Arrangement And Details

A6	C-27-785-000/MS-00	Comprehensive Maintenance Manual Fresh Water System Applicable To Halifax Class
A7	C-01-000-102/AG-000	Policy/Management Procedures And Guidelines National Defence Index Of Documentation (NDID) System
A8	C-03-007-145-MD-002	Illustrations for Electrical Equipment
A9	000001-03	Carpenter's Tool Box Drawing
A10	C-23-045-000/AG-001	Damage Control (Sea) Volume 1 Of 4 Shipboard Damage Control
A11	A-GG-040-010/AG-006	A Guide to WHMIS in DND and the CF
A12	C-82-007-001/AF-000	Reference Book on Petroleum and Associated Products
A13	A-PD-905-001/PT-001	Trade Study Manual for Hand Tools
A14	C-49-576-000/MS-001	Instruction Manual - Band Saw 14”
A15	C-49-569-000/MS-001	Instruction Manual Bench Grinder 8” Applicable to Halifax Class
A16	C-87-403-000/NY-001	Naval Preventive Maintenance Schedule Hydraulic Impact Drill/Wrench Applicable to FDUs
A17	C-03-010-542/TD-001	Bronze check swing valves
A18	C-03-010-589/MS-001	Technical Manual Bronze Gate Valves (Montreal Bronze)

A19	C-03-010-588/MS-001	Technical Manual - Bronze Globe Valves (Montreal Bronze)
A20	C-03-010-552/MS-002	Operating Instructions, Maintenance Instruction And Overhaul Instructions Pneumatic Operated Diaphragm Valves Sizes 15mm-150mmm Model 137/138
A21	C-03-010-543/MS-003	Operation, Maintenance And Overhaul Instructions 3-Way Ball Valves Model 36006 TF - Size 50 Mm
A22	C-03-011-007/MS-002	Handbook - Insulation Manual For HMC Ships
A23	D-23-003-005/SF-002	Canadian Forces Specifications Specification For Maintenance Painting Of HMC Ships
A24	C-27-867-000/MS-001	Comprehensive Maintenance Manual Service Steam System Applicable to Halifax Class
A25	C-03-005-033/AA-000	Naval Engineering Manual - NEM Direction And Guidance
A26	C-03-005-012/AM-001	NAMMS Policy And Procedures - Naval Materiel Management System Manual
A27	C-82-007-001/AF-000	Reference Book On Petroleum And Associated Products
A28	C-03-010-547/MS-001	The Oil and Coolant Condition Analysis Program (OCCAP)
A29		International Marine Organization_MARPOL 73/78, Annex IV

A30	C-27-952-000/NY-001	Naval Preventive Maintenance Schedule (Hull) Black And Grey Water Collection Equipment Applicable To Halifax Class
A31	C-27-952-000/MS-001	Operating And Maintenance Instructions Vacuum Sewage Collection System Applicable To Halifax Class
A32	C-27-890-000/MS-002	Installation, Operation, and Maintenance Instructions for Sewage Treatment Plant Omnipure Model 15MX3-1
A33		Environmental Management of Shipboard Waste MARCORD G-18
A34		Canada Shipping Act - Pollutant Discharge Reporting Regulations, 1995
A35	C-27-890-000/NK-001	Naval Preventive Maintenance Performance Test Omnipure 15 MX3-1 Sewage Treatment Plant Applicable To Halifax Class
A36	C-23-045-000/AG-002	Damage Control (Sea)Volume 2 Of 4 Shipboard Chemical Biological Radiological Nuclear (CBRN) Defence
A37	C-77-304-000/NK-001	Naval Preventive Maintenance Performance Test (Hull) CBRN Filtration Units Applicable To Halifax Class

A38	C-29-354-000/MS-001	System Manual - Main Refrigeration Applicable To Halifax Class
A39		CPF 85 Ton Draft Manual
A40		CFNES Refrigeration Rules and Regulations handout
A41		MARCORD G-19 Vol 2;
A42		Federal Halocarbon Regulations 2003;
A43		Environmental Code of Practice for Elimination of Fluorocarbon Emissions from Refrigeration and Air Conditioning Systems;
A44		Environmental Directive 4003 - 1/2003 Spill Reporting;
A45	A-GG-040-001/AG-001	General Safety Program Vol 1 - Policy and Program
A46	C-26-010-047/TS-001	Safety Bulletin - Arc Flash Hazard In Naval Environments
A47	D-03-003-005/SF-000	General Electrical Specification for Canadian Forces Ships
A48	C-01-000-000/DA-001	NDID Coding System
A49	C-03-007-181/ME-001	Cable and Cable Termination Data for Shipboard Installation (All Class Ships)
A50	C-03-010-000/MM-001	Canadian Naval Shipboard Techniques for Electromagnetic Compatibility

A51	C-28-422-000/NY-001	Naval Preventive Maintenance Schedule (Hull) Watertight/Airtight Doors/Hatches/Scuttles Applicable To Halifax Class
A52	C-28-010-329/MS-001	Watertight Closures Inspection, Maintenance And Minor Repairs For Ships Staff (Applicable To All Surface Ships)
A53	C-23-HFX-000/MS-001	Halifax Class Ship Information Book - Volume 1 - Hull And Marine Systems
A54	C-28-010-329/MS-001	Watertight Closures Inspection, Maintenance And Minor Repairs For Ships Staff (Applicable To All Surface Ships)
A55	C-26-411-000/MS-001	Engine, Diesel (850 KW Generator Set)
A56	C-26-411-000/NY-001	Naval Preventive Maintenance Schedule (Engineering) Diesel Engine, MWM TBD602V16K - Applicable to Halifax Class
A57	C-77-305-000/NY-001	Portable Engine Driven Fire Pump
A58	C-87-010-010/MS-003	Drivers' HP Air Compressor
A59	D-49-003-003/SF-001	Welding Specifications for HMC Ships and Auxiliaries
A60	C-02-040-009/AG-001	General Safety Program - Volume 2 - General Safety Standards

A61	C-03-040- 000/TS-001	Maritime Command Precautions and Safety Practices for Welding and Burning
A62	CFCD 132	RCN Damage Control Manual
A63	C-09-153-003/TS-000	Explosives Safety Manual - Volume 3 - Naval Vessels
A64	D-04-002-001/SG-001	Canadian Forces Standard Qualification Standards For Welders In Ferrous And Non-Ferrous Metals
A65	C-49-007-001/AF-000	Operator's Manual Welding Theory And Application
A66	C-28-240-L00/NY-001	Shipboard preventive maintenance schedule for cargo reversible eye pads
A67	C-28-240-000/NY-002	Naval preventive maintenance schedule (hull) associated deck and hull fittings (standing rigging)
A68	C-03-007-021/AF-000	Naval ships technical manual chapter 9200 winches and capstans
A69	C-03-010-545/MS-001	Description, Selection, Operation, Maintenance and Repair Teleflex Naval Technologies Remote Mechanical Valve Actuator System Applicable to Halifax Class
A70	C-03-010-545/NY-001	Naval Preventive Maintenance Schedule (Hull) Teleflex Naval Technologies Remote Mechanical Valve Actuator System Applicable to Halifax Class

A71	C-28-590-000/TB-001	Preparation of Polyethelene-Jacketed Aramid Fibre Rope Guard Rails & Lifeline
A72	C-28-460-A00/MS-001	Portlights and Windows Wynn Wipers, Tank Window Wash
A73	C-28-460-A00/NY-001	Portlights and Windows Reversible eye pads
A74	C-29-355-000/MS-001	Handbook (Comprehensive Maintenance Manual) and Parts List Heating, Ventilation and Air Conditioning System Applicable to Halifax Class Books 1 to 3 of 3
A75	C-03-015-003/AM-001	Survey and Repair of Steel Ships
A76	C-38-138-000/MS-001	Cathodic Protection Manual
Official Allied Force or Allied government publication		
Commercial publication (published textbooks)		
C1	**978-0131140806**	Electronic Devices - Conventional Current Version (7th Edition), Thomas L. Floyd
C2		Electrical Engineering and Science for Manufacturing Applications
C3		Modern Welding Book
C4		Show Nozzle and Stemp Type Regulator Fig. 12.22, 12.23

C5	https://manualzz.com/doc/en/55892585/esab-heating-and-welding-guide-instruction-manual https://www.northerntool.com/images/downloads/manuals/45398.pdf https://docplayer.es/5813086-Cutting-heating-and-welding-guide.html https://www.amazon.com/Victor-Cutting-Heating-Welding-Guide/dp/B008KAL0W4	Victor, Cutting-Heating-Welding Guide, 2012, Rev E
Other publications (data sheets, URLs, etc.)		
D1		1950324x: RayChem THERMO POLYWELD 3-Way Clamp Drawing
D2		1950324x: RayChem THERMO POLYWELD 3-Way Clamp Drawing
D3	http://www.new-line.com/information	New-Line Hoses & Fittings
D4		Marine Hose Correct Hose Use, Care & Maintenance Guide PDF
D5		The Gates Safety Guide PDF
D6		Festo Electrical Lab-Volt Manuals
D7		Festo Industrial Maintenance Pumps Training System LabVolt Series Manuals

Annex A

D8		International Marine Organization, Resolution A.774, (18), 26 November 1993
D9		Ozone Layer Protection Regulations, Schedule A, Chapter 1 of the Statutes of Nova Scotia, 1994-95, The Environment Act NS Reg 54/95;
D10		Halocarbon service logs;
D11		Mechanical refrigeration code B52-05 parts 1 and 2;
D12		Ozone-Depleting Substances Regulations, 1998;
D13		Safety Data Sheet E-6213 - Halocarbon R134A
D14	https://literature.rockwellautomation.com/idc/groups/literature/documents/wd/gi-wd005_-en-p.pdf	Allen-Bradley - Typical Wiring Diagrams
D15	Allaboutcircuits.com	
D16	https://www.rapidwelding.com/files/ArcairSLICEInstructionManual.pdf	Arcair - Instruction Manual For SLICE Exothermic Cutting Equipment
D17		Survey Cal-Pre DWP Hull above Water line

Annex A

ANNEX B - TASK LIST

The original QSP board was based on the JTAR.(Include in CDAR)

The QSP Review board held Jul-Aug 2019 performed its own task analysis, amending tasks to reflect the actual RQ-OS requirements on the job. The intention of the board was to have the master task list / JTAR reviewed and amended IAW recommendations in the Needs Assessment.

Currently, the task list is not reflected within the QSP. An updated task list will be added once a MTL/JTAR review is complete.

Task No	Task Statement	Train / No Train	Priority	Substantiation (Required for NT Decisions & High Priority Items)	PO

ANNEX C - QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

The original QSP writing board was convened 9 Jan – 3 Feb 17. The RoP for that board is found in this document:

Original_ANNEX C_RoP.docx

A review board was convened 22 Jul – 2 Aug 19. The RoP for that board is found in this document:

Annex C - ROP_v2.docx

A finalization group review was completed 04-08 April 2022.

1. The goal for this board was to confirm task breakdown and performance-oriented Performance and Enabling Objectives, and to ensure current content was properly accounted for in Learning Objectives and assessment.
2. It became very clear that this QSP was focussed on training delivery.

**Please note that no major changes were made to the QSP or content, the review was to finalize the document as best as possible to proceed to steady state and formalize any future changes iaw policies.

3. It is recommended that the next iteration contain proper PCs. The only PC in this course is PO 007. Again, as this QSP has been created, and designed around delivery and not the job, the EO's are broken into DL and F2F delivery, requiring EC's after each EO.
4. It is recommended that when a QSPWB is convened in the future, that the OJT packages be addressed at the same time, and as one large QSP, with OJT packages as a method of delivery vice their own QSP. This may include the Roundsperson packages.
5. It is also recommended that the structure of training delivery (DL, Theory, Practical, etc) **not** be the priorities when developing Performance Objectives. It is clear in this QSP that delivery methods shaped the objectives and grouping of tasks. This is not iaw CFITES analysis, and design doctrine and should be addressed in any future reviews.
6. Task numbers should be included within each PO and EO where it is located in the document.
7. PO's that try to group all administration, or theory should be avoid. The task breakdown should by followed iaw CFITES Analysis phase to identify where this sub-

tasks/skills/knowledge fit in the completion of a task. If they are grouped in an EO for training delivery or scheduled together for likeness and ease of delivery, that should only occur as a training methodology during Design iaw CFITES. This was clearly not followed in the development of this QSP.

8. The following PO titles and breakdown were amended to better capture the actual tasks and level that a S3 MAR TECH is completing on the job, and to improve verb and task breakdowns.

a. PO 001 – renamed Complete Technical Administration
b. PO 002 – renamed Maintain Pressurized Systems
c. PO 005 – Maintain Engines was renamed to “Maintain Internal Combustion Engines” to better capture the actual tasks and level that a S3 MAR TECH is completing on the job for small diesel and gas engines.
d. PO 006 - Maintain Propulsion Systems was renamed to “Inspect propulsion systems”. This was changed to reflect the actual tasks a S3 MAR TECH is completing on the job, which is not system maintenance. The S3 is completing basic inspection of components of the gearing and shaft lines, and steering systems. Because the tasks for this PO are contained in the OJT and Roundsperson packages, this QSP focussed on identifying the tasks, but training was focussed on theory/knowledge to support tasks in later training.
e. PO 007 – Repair Metal (using heat).

**Note: Any additional differences from the original QSPWB scalar were made unofficially between versions and was outside the scope of this finalization.

9. All POs were reviewed and amended to ensure (as best as possible without completing Analysis iaw CFITES) that standard statements were indicative of tasks completed on the job at the S3 level.

QSP Finalization Members:

Position	**Rank and name**	**Unit**	**Dates**
MAR TECH SME	MS L. Allan	NTDC(P)	04-09 Apr 2022
MAR TECH SME	MS A. Perkins	NTDC(P)	04-09 Apr 2022
MAR TECH SME	MS W. Kelly	NTDC(P)	04-09 Apr 2022
MAR TECH SME	MS L. Lee	NFS(P)	04-09 Apr 2022

MAR TECH SME	MS M. Savola	NFS(P)	08 April 2022
TDO Support	J. Cayer	Resource Support to NTDC(P)	04-09 Apr 2022

Board Members. Original QSPWB

Position	**Rank and name**	**MOSID / Unit**	**Dates**
MAR TECH TM	CPO2 Shane Foxe (MAR TECH)	00379 / NPTG HQ	Not in attendance
PRes TM	CPO2Tim Cotey	NPTG HQ	22 Jul (forenoon)
Chair	CPO2 Don Anderson	00379 / NTDC(P)	
SME– DNavP&T (Legacy MESO)	CPO2 Jeff Gourlay	00379 / DNavP&T PRes Occ Manager	22 Jul – 2 Aug
SME – MAR TECH (Legacy Electrician)	PO1 John Beaulieu	00379 / NTDC(P)	22 Jul – 29 Jul
SME – MAR TECH (Legacy Hull Tech)	PO1 Chris Switzer	00379 / NTDC(P)	22 Jul – 2 Aug
SME – MAR TECH (Legacy Mar Eng)	PO2 Randy Holowachuk	00379-05 / NFS(A)	22 Jul – 2 Aug
SME – MAR TECH (Legacy MESO)	PO1 Nicole Phippard	00379 / HMCS UNICORN	22 Jul – 2 Aug
SME – MAR TECH (Submariner / Legacy Electrical)	PO2 Dale Armstrong	00379-02 / NFS(P)	22-23 Jul (forenoon), 24 Jul, 29-30 Jul
SME – MAR TECH	MS Laura Allan	00379 / NTDC(P)	31 Jul-2 Aug

TDO	Lt(N) Sarah Valentine	00211 / NTDC(P)	22 Jul – 2 Aug
Secretary	SLt Charles Boyes	00211 / NTDC(P)	22 Jul – 2 Aug

Board Objective.

Decision/Event	Discussion / Detail
Introductions	Board introduced themselves in relation to their subject matter expertise.
Group Norms	Board created group norms to ensure standardized and respectful behaviours are maintained throughout the board process.
TDO Brief	Reviewed board mandate: ● Analyze job reqs based on Occ Spec, ● Design training to reflect job reqs and equipment with focus on ResF schedule / deviations, and ● modularize the course.
	Reviewed board roles and responsibilities
	Reviewed QSP structure. There will likely be an addition to the Annexes: M – Modularization.
	Introduced CFITES principles and QC cycle
	Listed Analysis phase steps: 1 Review Documentation 2 Perform Task Analysis 3 Review POs
PRes TM visit	TM inquired whether course critiques and AARs have been reviewed prior to review board. Board confirmed this was completed and captured in the Program Evaluation.
Document Review – Board	Board reviewed documentation (FCRs from PE and NA, Occ Spec, Job Descriptions, NavResOrd 5031-8 “Class “A” Remuneration for DL” and developed their expectations for the resulting QSP (***subsequent additions and emendations are in bold italics***):

Expectations for QSP	1 Restructure the QSP to reflect OS job, including RegF and ResF requirements *a. Discussion of how at ResF, OSs are handcuffed by rank (can't do RIB maintainer course and trailer, etc). Risk of attrition if progression is to slow* *b. Discussion of how to modify OS training in order to give OSs increased responsibility and opportunity for advancement* c. Create standardized list of instructional strategies to use d. Ensure strategies match the target audience e. Ensure strategies add flexibility; investigate ROI of 2nd IS 2 Create clear Assessment Plan (CEC, EC, PC) a. Clear directive for safety 3 Ensure clear sequence for training and modules a. Mandatory vs preferred 4 Distinct milestones 5 Include standardized assignments and handouts *a. Discussion of the value of homework* *b. Discussion as to whether homework would be mitigated with AIT* *c. Discussion of the difference between reinforcement/refresher and homework* 6 Modify tasks as appropriate based on defined performance of an OS (may not reflect JTAR since JTAR needs amending) 7 Create baseline scaffold for performance progression through all MARTECH coursing 8 Meet platform non-specific intent a. Ensure flexibility with assessments / practical elements based on TEL / equipment capabilities ***(acceptable vs ideal)*** 9 Prioritize TEL ***(acceptable vs ideal)*** 10 Identify TE(s) / TS for delivery of each IS 11 Clearly define what TPts are covered by what IS, TE
ResF Progression	Robust conversation about the progression from Basic through OFP for ResF MAR TECHs. Direction from the command is that all recruits will be OFP within 36 months IAW letter. Current timeline has recruits OFP during the third summer, when they will be OS/AB, but hold no RIB qual, be precluded from domestic operations, and not be available to act as roundsperson. Reviewed RQ-OS for RegF and ResF will produce basic maintainer, with the possibility that RIB quals for ResF units. If PRes personnel follow shore maintainer route, they would only be able to serve as shore maintainer if they transfer RegF unless they meet the qual for new position (i.e. would have to go back to do the Roundsperson, USQ-AB, etc). Concern raised over the fact that ResF are only

	training pers up to LS, thereby creating a rift in progression between ResF and RegF. A roundsperson NRD package could be an option. This was added to "Tiller Flats" to discuss throughout the board.
Day 2: 23-Jul	
Maintenance Levels - Definitions	The board decided on the following levels of expected competency as they relate to the term "maintenance": ● Basic Maintenance – OS ● Intermediate Maintenance – AB ● Advanced Maintenance – LS Where many POs and EOs previously read "Maintain" they will now read "Perform Basic Maintenance" with the understanding that they meet the above threshold. The board further distinguished between "Maintain" and "Perform Maintenance," where "Maintain" relates to maintenance being conducted by the OS on the entirety of a system, where qualifiers relate to the complexity of the system. (e.g. "Maintain basic pumps" would mean they do maintenance on the entire pump, but will not do maintenance on "complex" pumps). "Perform basic maintenance" refers to the maintenance itself (the action) being basic as they will not be expected to do all the maintenance required for complex systems (e.g. "Perform basic maintenance on engines" would mean the OS would only do very basic maintenance on the engine with more complex maintenance tasks coming during later training).
Manufacture Materials / Welding	Discussion as to the value of learning welding during RQ-OS and whether or not the ships expect OSs to be able to fabricate, flash up arc welder, etc. on day one. TDO read out job description, as it pertains to welding. Current PO does not link to the entire fabrication/metal work process which is at the USQ-AB level, therefore may be introduced too early if stays on RQ-OS since they won't use it until USQ-AB. If ships *do not* expect fabricating, etc, ~~PO 010~~ "Conduct cutting and welding" could be shifted up to USQ-AB. It was noted that the Occ Spec would need to change to remove the PO as currently RQ-OS/Roundspersons are required to weld to a small degree. Discussion about whether it would also make more sense to just remove the "Arc Welding" portion of the PO but maintaining "Soldering, brazing" and "Exothermic torch". **The board voted to remove ~~PO 010~~ Manufacture Materials (Cutting and welding) in its entirety** *pending approval of DNavP&T***.**
RIB maintainer qualification	Discussion raised as to whether RIB maintainer could be moved to RQ-OS from USQ-AB to help ResF personnel achieve the qual (PRes not required to take USQ-AB). Some RIB maintainer EOs are already embedded in current RQ-OS.

	Possibility of time issue and a discrepancy in progression between RegF and ResF. Legacy MESO had RIB maintainer as a requirement for OFP. **The board voted to include RIB Maintainer in RQ-OS (remove from USQ-AB)** *pending approval of DNavP&T*.
PO 001 – Tech Admin	**The board changed the name of PO 001 from "Perform Tech Admin" to "Perform Basic Tech Admin".** **PO 001 must be the first PO in sequence.** There is sufficient foundational material (especially math) within the PO to justify its being mandatory that it appear first in sequence.
	EO ~~001.03~~ 001.05 encompasses DRMIS Level 1. Discussion on whether DRMIS can be moved until later in the course as it is a perishable skill. Could list DRMIS as preferred/mandatory toward the end of the course. Discussion on whether to break down the EO throughout the course to bring individual parts up at relevant times. Fleet School cannot meet this because of a lack of training aids. It was offered that DRMIS could be a standalone PO. Current EOs would not fall under DRMIS. Clarified that EOs are their own standalones but grouped under an overall performance. **The board voted to keep DRMIS under PO 001.** **The board changed the name of EO ~~001.03~~ 001.05 from Conduct Maintenance Admin to Utilize DRMIS portal as a Maintainer** (in line with DRMIS QSP)**.**
	In order to emphasize the re-production of drawings vs their analysis: **The board changed the name of EO ~~001.02~~ 001.04 from "Analyze Drawings and Schematics" to "Produce Basic Drawings and Schematics".** Producing drawings will include basic analyzing of drawings and schematics as teaching points.
	Discussion of EO 001.02 and whether to front-end load all drawings and schematics in PO 001. It was decided to teach drawings / how to draw in PO 001 and then have specific drawings/schematics appear in subsequent POs/EOs.
	The board changed the name of ~~EO 001.04~~ EO 001.03 from Apply General Mathematic to Apply Basic Mathematics to maintain the same "basic" qualifier for the "basic maintainer"**.**
	Discussion as to whether EO 001.02 (Produce Basic Drawings and Schematics) and EO 001.04 (Apply General Mathematics) should swap places based on the requirement for math to produce drawings. It was decided that the basic math skills are required prior to the production of drawings and schematics. **The board voted to reorder EOs. The order is now as follows:**

	EO 001.01 – ~~Identify Safety Requirements for Engineering Operations~~ Describe Safety Requirements for Engineering Operations **EO ~~001.02~~ 001.03– Apply Basic Mathematics** **EO ~~001.03~~ 001.04– Produce Basic Drawing and Schematics** **EO ~~001.04~~ 001.05- Utilize DRMIS portal as a Maintainer**
PO 002 – ~~Liquids~~ Fluid Systems	**~~The board voted to change the name of PO 002 from Maintain Liquid Systems to Perform Basic Maintenance on Liquid Systems~~.** **PO 002 must be second in sequence.**
	It was noted that ~~EO 002.01 (Utilize Hand Tools and Measuring Equip)~~ is a skill, not a task. Discussion as to whether it would more accurately fit as a TPt under other EOs where the tools will be used. If all tools are introduced at once, they will have to be reviewed when relevant tools are needed. It was decided that basic tools will be introduced in this EO and that speciality tools will still need to be covered off as TPts when relevant (ex, electrical tools in PO 006). Discussion as to whether scope is big enough to make it a stand-alone PO as "maintain tools". Board decided scope too small. **~~The board changed the name of EO to Perform Basic Maintenance on Shop Tools and Measuring Equipment~~** ~~as per the definitions of "Perform Basic Maintenance" above.~~
	The board changed the name of EO from Maintain Valves to Maintain Basic Valves as per the definitions of "Perform Basic Maintenance" above.
	The board changed the name of EO from Maintain pipes and hoses to ~~Maintain Pipes, Hoses, and Insulation~~ because insulation is part of maintaining piping and hoses and as per the definitions of "Perform Basic Maintenance" above.
	The board changed the name of EO Maintain Pumps to Perform Basic Maintenance on Pumps as per the definitions of "Perform Basic Maintenance" above.
	The board changed the name of EO from Apply Liquid Engineering Theory to Shipboard Systems to Perform Basic Maintenance on Common Shipboard Liquid Systems as per the definitions of "Perform Basic Maintenance" above.
	PO 002 is largely theoretical. As such, "maintain" might accurate for EO 002.06 (Maintain Liquid Specialty Equip). Discussion as to what equipment OSs were likely to be expected to handle. As they are OS, we built scaffolding for their progression through follow-on training. **The board removed ~~EO 002.06~~ Maintain Liquid Specialty Systems.** It will become a TPt under EO 002.005 (Perform Basic Maintenance on Common Shipboard Liquid Systems) because it is simply theoretical and small scope.

	The board removed ~~EO 002.07~~ Perform Liquid System Testing. It was recommend that EO 002.07 should be reformatted as a TPt, where applicable throughout other EOs.
~~PO 003~~ – Compressed Air Systems	**The board changed the name of PO 003 to ~~Perform Basic Maintenance on Compressed Air System~~** because the inclusion of "compressed" in the title of the PO more accurately captured the nature of the EOs. It is a defining line between compressed air systems and HVAC/R and as per the definitions of "Perform Basic Maintenance" above.
	With reference to EO 003.03, the question was asked whether FF equipment falls under breathing air compressors or DC systems. The point was then raised that operating compressors could fall under EO 003.01 as a TPt**. It was decided that ~~EO 003.03~~ should be removed and appear as a TPt under air systems** because they are the same principle of operation.
	The board voted to reorder EO 003.01 and EO 003.02 because high pressure supplies low pressure systems**. It now reads:** **~~EO 003.01 – Perform Basic Maintenance on High Pressure Air Systems~~** **~~EO 003.02 – Perform Basic Maintenance on Low Pressure Air Systems~~**
~~PO 004~~ ~~EO 004.02~~ ~~PO 005~~ – ~~HVAC/R~~	**~~The board changed the name of PO 004 from Maintain Heating and Cooling Systems to Perform Basic Maintenance of HVAC/R Systems~~** because HVAC/R as a catchall term for the PO and better aligns the title with industry definitions and phrasings and as per the definitions of "Perform Basic Maintenance" above.
	EO 004.01 (Perform Safety with ODS) is an industry standard (ODS); as such, though there was some discussion to changing the name to better reflect the actual expectation for OS, there was uncertainty as to whether that can actually be done. This EO also comes with a MITE code 109133. The qual name is Ozone Depleting Substance Awareness Course. **Decision by the board to title the EO "Ozone Depleting Substance Awareness" to reflect industry standard and the actual qualification.**
	The board changed the name of EO from Operate Refrigeration Systems to Perform Basic Maintenance on Refrigeration Systems as per the definitions of "Perform Basic Maintenance" above.
	Discussion of the relation between HVAC and CBRN (EO 004.04) and whether CBRN components should be moved to DC systems. Board decided CBRN will remain as part of HVAC/R because it fits within the HVAC/R system.
Day 3: 24-Jul	
PO 005 - Engines	Discussion as to whether "internal combustion engine" is platform-neutral. Discussion as to whether the title includes propulsion motors, Z drives, and other unique machines. Focus should be on systems we currently have and not project into the future. Prime mover isn't captured under ICE. Inclusion of "Propulsion component" to title might better capture all EOs. Removing

	"internal combustion" might meet end as well, with the added benefit of capturing possible future acquisitions that include non-internal combustion engines (i.e. steam). **The board changed the name of ~~PO 006~~ 005 from Maintain Internal Combustion Systems to Perform Basic Maintenance on Engines**
	~~**The board changed the name of EO from Maintain Marine Diesel Engines to Perform Basic Maintenance on Marine Diesel Engines** as per the definitions of "Perform Basic Maintenance" above.~~ Discussion about removing "Marine" however this is in line with industry standard.
	~~**The board changed the name of EO from Maintain Small Gasoline Engines to Perform Basic Maintenance on Small Gasoline Engines.**~~
	The board discussed the viability of the EO being a PO. Board enumerated possible elements under a provisional PO and decided it was viable. ~~**The board removed EO 005.04 Inspect Propulsion Systems. It will become PO 006 Perform Basic Maintenance on PBM on Propulsion Systems**~~ because the content did not align with the engines and the scope was large enough to justify it as a performance on its own.
PO 006 – Propulsion System	Only a stern drive is available for training. Discussion as to whether trainees will be performing basic maintenance or simply describing propulsion systems. **The board created PO ~~Perform Basic Maintenance on~~ Describe Propulsion Systems.**
	RQ-OS can assist with steering trials. The board discussed bringing the level of expectation down to the RQ-OS level. EO must also capture small boat steering. Discussion as to how to order EOs. It was determined steering and gearing/shaft lines are equally important and the order irrelevant. ~~**The board voted to have Steering precede Gearing/Shaft Lines**~~**.** Must also include RIB steering system teaching points to achieve RIB maintainer qual. **The board created EO ~~Perform Basic Maintenance~~ Describe Steering Systems**
	A Shaft line EO would include gearbox, full shaft line, thrust block, bulkhead seals, shaft seal, propellers, CRPP, brakes, Z drive and stern seal. TPts must touch on small boat stern drive. Must be included to be RIB maintainer qualified. **The board created EO ~~Perform Basic Maintenance~~ Describe Gearing and Shaft lines**
RIB Maintainer	There are three options going forward as it relates to the RIB Maintainer Course: 1 Embed the RIB Maintainer course as a standalone entity.

	2. Breakdown the entirety of the course as TPts through RQ-OS. Problem: any changes to RM QSP would mean going through all TPts to ensure compliance. 3. Offering all (or most) TPts, but no MITE code The MITE code is required for further training. *DNavP&T* does not want the MITE code embedded in the course because it is currently in the USQ-AB and that would be changing the current configuration. The RIB Maintainer course has not been updated since 2002. Significant risk of duplication of instruction if MITE code is not given out when pers go on to RM course. For ResF: risk of course being lost, or (even worse) being pushed up to AB. Question as to whether RIB maintainer MITE code makes RQ-OS class specific. By breaking up RM course into TPts, there is more flexibility to maintain class neutrality. Taking the TPts from RM course and compare with existing TPts from RQ-OS and train to the higher one. Discussion as to whether the creation of a PO called Small Boats would capture many of the RM course TPts. ~~**The board voted to embed the TPts from RIB Maintainer course as TPts, with the caveat that trainees will receive the RM MITE code.**~~
Small boat Trailers	Trailers: currently covered as its own course, independent of RM course (it is run by TEME). The content and basic maintenance skills required for maintaining a trailer are already covered in RQ-OS course but without a trailer focus. As such, a standalone EO is not required. TEME are responsible for "maintaining brakes" therefore it is not an OS task.
PO ~~007~~ 003 – ~~Power Generation and Distribution~~ / Electrical syst	OSs operate PG&D before roundsperson. However, it is at a performance level. Perform is the same level as operate – performing maintenance encapsulates operation. **The board voted to change the name of PO from Maintain Electrical Service Systems to Perform Basic Maintenance on Power Generation and Distribution System** as per the definitions of "Perform Basic Maintenance" above.
	With regards to battery maintenance, there are a diversity of skills required that are not adequately captured with "Apply … theory". Breaking the EO into multiple EOs adequately allow for emphasis on theory and practical aspects. AC machines are operated upon prior to DC machines because DC machines are AC machines with extra parts. However, DC theory precedes AC because DC theory is the fundamentals. **The board split EO "Apply DC Theory to Battery Maintenance" into:** **1 ~~EO 007.01~~ 003.01 Analyze DC Circuits** 4. **EO ~~007.02~~ 003.02-Maintain Battery Systems** Discussion as to day-one reqs as they relate to batteries and how much more will be learned in OJT. Question about the relative complexity of ship batteries. With the exception of class specific batteries, OSs should be fully ready to Maintain Batteries.

	5. ~~EO 007.03~~ 003.03 Analyze AC Circuits
	The board changed the name of ~~EO 007.06~~ EO 003.06 from Maintain Control Devices to Maintain Basic Control Devices as per the definitions of "Perform Basic Maintenance" above.
	~~The board changed the name of EO 007.07 EO 003.07 from Apply PG&D Theory to Shipboard Systems to Describe PG&D Shipboard Electrical Systems~~ ~~as per the definitions of "Perform Basic Maintenance" above.~~
~~PO 008~~	System itself will be basic (i.e. not the skill required). As OS progresses, systems will become more advanced (fibre optics). Debate about whether the services or connectors become more advanced with progression. Discussion of the relationship between PO 008 (Electrical Services) and PO 007 (PG&D) and whether these can be consolidated into one. All EOs in both POs could be captured under Maintaining PG&D. **The board removed ~~PO 008~~ Maintain Electrical Service Systems. All EOs will be captured under ~~PO 007~~ PO 003 Perform Basic Maintenance on ~~Power Generation and Distribution System~~ Electrical Systems** because PO 007 only covered the "generation" portion of PG&D with the "Distribution" portion being in PO 008. Generation and distribution are all one system so fall under the same overall performance.
	Former EOs from ~~PO 008~~ will be re-serialized as: **EO ~~007. 07~~ 003.07 Maintain Ships Service Lighting** **~~EO 007.08 003.08 Maintain Ships Cabling~~** **~~EO 007.09 003.09 Maintain Basic Electrical Connectors~~** **Describe PG&D Shipboard Electrical Systems will appear as ~~EO 007.10~~ ~~EO 003.10~~ EO 003.09**
~~PO 009~~ PO 006 – Shipboard Structures	Encompasses hatches, hull, cranes, etc. Occ Spec mostly focuses on hatches. PO will most likely require more than the single EO it currently houses. Alternatively, two POs – maintain hull and maintain closures. Discussion as to whether RAS falls under structural fitted equipment and who maintains the equipment Cranes = CSC. Fueling Stns = Upper Deck Engineers. Discussion as to whether EO could serve as an introduction to upper deck fittings or if this is covered off in NETP. Focus must be on maintenance. Proposal for EOs: Hull (including cathodic, reference to shaft grounding) as a theoretical base, followed by Closures (ex hatches), and fitted equipment. Performance of completing ship surveys is at the LS level (class specific) so RQ-OS is just a generic introduction to surveys (what they are looking at and what they are looking for). **The board changed the name of ~~PO 009~~ ~~PO 006~~ PO 004 Maintain Shipboard Structures to Perform Basic Maintenance on Shipboard Structure** as expectations are at a low level (understand). However, there is

	some basic engagement with structure: ex, cathodics, erosion, visual inspection of crane, deck survey.
	The board created ~~EO 009.01~~ ~~EO 006.01~~ EO 004.01 Perform Basic Maintenance on Hull Structure because of the maintenance on cathodics.
	Occ Spec identifies need for knowledge and skill in RQ-OS. Closures encompasses water-tight and non-water-tight closures. OS are expected to have full knowledge in RQ-OS. Discussion as to how this applies to more advanced closures, like the hanger door. **The board created ~~EO 009.02~~ EO ~~006.02~~ 004.02 Maintain Basic Closures.**
	Includes RAS, heavy equip, capstan. OS would be expected to grease, so there is some engagement. Better suited being captured under EO 009.01. Hull structures includes deck fittings. A lot of this is covered in NETP. "Upper deck" is more specific than "fitted equipment". In that case, it is easily encompassed by EO 009.01. **The board removed ~~EO 009.03~~ EO 006.03 Maintain Structural Fitted Equipment. It will appear as TPts under ~~EO 009.01~~ EO 004.01 Perform Basic Maintenance on Hull Structure.**
~~PO 010~~	Basic DC is covered in NETP. Maintaining DC system is covered extensively in other POs, ex fire main is a liquid system. Must ensure systems are general and not class specific. Maintenance of fitted protection systems are fully covered off in multiple previous EOs. RQ-OS will cover how to maintain system (ex AFFF), which NETP would only identify. Identification review would only take a moment and not necessary to maintain. Where applicable, we will now mention DC systems as teaching points within the EOs of systems they are a part of (e.g. fusible links covered in compressed air systems). EO 010.02 will be covered off in Roundsperson Training. Given the above, **the board voted to remove the entirety of ~~PO 010~~ Maintain Damage Control Systems.**
Perform Basic Maintenance / Cumulative PO / Pre-Start Checks	Possibility of a final PC that encompasses multiple system maintenance skills (prestart checks, etc) for RIB consolidation If someone has already been tested, this PC may be redundant. A cumulative assignment would draw on many skills, increasing retention. Lack of retention is not a failure of the training system, but a trainee and/or instructor issue. If they have already been tested, students will be expected to maintain that base of knowledge. Another option: break down the steps of maintaining a system into different POs. Such that, when students reach final PO, they will have completed a cumulative project, simply spread out over several POs. PC can be all links of previous PCs. A new PO could consolidate these in a PC. Goal of the PO is not RIB/platform specific. It could work on any system. <u>COA 1</u>

	Have a std system/media (ex, RIB) for working on / assessment throughout all relevant POs. *Propulsion PC would finish off the basic maintenance of system/media.* *Add EO in Basic Maintenance on Propulsion Systems for Pre-start* <u>COA 2</u> *Create a "final PO" to include TPts that didn't fit anywhere else and will link all systems together.* *PC will use a system/media to assess overall understanding for all systems* ~~**The board decided on COA 1. An EO Perform Pre-Start Checks will be added to PO 008.**~~ This decision was nullified by the creation and inclusion of ~~**PO 009 Perform Marine Engineering System Rounds**~~. The EO Perform Pre Start Checklist will now appear as a TPt under EO 009.01 Perform the Duties of a MAR TECH Roundsperson.
Day 4: 25-Jul	
Tiller Flats Review	The following points were in Tiller Flats and reviewed by the board to see if they were covered off throughout the Analysis phase.
	<u>Define "Basic Maintainer"</u> Before moving on to design, BM needs to be defined. The other option is to use the Occ Spec definition. Occ Spec combines Roundsperson and Basic Maintainer. Occ Spec definition does not reflect what is necessary for someone to complete RQ-OS. If definition is changed in QSP, a recommendation would be made to change the Occ Spec. Working definition: ~~*BM is an OS able to perform minor corrective and preventative maintenance with little or no supervision.*~~ Shown a task once or twice can then work independently. Basic maintainer must be independent of OJT. Definition must reflect level of expectations. As such, "perform basic maintenance" would be more accurate. Definition must also specify what OS will work on. *This would include shipboard equipment and systems managed by MSE div. Equipment must be qualified.* Discussion as to what exactly "basic" entails. Term has been used throughout QSP, but the term has not been fully defined. Including examples would mitigate this issue. From the occupation manager: expectation from an OS is that of an apprentice's helper and in preparation for roundsperson qual. Apprentice's helper and supervision are at odds. This is an individual, ship issue as to how they want to direct trainees. Definition of apprentice's Helper - assists the journeyman. However, whereas journeyman is logging hours toward apprenticeship, helper is just a labourer. This does not capture OS as they are skilled and in a progression. As such, "apprentice's helper level" is more accurate and better reflects the level of training. Helper is inaccurate because much of the OS's job is done independently. OS may be a better term. "Assigned" would better capture engagement with systems than "managed".

	Working definition: ~~*BM is an OS who performs preventative and corrective maintenance at a basic level on equipment and skills assigned to MSE div.*~~ The expectation is that an OS will have *introductory engineering knowledge and skills required for working toward roundsperson qual on any class of ship. OS is the basis for the occupation.* Working Definition: ~~*A BM is an OS who can perform preventative and corrective maintenance at a basic level on equipment and systems assigned to MSE div. They will have the skills and knowledge to work towards engineering roundsperson qual on any class of ship.*~~ The ashore path is being ignored in this definition. Including "at sea or shore" would mitigate this. However, the ashore path remains to be developed. Basic Maintainer working at a "basic" level is ambiguous. Entry-level would pose the same problem as it would need to be defined. Entry would capture those who do not enter system as OS. **The board decided on the following definition of a Basic Maintainer:** **A BM is an OS who can perform preventive and corrective maintenance at an entry level on equipment and systems assigned to MSE div at sea or shore. They will have the skills and knowledge to work towards engineering roundsperson qual on any class of vessel.**
	"Basic Maintenance" is entry level maintenance that will be further defined based on the teaching points found in each specific systems' EOs as the term "basic" is different for each system.
	"Maintain Small Boat Components" Must be sure to cover off everything in the RIB maintainer qual in TPts. This will stay in Tiller Flats to be reviewed at the end of the board to ensure all the TPts are covered.
Expectations	The board reviewed expectations established on day 1. No expectations have yet been fully met, but several are underway or nearing completion. Changes as follows <u>(in italics)</u>: 6. Meet platform non-specific intent a. Ensure flexibility with assessments / practical elements based on TEL / equipment capabilities *(acceptable vs ideal)* 7. Prioritize TEL *(acceptable vs ideal)* At this juncture, it is unlikely board will produce a full QSP by the time the board wraps up; however, it will have a clear modularization and DL plan. The QSP will be completed the following week.
Instructional Strategies	An instructional strategy incorporates the method, media, and environment. 1 Interactive Lecture (IL) or Face to Face (F2F) as the most basic mode of classroom instruction. "Interactive" can be misleading as it simply denotes a physical interaction between instructor and student. **The board voted to**

	use F2F to define classroom-based instruction with an instructor present. 2 Explain-Demonstrate-Imitate (EDI). The board debated whether to include Practice. It was decided that imitate adequately captures this. 3 Distributed Learning (DL). If synchronous instruction is required, this must be marked. CBT or not. 4 Self Study. This is completed outside course time and includes only learner-content interaction.
Instructional Methods	The board discussed possible methods of instruction, which include, but are not limited to: 1 Homework (link to F2F/Self-Study; no time associated) 2 Assignment (F2F or DL; time associated) 3 Case Study (F2F or DL) 4 Lecture/Coach (F2F or synchronous DL) 5 Group Work (F2F, EDI, DL) 6 Tour (F2F) 7 SME Lecture (F2F or synchronous DL)
CRCN Intent	Board reviewed letter with CRCN intent dictating PRes MAR TECHs must be OFP prior to start of third summer. Discussion about how at least part of third summer will be needed for residential based on current course timings and requirements since training cannot end on DL (too much practical). Discussion as to whether NETP is needed to be OFP and where that might fit in to the ResF plan. Time of year pers joins would also impact scheduling (i.e. will they have a full summer of training) Board decided to continue with the design phase of the board to see how much is required for DL versus residential to see if it is possible to hit CRCN intent (improbable).
Roundsperson / PO 009	It has been proposed that general duties of being a roundsperson be covered in RQ-OS as one PO that would cover all platform-specific information for each system. There would be an OJT component (with variations depending on platform) added to the RQ-OS but Roundsperson training/OJPRs would be removed. Essentially, would serve as a culmination for RQ-OS and link to Roundsperson. New ship platforms would have to be added to RQ-OS QSP once in place. If adopted, the Roundsperson PO would appear as: **~~PO Perform Marine Engineering System Rounds~~** **~~EO 1 Perform the Duties of a MAR TECH Roundsperson~~** **~~EO 2 Perform Auxiliary Systems Rounds~~**

	~~**EO 3 Perform DC Systems Rounds**~~ ~~**EO 4 Perform Electrical Systems Rounds**~~ ~~**EO 5 Perform Propulsion Systems Rounds**~~ EO 3 might be better suited prior to EO 2. Board decided to wait until TPts are defined to choose order of EOs. NTDC(P) MSE Division already approved this format if it suits the board. Could replace the Pre-Start Checks in PO 007 Engines. Perform Pre-Start Checklist would become a TPt in the PO under EO 1. Discussion of whether you need an NQual or simply MITE to achieve OFP. This would impact where the NQual would be given since OFP is supposed to come before any OJT. ~~**The board approves the creation of PO 009 Perform Marine Engineering System Rounds**~~ to consolidate RQ-OS training and prepare for roundsperson OJT (platform specific) package.
EO 001.01 ~~Identify~~ Describe Safety Requirements	Discussion that "Identify" did not adequately capture the level of training required. Described was proposed. **The board changed the name of EO 001.01 to Describe Safety Requirements for Engineering Operations** to better capture the expectation required.
Day 5: 26-Jul	
Distributed Learning	Krista Frate (NTDC(P) LSC RCN DLN Helpdesk) addressed the board about DL and the DLN capabilities. New generation of OS will be more adept / ready to engage with DL based on prior experience (have seen it during high school and possibly elementary). DL addresses multiple types of learning and allows for more flexibility in pace. Social element to DLN "Community" feature: Opportunity to interact online for new trainees before meeting in person. Will be more prevalent in DLN 3.0 to allow more social learning. Material will always be available for them to review before, during, and after the course. Questions from the board about problems with students helping each other out on ECs and PCs. There are time sensitive, remotely or in-person invigilated exams available through the DLN and other DL programs. There are flexibilities to creating and invigilating exams built into the DLN, including permissions, question banks, etc. to mitigate the sharing of exams between trainees.
~~EO 002.03~~ EO 002.08 Pipes, Hoses, and Insulation	When reviewing teaching points for the EO, the board determined there was more advanced concepts with relation to pipes, hoses, and insulation (like hydraulics) that the MAR TECH would learn after RQ-OS. **The board changed the name of ~~EO 002.03~~ EO 002.08 from Maintain Pipes, Hoses,**

	and Insulation to Perform Basic Maintenance on Pipes, Hoses, and Insulation.
Nomenclature of Fittings	EOs will read "male" and "female" in reference to the opposing ends on a hose connection to maintain industry terminology and avoid confusion. It is acknowledged that these may have to change in the future to be more correct.
PO 003 Air Systems	**~~The board changed the name of PO 003 from Perform Basic Maintenance on Compressed Air Systems to Perform Basic Maintenance on Compressed Gas Systems~~**
	~~The board changed the name of EO 003.01 from Perform Basic Maintenance on High Pressure Air Systems to Describe Compressed Gas Systems~~
	The board changed the name of ~~EO 003.02 from Perform Basic Maintenance on Low Pressure Air Systems to Perform Basic Maintenance on Compressed Air Systems~~
Amalgamation of systems into PO 002 Fluid Systems	While working on the EOs, the board discussed the actual differences between each system. It was mentioned that most systems (air, liquid, HVAC) have the same basic components (valves, pumps/fans, pipes/hoses) therefore basic maintenance of all these systems would be pretty much the same. Though there are specific elements from each type of system that would be different, these were already included as TPts under the main EOs. Discussion about whether refrigeration was also the same type of system; it was determined that refrigeration was specialized enough to stand on its own. **The board decided on amalgamating "Liquid Systems" "Compressed Air Systems" and "HVAC systems" under PO 002 "Perform basic maintenance on Fluid systems".**
Reordering POs	Discussion on the ordering and construct of POs as they relate to industry standards after the amalgamation. PG&D was seen to be a pre-requisite of HVAC under the original construct; with HVAC becoming part of Fluid Systems, it was discussed how the pre-requisite would work. The board decided the elements of electrical in HVAC could wait until the PG&D phase. Refrigeration was a specialty of Fluid systems therefore it made sense to go in sequence. The board voted to reorganize the POs as follows: **~~PO 001 – Perform Basic Tech Admin (mandatory)~~** **~~PO 002 – Perform Basic Maintenance on Fluid Systems (mandatory)~~** **~~PO 003 – Perform Basic Maintenance on Refrigeration Systems~~** **~~PO 004 – Perform Basic Maintenance on PG&D Systems~~** **~~PO 005 – Perform Basic Maintenance on Shipboard Structures~~** **~~PO 006 – Perform Basic Maintenance on Engines~~** **~~PO 007 – Perform Basic Maintenance on Propulsion Systems~~**

	~~**PO 008 - Perform Marine Engineering System Rounds (mandatory)**~~ The matter of consolidating PO 006 and PO 007 was discussed but deferred until Monday.
Day 6: 29-Jul	
"Basic Maintainer" definition	Board reviewed "Basic Maintainer" definition from Friday as part of RoP review (specifically for PO2 Armstrong who was not present for the discussion). A quantifier based on DRMIS may help narrow and define the scope of the basic maintainer. Can't base definition on, for example, a weekly or monthly, because senior staff would be handcuffed by that progress. All maintenance is not based on a maintenance schedule. "IAW DRMIS" would not work because not all ships are on DRMIS. They all have access to hard copy maintenance schedule. "IAW maintenance schedule" might be better then. The Occ Spec captures the level of maintenance expected from an OS. **The board voted to keep the previously agreed upon definition (see above)** because not all maintenance is based on a maintenance schedule.
Roundsperson	Need to ensure all platforms have an OJT package (thinking specifically of submarines).
PO 002 - Perform Basic Maintenance on Fluid Systems	The board revisited the title of PO 002 - Perform Basic Maintenance on Fluid Systems. Discussion as to whether "fluid" is too broad and whether it captures all subjects to be covered in the EO. It was proposed that "Liquid and Air" or "Liquid and gas" or some combination therein would clarify the objective. However, "fluid dynamics" is used throughout and the meaning would be lost/obscured if the name is changed. The definition of fluid reads: "a substance (such as a liquid or gas) tending to flow or conform to the outline of its container". Adding these qualifiers would help clarify "fluid" without changing the meaning. **The board voted to change the name of PO 002 from Perform Basic Maintenance on Fluid Systems to Perform Basic Maintenance on Fluid ~~(Liquid / Gas)~~ Systems** to clarify for instructors / trainees what is to be covered.
Merging PO 006 and PO 007	Discussion on merging Engines and propulsion systems POs to cover prime movers, what moves prime movers, then steering, in that order. However, not all engines lead to propulsion, and these would need to be kept separate. Not all prime movers lead to propulsion either (e.g. alternator). Similar to liquid systems, diesel and gas are standalone and should be taught separately.

	The argument for merging the POs is primarily about PO/EO layout for future courses, not based on performance at the OS level. "Engines and Propulsion systems" would capture everything; however, they are two different performances. Discussion as to where electrical prime move on a submarine would fit. Keeping engines distinct makes sense because of the relationship between prime mover and propulsion (propulsion does not begin with prime mover; prime mover is just something that spins). Motors are covered in PG&D. **The board voted to keep PO 006 ~~Perform Basic Maintenance on Engines~~ Describe Propulsion and PO 007 ~~Perform Basic Maintenance on Propulsion Systems separate~~ Perform Maintenance on small Boats** because they are distinct systems.
CRCN intent – Discussion with NTDC(P) MSE Div Cdr	The NTDC(P) MSE Div Cdr (LCdr Deutekom) addressed the board. It was confirmed for the board that DNavP&T intends trainees to be OFP by the start of the third summer, however, understands that if this is not possible, the board is invited to make a different recommendation. As of yet, there has been no pushback on the (informal) idea of completion by the end of third summer. It was reaffirmed that ending on DL (OFP before the 3rd summer) is not ideal. The question of ensuring reservist engagement was discussed. Intent is to avoid losing interest of reservists by incorporating hand's on and shipboard training. It was proposed that an Orca roundsperson qual in summer 2 would be an incentive. This may not be feasible depending on scheduling, including completion of NETP.
DL	The board was visited by Ms. Colleen Armstrong, working as the DLN 3.0 SME at NPTG HQ (EDS 01 employed at NTDC(P) LSC). She discussed: Technology cannot replace a good teacher. DL is not aiming to do that. What it does is add a different element of instruction and gives trainees ownership of their learning. There are different types of learners, who work at varying paces, and design needs to take this into account. This is done via universal design. Meet the mark of as many different learners as possible and offer multiple points of entry/assessment with the material. Instructors and designers will be the key to success. Finding new ways to engage and keep engagement of trainees. Older learners with a sense of continuous education are often the best DL consumers. Anyone can be the audience for DL. They need to see the added value though. Online regional delivery are especially useful for math and science-based courses. This could mitigate isolation and educational challenges. DLN 3.0 will offer more engagement than previous versions.

	DLN modules can also be used in-class. The board asked how their recommendations will be implemented. Main scenarios would be identified, and developer would receive notes on how to group EOs so they can be covered by single activities. At this point, broad activity suggestions are best route, so development is not handcuffed later in the process. The board has been using generic, catchall terms (image, video); however, other options exist (branch scenario, challenge, etc). Presenting options will be crucial for reserve units. This would mitigate issues around lack of available resources or inconsistency of resources from unit to unit and between RegF and ResF. Quantifying this becomes a challenge. Deviations could be weighed on a case-by-case basis.
PO 003 - ~~Perform Basic Maintenance on Refrigeration Systems~~	Discussion of whether the ODS could be moved to PO 001 Perform Basic Tech Admin. Training hours are approximately 100 hours. It is a national qual and comes with a MITE code. Refrigeration could then be moved to PO 002. ODS and refrigeration would have to happen back-to-back because of instructor availability. DL portion could mitigate staffing issue. Splitting them only matters for in-house students. ODS and refrigeration do not link thematically, and refrigeration is largely theoretical. Therefore, the two can be separated. The board decided that Refrigeration Systems cannot stand on its own as PO. **The board voted to move EO 003.01 ~~ODS Awareness~~ Describe ODS Regulations to PO 001 Perform Basic Tech Admin. It will become EO 001.02.** **The board voted to move EO Describe Domestic Refrigeration Systems to the final EO of PO 002.** As a result of the above **PO 003 – Perform Basic Maintenance on Refrigeration Systems was removed.**
PO 002 - Perform Basic Maintenance on Fluid Systems	**~~The board created EO Describe Fluid Dynamic Fundamentals~~** to introduce trainees to the basics early in the PO. The board discussed EO organization after the earlier redesign of the PO. Discussion of whether to put tools before fluid dynamic fundamentals. It was decided that tools should precede fluid dynamics. It was decided that pipes/hoses will come before valves, followed by pumps. These three should be taught together, but the order does not matter. The board discussed the lack of focus on DC systems in the PO. A dedicated EO would address this deficiency. There is enough to be a standalone. Fitted DC systems was too narrow / confusing. NETP might cover these TPts.

	Operation should be left to NETP, RQ-OS would focus on “describe” level expectations. The board decided to create Describe Liquid-Based DC Systems as a TPt rather than EO. Board discussed whether or not to create an EO dealing with liquid and air (i.e. fluid) or to treat them separately. The board decided on a general approach: introduce the theory (tools and fluid dynamics), followed by the main components (valves, pipes, pumps) then specialty equipment, and finally the gas systems. The board voted to break steam systems into components and turn them into TPts within other EOs. Discussion of whether generic and specialty systems could be combined. Depends on how they will be assessed. If all one diagram (for example), they should be combined. If specialty systems will be taught after, they must be separated. Related discussion of whether to use “equipment” or “systems” or both. “Component” might be good. Component describes parts of system. Compressed Air systems would include gas systems. It could be called that. It should be changed because it includes, for example, TAU. **As a result of the above conversation, the board voted to reorder / rename the EOs as follows:** **~~EO 002.01 Perform Basic Maintenance on Shop Tools and Measuring Equipment~~** **~~EO 002.02 Describe Fluid Dynamic Fundamentals Mechanical Engineering Concepts~~** **~~EO 002.03 Maintain Basic Valves~~** **~~EO 002.04 Perform Basic Maintenance on Pipes, Hoses, and Insulation~~** **~~EO 002.05 Perform Basic Maintenance on Pumps~~** **~~EO 002.06 Describe Common Shipboard Liquid Systems~~** **~~EO 002.07 Perform Basic Maintenance on Specialty Liquid Components~~** **~~EO 002.08 Perform Basic Maintenance on Compressed Gas Air Systems~~** **~~EO 002.09 Perform Basic Maintenance on Shipboard Ventilation Systems~~** **~~EO 002.10 Describe Domestic Refrigeration Systems~~**
PO Ordering	**The POs are now ordered as follows** with the understanding that some of the “mandatory” may shift based on the DL/in-house split (e.g. In-house portions of PO 002 may come after DL portions of PO 003)**:** **PO 001 – Perform Basic Tech Admin (mandatory)**

	PO 002 – Perform Basic Maintenance on Fluid ~~(Liquid / Gas)~~ Systems (mandatory) **PO 003 – Perform Basic Maintenance on ~~PG&D~~ Electrical Systems** **PO 004 – Perform Basic Maintenance on Shipboard Structures (prereq PC 003)** **PO 005 – Perform Basic Maintenance on Engines** **PO 006 – ~~Perform Basic Maintenance on Propulsion Systems~~ Describe Propulsion ~~(mandatory)~~** **~~PO 007 - Perform Marine Engineering System Rounds (mandatory)~~**
Classroom visit from NTDC(P) CO	Classroom visit from NTDC(P) CO (Cdr Sauvé). He discussed his vision of what a future classroom will look like: involve instructors acting as coaches and not instructors in the traditional sense. Instructors will serve as experts to guide students who will otherwise engage with different modes of DL and will not have "Stand and deliver" type format. This will allow for self-learning and ownership of their training to increase retention while still allowing for input as needed by the instructor.
EO 001.02 – ODS	The board discussed the best phrasing for the ODS EO. As per the qual, OSs will not manage the system. "Describe ODS" was proposed but the TPts would be more than just describing the substances. Legislations and regulations were both proposed. **The board voted to name EO 001.02 Describe ODS Regulations to** as this is what would be required of the OS on the job.
Day 7: 30-Jul	
EO 002.02 – ~~Fluid Dynamics~~ Mechanical Engineering	Discussion as to whether "Describe fluid dynamics" was the best title for the EO. "Basic Engineering concepts" would better capture the broader themes: fluids, thermal, refrigeration, etc. Engineering is very broad. Physics might narrow the scope. Physics doesn't include torque, rotation, etc. Qualifier of "mechanical" could work as a distinction from electrical. It also serves as better scaffolding for future training. **The board voted to change the name of ~~EO 002.02~~ EO 002.01from Describe Fluid Dynamics to Describe Basic Mechanical Engineering Concepts to** better capture entirety of EO.
PO 003 – ~~Power Generation and Distribution~~	It was noted that PG&D is captured in a single EO and the other EOs would be better captured under the catchall "electrical". Including the term "electrical" would also assist with scaffolding. **The board voted to change the name of PO 003 Perform Basic Maintenance on PG&D Systems to Perform Basic Maintenance on Electrical Systems** to better capture entirety of EO and assist with scaffolding.

Electrical Systems	
EO 003.08 – Cabling EO 003.09 – Connectors	**The board voted to combine EO 003.08 Maintain Ship Cabling and EO 003.09 Maintain Electrical Connectors to form EO 003.08 Maintain Cabling and Connectors** to avoid redundant training.
~~EO 002.08~~ – EO 002.11 ~~Compressed Gas~~ Air Systems	It was noted that "gas" Air Systems no longer needed to be covered in this EO, as extinguishers are covered off in NETP and gas canisters are covered off in EO 001.01 – Describe Safety and EO 002.02 – Describe Mechanical Engineering Concepts. **The board voted to rename ~~EO 002.08~~ EO 002.11– Perform Basic Maintenance on ~~Compressed~~ Air Systems** to more accurately reflect material covered in the EO.
Inclusion of Breathing air compressors	Discussion was raised as to the validity of including breathing air compressors as part of this QSP or whether it should be moved to Roundsperson packages. Operating these compressors is covered off in the roundsperson qual during OJT and that it would be redundant to cover it in this EO. Further, no other DC operation is covered at the RQ-OS level. Removing these TPts would shorten the course and remove redundancies with other compressors. Also, would avoid the need for deviations if units did not have access to breathing compressors. Breathing compressors could be covered off as an example under the larger heading of Air Compressors to make trainees aware of the presence of breathing air compressors on board ships. **The board voted to remove the TPts covering the operation and description of breathing air compressors** as these topics will be covered during the roundsperson phase. The TPts will be mentioned while covering generic Air Compressor elsewhere in the EO. PO2 Randy Holowachuk voted against the above motion on the following grounds: 1 Filling the bottles at this level is a good training opportunity and will only serve to make trainees more valuable when they arrive on ship 2 Filling bottles is valid because it gives trainees in the classroom a sense of accomplishment 3 The training is part of the MARTECH trade and it is therefore valid to, even briefly, introduce it at this level.
Day 8: 31-Jul	
~~PO 007 008 – Engineering Sys Rounds~~	The matter was raised that PO ~~007~~ 008 might be overwhelming for trainees and lead to training that is not required because this is the only PO that is not platform-neutral. Also the risk that training in the PO will be too vague.

	Trainee's choice of platform will already be submitted by the time they are introduced to platforms. This might require flexibility on the part of the career manager.
~~EO 002.02 Mech Engineering~~ ~~EO 001.01 Tools~~ EO 002.06	**The board voted to move EO ~~002.01~~ ~~002.02~~ EO 002.06 Perform Basic Maintenance on Tools and Measuring Equipment to after EO ~~002.02~~ 002.01 Describe Mechanical Engineering Concepts** because topics covered in mechanical engineering are needed for measuring.
EO 002.06 Shipboard Liquid ~~EO 002.07~~ 002.10 Specialty Components	The initial idea behind Specialty Components was unique systems. In point of fact, many of these are subsystems of the common systems (EO 002.06). The remaining elements are components of systems and should be accurately labelled as such. **The board votes to rename ~~EO 002.07~~ EO 002.10 from Perform Basic Maintenance on Liquid Specialty Components to PBM on Miscellaneous Fluid ~~System~~ Components.** **The board swapped PBM on Shipboard Liquid Systems and PBM on Miscellaneous Fluid Systems Components** so that the later will work in conjunction with the previous EOs and culminate in an understanding of shipboard liquid systems.
Small Boats	**The board voted to create PO 007 Perform Basic Maintenance on Small Boats** in order to consolidate all EOs concerning small boats. As a result of the above, ~~**Perform Marine Engineering Systems Rounds becomes PO 008**~~.
Day 9: 1-Aug	
~~PO 008 - Roundsperson~~	Discussion as to the nature of the intended Roundsperson PO. The idea is that it would serve as a summation of other POs, a consolidation. As such, it is possible to divide the EOs and place them in the corresponding POs. PO as it is currently envisioned is all theory-based. OJT will introduce practical side. As such, including roundsperson is not a great title. <u>COA 1</u> ● Class specific roundsperson in-house (details of equipment for class of ship [3-4 days]) ● Class specific Rndp OJT package (walkthrough of systems, etc)

	● At course conclusion, trainees separate to class-specific course, based on where they are posted
	  ● Cons: Trainees receive posting msgs prior to OFP
	  ● Packages are not developed (except for CPF)
	  ● In-house not developed)except for CPF)
	  ● NFS lack assets for in-house concurrent training
	
	<u>COA 2</u>
	
	● Compare all different classes of vessels during rndp in-house (7-10 days)
	● Modularization of in-house (DL) to potentially take it after phase 3
	
	The board voted to de-link and remove PO 008 Perform Marine Engineering Systems Rounds because it does not fit with the structure of other POs.

~~002.09 – Ventilation~~	**The board voted to remove ~~002.09 PBM on Ventilation Systems~~. The EOs will now appear in EO 002.08.**
PO 006 – Propulsion	**The board voted to change the name of PO 006 – PBM on Engines to Describe Propulsion because the practical EOs were removed and "Describe" better captures expectations.**
Day 10: 2- Aug	
TEL	With reference to TEL, it was noted that NRDs have Kingston-class simulators that are good training aid for PG&D and ship's systems.
EO 002.09 POL	**The board voted to remove all POL-related TPts from EO ~~002.09~~ EO 002.02 Describe Domestic Fluid System components and created a new EO ~~002.02~~ EO 002.05 Describe POL Systems** because there is enough material for a distinct PO and the material should be front end loaded.
Cathodics	**The board voted to move TPts on Cathodics and Degaussing to EO 003.09 Describe Shipboard PG&D**
Shaft Grounding	**The board voted to move the TPt concerning shaft grounding to EO 003.09 Describe Shipboard DP&G.**
Roundsperson	Suggestion is to take existing PO titles and have a corresponding EO under a roundsperson PO. This would serve as an overview for the whole course. Roundsperson TPts would stay. The overview is on top of those. Another suggestion is to treat this as a review, completed at NFS just prior to posting. This has the benefits of reviewing all material so it's fresh (might have been a while since trainees saw some of the material).

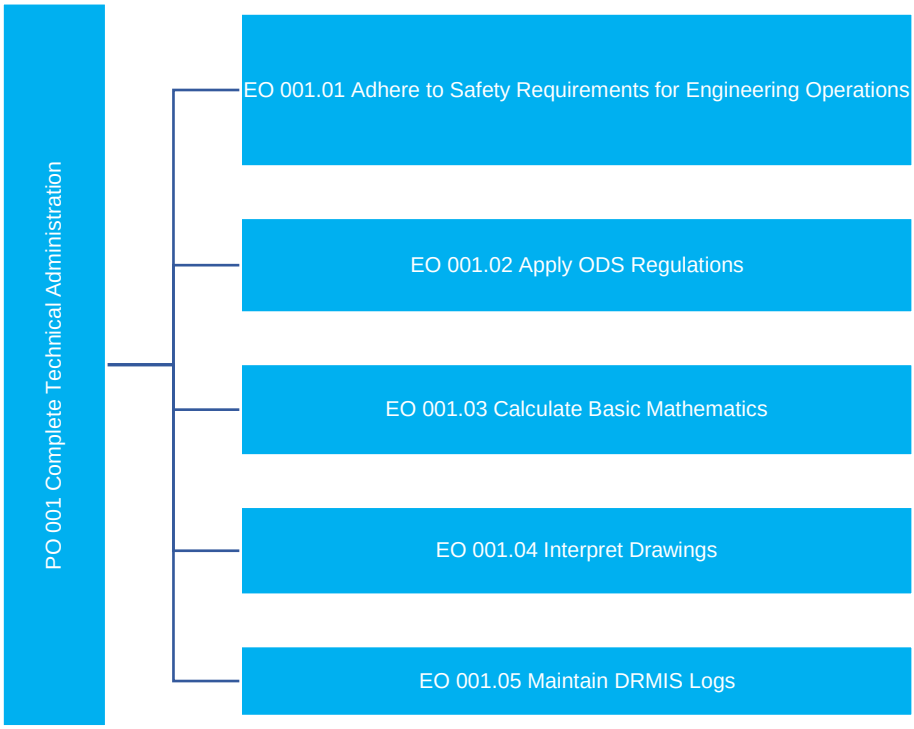
	Problem with the above is scheduling. Trainees would not be OFP until they can return to unit. Discussion as to whether Roundsperson should be its own PO or whether it should be split into corresponding POs. New EOs would have new names, encompassing comparison. “Rounds” would be removed **The board votes to distribute previous EOs under Roundsperson PO to corresponding POs (ex engines, fluid systems, etc) as either EOs or TPts** because this will allow trainees to compare platforms/systems. Review will be needed to confirm EO/TPt. EO Perform the Duties of a MARTECH Roundsperson will be included as a rationale in each module. **Developers are to ensure the reason for course is included in each module.
Expectations	The board reviewed expectations, to ensure they were met. Highlighted expectations were met. 1 Restructure the QSP to reflect OS job, including RegF and ResF requirements 2 Create standardized list of instructional strategies to use a. Ensure strategies match the target audience b. Ensure strategies add flexibility; investigate ROI of 2nd IS 3 Create clear Assessment Plan (CEC, EC, PC) a. Clear directive for safety 4 Ensure clear sequence for training and modules a. Mandatory vs preferred b. Distinct milestones 5 Include standardized assignments and handouts (partial) 6 Modify tasks as appropriate based on defined performance of an OS (may not reflect JTAR since JTAR needs amending) 7 Create baseline scaffold for performance progression through all MARTECH coursing 8 Meet platform non-specific intent a. Ensure flexibility with assessments / practical elements based on TEL / equipment capabilities ***(acceptable vs ideal)*** 9 Prioritize TEL ***(acceptable vs ideal)*** 10 Identify TE(s) / TS for delivery of each IS

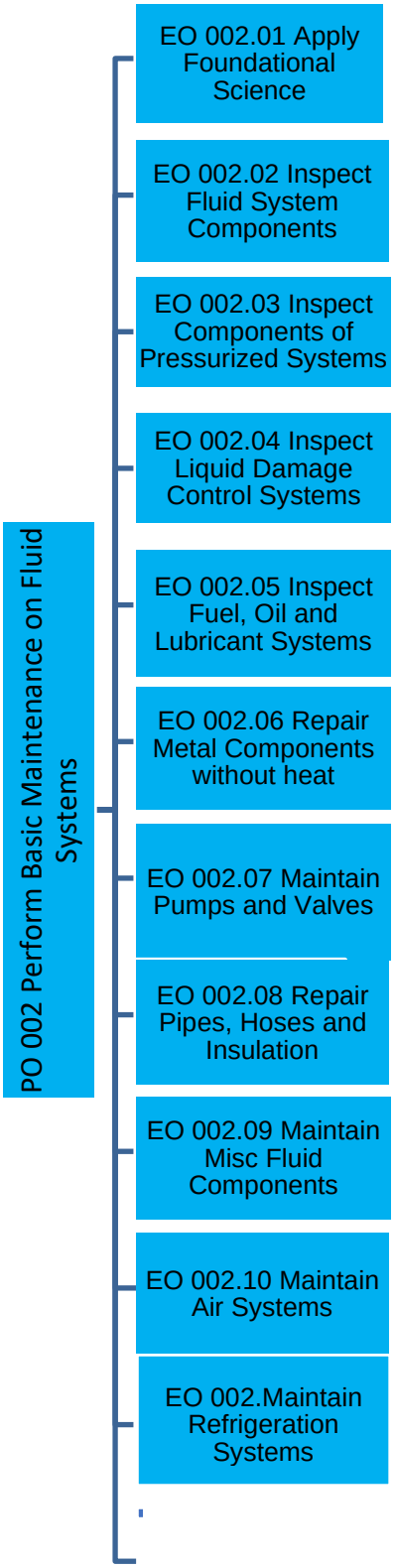
	11 Clearly define what TPts are covered by what IS, TE Outstanding expectations will be met at a later date, with developers keeping in mind board input.
Adjournment	The board adjourned at 1340.

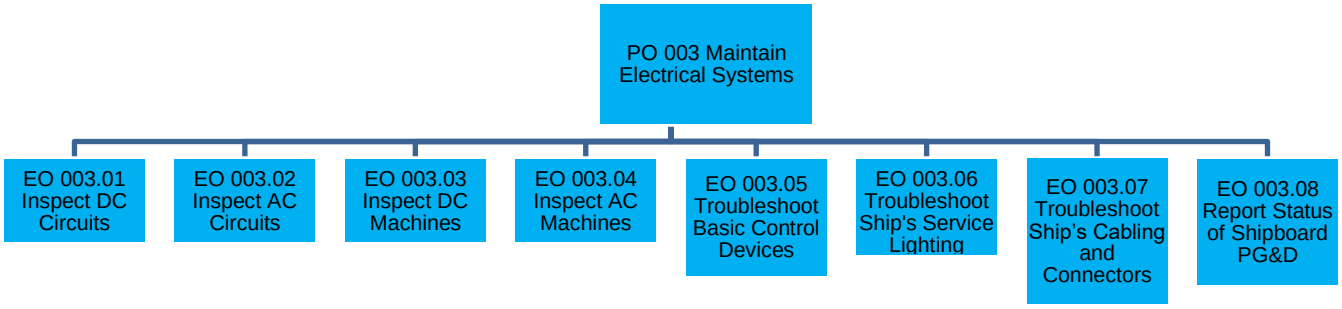
ANNEX D - QSPWB SCALAR

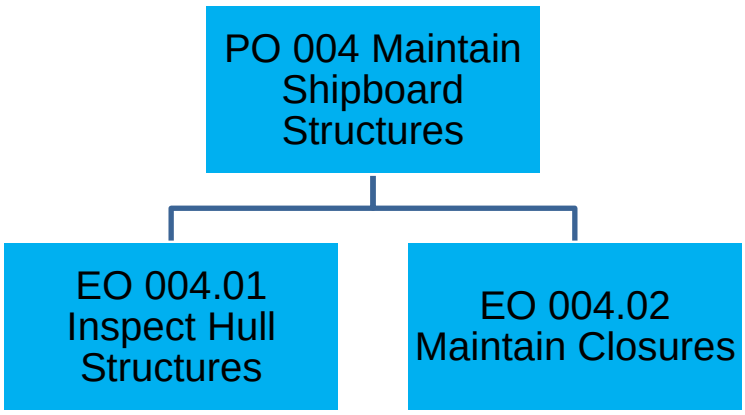
The mapping of the task analysis from the QSPWB is detailed in the standard paragraphs of the POs in Chapter 3 of this QSP, therefore no scalar is inputted here.

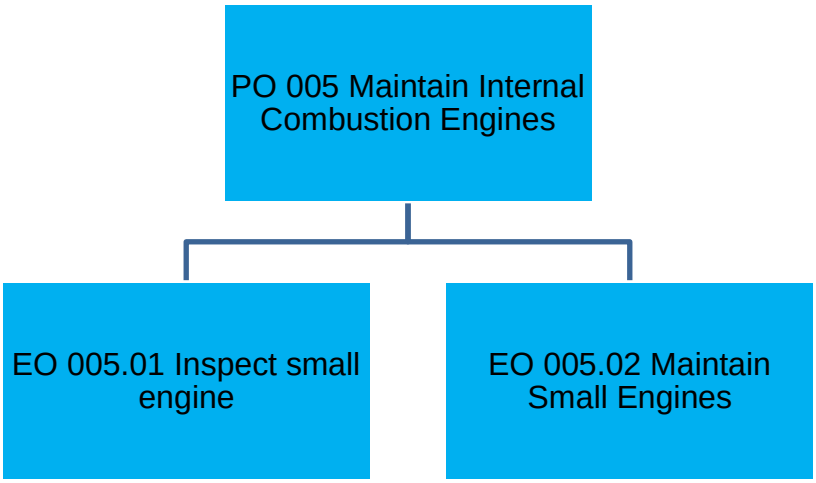
The mapping of the instructional analysis from the QSPWB is detailed in the standard paragraphs of the EOs in Chapter 3 of this QSP, therefore no scalar is inputted here.

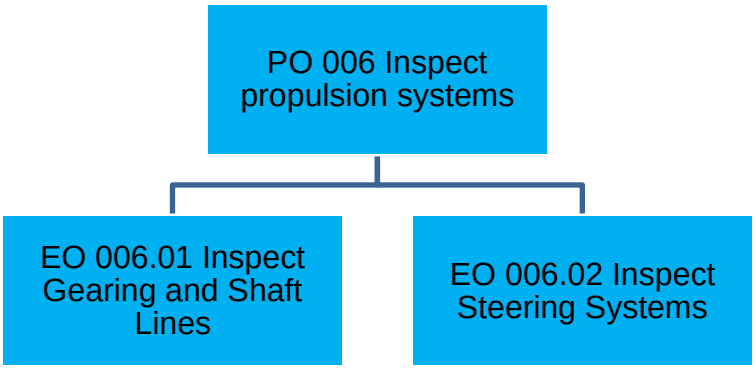


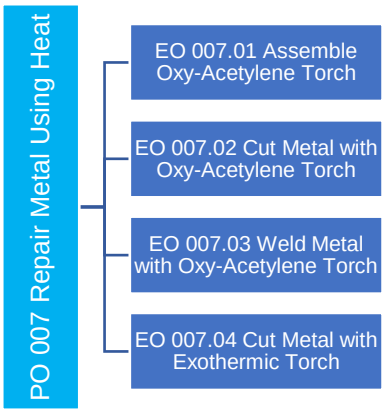

PO 002 Perform Basic Maintenance on Fluid Systems
EO 002.01 Apply Foundational Science
EO 002.02 Inspect Fluid System Components
EO 002.03 Inspect Components of Pressurized Systems
EO 002.04 Inspect Liquid Damage Control Systems
EO 002.05 Inspect Fuel, Oil and Lubricant Systems
EO 002.06 Repair Metal Components without heat
EO 002.07 Maintain Pumps and Valves
EO 002.08 Repair Pipes, Hoses and Insulation
EO 002.09 Maintain Misc Fluid Components
EO 002.10 Maintain Air Systems
EO 002.Maintain Refrigeration Systems


PO 003 Maintain Electrical Systems
EO 003.01 Inspect DC Circuits
EO 003.02 Inspect AC Circuits
EO 003.03 Inspect DC Machines
EO 003.04 Inspect AC Machines
EO 003.05 Troubleshoot Basic Control Devices
EO 003.06 Troubleshoot Ship's Service Lighting
EO 003.07 Troubleshoot Ship's Cabling and Connectors
EO 003.08 Report Status of Shipboard PG&D


PO 004 Maintain Shipboard Structures
EO 004.01 Inspect Hull Structures
EO 004.02 Maintain Closures


PO 005 Maintain Internal Combustion Engines
EO 005.01 Inspect small engine
EO 005.02 Maintain Small Engines


PO 006 Inspect propulsion systems
EO 006.01 Inspect Gearing and Shaft Lines
EO 006.02 Inspect Steering Systems


PO 007 Repair Metal Using Heat
EO 007.01 Assemble Oxy-Acetylene Torch
EO 007.02 Cut Metal with Oxy-Acetylene Torch
EO 007.03 Weld Metal with Oxy-Acetylene Torch
EO 007.04 Cut Metal with Exothermic Torch

ANNEX E - ON JOB PERFORMANCE RECORD (OJPR)

There is no OJPR for MAR TECH RQ-OS Basic Maintainer. Upon qualification, basic maintainers have achieved Operational Functional Point (OFP) and are to complete the classroom/DL portion of an applicable Roundsperson Class Package at which point they will do Roundsperson OJT.

RE RQ-OS - ANNEX E - OJPR .msg

ANNEX F - ACRONYMS AND ABBREVIATIONS

Acronym / Abbreviation	Term or Definition
A/C	Air Conditioning
AC	Alternating Current
ACOS	Automatic Change Over Switch
ACR	Air Conditioning and Refrigeration System
AFFF	Aqueous Film Forming Foam
B/W	Black Water
CBRN	Chemical Biological Radiological Nuclear
CBT	Computer Based Training (example is the DLN LMS)
Class	A class of ship is a group of ships of a similar design. Currently, in the RCN, this includes surface vessels, submarines, and auxiliary vessels, both commissioned and non-commissioned, including: 1. Halifax-class frigates; 2. Kingston-class coastal defence vessels; 3. Victoria-class submarines; 4. Orca-class patrol vessels; 5. HMCS Oriole; 6. NRU Asterix; and 7. Harry DeWolf Class AOPV
D/C	Damage Control
DC	Direct Current
DDFP	Diesel Driven Fire Pump
DFO	Diesel Fuel Oil
DLI	Distributed Learning Instructor Qualification
DLO	Diesel Lube Oil
DVM	Digital Voltmeter
EMF	Electromotive Force
F/W	Fresh Water
FA	First Aid
FOS	Fuel Oil System

FOT	Fuel Oil Transfer
G/W	Grey Water
GT	Gas Turbine
HAZMAT	Hazardous Materials
HMI	Human Machine Interface
HP	High Pressure
HVAC	Heat and Ventilation System
ICE	Internal Combustion Engine
JP5	Jet Propellant 5
LP	Low Pressure
LOTO	Lock Out Tag Out
MCOS	Manual Change Over Switch
MLO	Main Lube Oil
MSDS	Material Safety and Data Sheets
MVAC	Mobile Cooling System
NFS	Naval Fleet School
NRD	Naval Reserve Division
ODS	Ozone Depleting Substances
OCCAP	Oil Coolant Condition Analysis Program
PDE	Pielstick Diesel Engine
PG&D	Power Generation & Distribution
PPE	Personal Protective Equipment
PLC	Programmable Logic Controller
POL	Petroleum, Oils and Lubricants
PRes	Primary Reserve
PVC	Polyvinyl Chloride
RADHAZ	Radiation Hazard
RHIB	Rigged-Hull Inflatable Boat
RMS	Root Mean Square
S/W	Seawater
SFC	Static Frequency Converter

SWBD	Switchboard
SSS	Synchronous Self Shifting
TAU	Twin Agent Unit
TDS	Total Dissolved Solids
TE	Training Establishment
TS	Training Site
UPS	Uninterruptable Power Supply
Vv	Valve
W/T	Watertight Door

ANNEX G - SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used by Campus Support to properly schedule all required classrooms, trainers, ships, personnel, etc for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).
8. The tables are completed in the preferred sequence for lessons and resources. They are to be read in conjunction with the proposed course schedule (when available).
9. For modularized courses, the tables shall be divided based on module (see the modularized course template).
10. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module and can include:
    a. Elements which must be scheduled in a particular order;
    b. Elements which can be scheduled in any order;
    c. Common scheduling conflicts or issues with the phase; and
    d. Changes to required resources based on number of personnel on course; etc.
11. The following excel document is available to use as a template for calculations. Simply "Save as" upon opening and input data as the board progresses. Once complete, insert a copy of the table for printed versions with an embedded version of the file included for use after course amendments.

Mod	EO #	PO / EO / LO Title	EO	EC	CEC	PC	Scheduler Notes
A			DL				Class A = 4 days
	001.03	Perform Basic Mathematics	1140	60			
B			DL				Class A = 12.5 days
	001.01	Maintain Safety Requirements for Engineering Operations	410	60			
	001.04	Interpret Drawings	150	60			
	002.01	Apply Foundational Science	400	60			

	002.02	Inspect Fluid Systems Components	550	60			
	002.03	Inspect Components of Pressurized Systems	575	60			
	002.04	Maintain the Liquid Damage Control Systems	175	60			
	002.05	Maintain Petroleum, Oils and Lubricant Systems	325	60			
	002.06.01	Utilize Shop Tools	150				
	004.01	Inspect hull Structures	440	60			
C			**Training Establishment**				9 Days
	001.05	Maintain DRMIS as a maintainer	90	10			60 min BAT 60 min TAT Req minutes : Workshop 2.06.02+03 = 650; Workshop o/s Control shop 2.07.01 = 1200; Pump Trainer 2.07.02 = 750.
	002.06.02	Utilize Shop Tools	200	425			
	002.06.03	Install Metal Shapes and Components	450				
	002.07.01	Maintain Valves	1200	150			
	002.07.02	Maintain Pumps	750				
D			**DL**				Class A = 3.5 days
	001.02	Apply ODS regulations via HRAI Canada	350	90			
	002.11	Maintain Refrigeration Systems	350	200			
	007.01.01	Describe Oxy Acetylene Safe Work Practices	100				

D			**Training Establishment**				25 Days
	002.08	Repair Pipes, Hoses and Insulation	1320	250			BAT 120 TAT 120 TTT 600 PT 720
	002.09	Maintain Miscellaneous Fluid Components	740				
	002.10	Maintain Air Systems	625	150			
	004.02	Maintain Closures	1150	100			
	007.01.02	Perform Oxy Acetylene Safety	100			550	
	007.01.03	Perform Oxy Acetylene Equipment Set-up	100				
	007.02	Perform Oxy Acetylene Cutting	250				
	007.03	Perform Oxy Acetylene Welding	1200				
	007.04	Perform Exothermic Cutting	1050				
E			**DL**				Class A = 4.5 days
	005.01	Inspect Small Engines	525	60			
	006.01	Inspect gearing and shaft lines	405	60			
	006.02	Inspect steering systems	225	60			
E			**Training Establishment**				3 days
	005.02	Maintain Small Engines	930	260			BAT 60 TAT 60
F			**DL**				Class A = 10 days
	003.01	Inspect DC Circuits	810	60			
	003.02	Inspect AC circuits	330	60			
	003.03	Inspect AC machines	500				
	003.04	Inspect DC machines	150				

	003.05	Maintain basic control devices	200				
	003.06	Maintain ship's service lighting	120				
	003.07	Perform basic maintenance on ship's cabling and connectors	125				
	003.08	Inspect shipboard PG&D	600	60			
F			**Training Establishment**				19 Days
	003.03	Inspect DC machines	750	60			BAT 120 TAT 120 TTT 450 PT 540 Req. Lab-Volt minutes: 3.03 = 750 3.04 = 700 3.05 = 1800
	003.04	Inspect AC machines	700	60			
	003.05	Maintain basic control devices	1900	720			
	003.06	Maintain ship's service lighting	200	140			
	003.07	Perform basic maintenance on ship's cabling and connectors	1355	320			
	Sub-total	TE = 56 Days, DL = 34.5					
	Total	90.5 Days					

*** EXAMPLE: To be deleted from final QSP version***

EO ###.##	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
PO 001: (PO Performance Statement)						
PO ATT:	0					
EO 001.01	(EO performance statement)	Instructor	Classroom w/ DWAN			
etc…	(EO performance statement)		Trainer			
EO 001.0#	(EO performance statement)	Instructor	Classroom w/ DWAN			
PO 002: (PO Performance Statement)						
PO ATT:	0					
EO 002.01	(EO performance statement)	Instructor	Classroom w/ DWAN			
etc…	(EO performance statement)		Trainer			
EO 002.0#	(EO performance statement)	Instructor	Classroom w/ DWAN			

ANNEX H - ADDITIONAL TRAINING RESOURCES

Critical Training Resources	
Stationary Per Trainee at start of course:	- Pen - Pencil - Binder - Paper - Eraser
Lab Volt Equipment	- Accessory Bay LabVolt - Power Supply 8821-20 - Power Supply 8840-00 - DC Machine 8211-00 - Dynomometer 8960-10 - Synchronous Machine 8241-00 - Component Panel 9127-10 - Contactor 9106-20 - Control Relay 9108-10 - DC Volt/Ammeter 8412-10 - AC Voltmeter 8426-00 - Tachometer 8931-00 - Control Bay 8846 - DC Motor/Generator 9318-00
Basic Tools	- Wrenches/sockets; - Pliers; - Pincers; - Screwdrivers; - Files; - Hack saw; - Hammers; - Pry bars; - Vise-grips; - Drill bits; - Taps & dies; - Punch; and - Chisels
Power Tools	- Drill; - Grinder: - Impact wrench; - Band Saw; and - Pneumatic tools
Testing Equipment	- True RMS Multimeter - Multimeter - Insulation / Megometer - Clamp meter

	- Tachometer
LOTO Equipment	- Locks - Lock out devices
Electrical Tools	- Soldering Kit - Wire Strippers - Wire Cutters - Heat Gun
Safety Equipment / PPE	- Safety Glasses

ANNEX I - COSTING INFORMATION

Include information about cost in whichever format is applicable (Not to be deleted).

Template Annex I Costing.xlsx

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

**5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.

Course:		**Serial:**				
TD	**No. Of Students / instructors /**	**Cost per each**	**# Days (if applicable)**	**Initial TE Set-up**	**Each successive course**	**5 Year total**
Airfare (Return)	0	$0.00		$0.00	$0.00	$0.00
R&Q	0	$0.00		$0.00	$0.00	$0.00
Incidentals	0	$0.00		$0.00	$0.00	$0.00
Subtotals				$0.00	$0.00	$0.00
Material	**Quantity**	**Cost per each**	**# Days (enter 1 if not applicable)**			
PP&S (Estimate)	0	$0.00	0	$0.00	$0.00	$0.00
Classroom (square meters)	0	$0.00	0			
Computer requirements	0	$0.00	0	$0.00	$0.00	$0.00

IT Equip (Proj, Smart board etc)	0	$0.00	0	$0.00	$0.00	$0.00
DWAN connection (Set up)	0	$0.00	0	$0.00	$0.00	$0.00
Link Toolkit (on DVD)	0	$0.00	0	$0.00	$0.00	$0.00
Instructional Materials (Printed / Handouts etc)	0	$0.00	0	$0.00	$0.00	$0.00
Subtotals				$0.00	$0.00	$0.00
PYs (only if course includes incremental on SWE, or course to be delivered off site	**Quantity**	**Cost per day* (SWE if necessary)**	**# Days (if applicable)**			
Instructor (PO1/WO) (incremental)2	0	$0.00	0	$0.00	$0.00	$0.00
Incremental Civilian Instructor	0	$0.00	0	$0.00		
Subtotals				$0.00	$0.00	$0.00
			Total Estimated Costs	$0.00	$0.00	$0.00

ANNEX J - COURSE HISTORY

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

08 May 18 – The following PO's are approved by NPTG to be changed within the QSP:

PO 002	Maintain Liquid Handling Systems	RDMIS #
EO 002.01	Utilize Hand Tools and Measuring Equipment	870592 / 871652 / 871948
EO 002.02	Maintain Valves	870592
EO 002.03	Maintain Heat Exchangers and Coolers	870592
EO 002.04	Maintain Lagging and Insulation	870592
EO 002.05	Maintain Pipes and Hoses	870592
PO 007	Maintain Internal Combustion Engines	
EO 007.01	Operate Small Gasoline Engines	870601
EO 007.02	Maintain Marine Diesel Engines	870601
EO 007.03	Maintain Marine Diesel Engine Coolant	870601
EO 007.04	Maintain Marine Diesel Engine Oil	870601
EO 007.05	Maintain Gas Turbines	870601
PO 010	Maintain Damage Control Equipment	
EO 010.01	Inspect Fitted Protection Systems	
EO 010.02	Maintain First Aid Extinguishers and hose components	
PO 013	Maintain Shafts and Gearing	
EO 013.01	Maintain Shaft and Gearing Systems	870594
EO 013.02	Maintain Machinery Brakes	870597
EO 013.03	Maintain Steering System	870598
PO 014	Maintain Refrigeration and A/C systems	
EO 014.01	Perform Safety with ODS	870602
EO 014.02	Operate Refrigeration Systems	870602 / 871661
EO 014.03	Operate Chillers	870561 / 871662

EO 014.04	Operate Chilled Water Systems	870577

Due to the impact of these PO's being changed, NPTG has requested that the current version of this QSP with the above-mentioned changes implemented will be version 2.0 of the Marine Technician Basic Maintainer (RQ-OS) (00379) QSP.

03 Oct 18 – The following PO's are approved by NPTG to be changed within the QSP:

PO 001	Maintain Battery Charging Systems	RDMIS #
EO 001.01	Identify Safety Requirements for Engineering Operations	870587 / 870589
EO 001.02	Analyze Drawings and Schematics	871621
EO 001.04	Maintain Battery charging systems	
EO 001.05	Inspect Battery Assemblies	871645
EO 001.06	Observe HAZMAT/WHIMS	870591
PO 006	Conduct Cutting and Welding	
EO 006.02	Perform Welding (Arc SMAW)	871653 / 870600
PO 008	Operate Power Generation and Distribution	
EO 008.02	Operate ACOS/MCOS	870593
PO 011	Maintain Domestic Service Systems	
EO 011.03	Maintain Black and Grey Water Collection Systems	871659
PO 016	Maintain Steam Systems	
EO 016.01	Maintain Feedwater systems	870599
PO 017	Maintain Compressed Air Systems	
EO 017.01	Maintain Low Pressure Air Systems	
EO 017.02	Maintain High Pressure Air Systems	
EO 017.03	Operate Breathing Air Compressors	
PO 018	Perform the duties of a Roundsperson	
EO 018.01	Perform the duties of a MARTECH Roundsperson	873220

Numerous formatting corrections were also applied, some examples of these formatting changes are as follows:

1 Chapter 1 and 3 changes to reflect the updated NPTORD 4500-22 (CAR RDMIS# 870559);

1. The removal of DRMIS as a requirement when not applicable to the subject;

2. Examination descriptions; and
3. Minor changes to wording within the teaching points to add clarity.

Due to the impact of these PO's being changed, NPTG has requested that the current version of this QSP with the above-mentioned changes implemented will become version 3.0 of the Marine Technician Basic Maintainer (RQ-OS) (00379) QSP.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex B

NPTORD 4500-22
Annex B_QSP Amend

Date	Change made in:	Substantiation	
09Sep22	RDIMS v8	This version has been updated for the last 3 years on the Share Point.	
09Sep22	RDIMs v8A	No change.	
09Sep22	RDIMs v9 And 9A	Addressed comments from TDO Jayna Cayer	
09Sep22	RDIMs v9B	Populated Annex D to match QSP structure.	
09Sep22	RDIMs v9C	Changed method of Delivery by switching method 1 and method 2. This was to facilitate students loading to DL prior to the F2F portion of the course.	
09Sep22	RDIMs v9D	Populated Annex G with sequencing and timings.	
09Sep22	RDIMs v9E	Adjusted Timings- Course Duration and Annex G	
09Sep22	RDIMs v9F	Mod 0 added by Jayna Cayer –Intro to RCN	
09Sep22	RDIMs v9G	Completed Annex F, Annex H, Chapter 1 Resources	
09Sep22	RDIMs v9H	Updated Annex O	

09Sep22	RDIMs v9I	Welding PC timing discrepancy corrected. Originally PC 007 was 300 minutes in the PO but 550 minutes in Annex L and EO00 7.04. Updated Annex L and EO to match PO timing.	
09Sep22	RDIMs v10	RDIMs Version error. (Accidental Save)	
09Sep22	RDIMs v10A	PO 007 Re-Work - Added in SMAW 7.05. 2050 minutes were reinstituted to correct the SMAW removal error. Changed Weld Oxy/Ace before cutting Oxy/Ace.	
09Sep22	RDIMs v10B	Corrected Annex G - Switched Mod E and Mod F Sequence. - Updated QSP throughout.	
09Sep22	RDIMs v10C	Changed Titles - EO 5.01 - Inspect Small Engine from Complete Small Engine Famil. PO 006 - Inspect Propulsion Systems from Complete Propulsion Systems Familo. And populated Annex A.	
09Sep22	RDIMs v10D	Updated by switching Mod E and F through the entire QSP to match Annex G change, this was for Elec. content to be the last EO learned.	
09Sep22	RDIMs v10E	Addressed/Captured comments. Comments Deleted.	
09Sep22	RDIMs v11	Updated Annex J, Course Duration added Note: for delivery methods; Mod A B and ODS (1.02) to be taught prior to all other Mods.	
14Sep22	RDIMs v 11a +b	RDIMs v11a – was caused due to program crash. RDIMs v11b; -Corrected Footer Format. 1.05 – DRMIS - changed EC to PC 2.06 – repair metal without heat - changed from EC to CEC 2.11 – refrig leak testing - changed from EC to PC 002 3.07 – Elec - changed from EC to PC 003 4.02 – Maintain basic closures – changed from EC to PC 004	

		5.02 – Maintain small engines - uses an EC then requires OJT 6.02 – Inspect Prop systems - uses an EC then requires OJT This was done because all assessments were EC's with the exception of PC 007. Aligning to CFITES.	
14Sep22	RDIMS v11c +d	Corrected 2.06 CEC changed to PC. PO 001- safety, will be assessed in PC 003 for trainee's to perform a physical practice of a LOTO form and lock before the Cabling PC Project.	
26Sep22	RDIMS v11eg+h	Populated References in Ch.3 EO's. PO 003; 2.06 CEC changed to PC. PO 001 will conduct LOTO in PC 003; Inserted EO references for PO 002 series; Finished inserting EO ref codes; Changed Lab-Volt sequence, Safety labs put to the first time trainee's will be in the lab-volt (3.03 EDI); and Corrected Annex G to match EO's in Ch 3.	

ANNEX K - ASSESSMENT PLAN

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.01	EO 001.01	Adhere to Safety Requirements for Engineering Operations	Written assessment on CBT; 30 questions from bank	80%	Remedial Training / Re-test
PC 001.02	EO 001.02	Apply ODS Regulations	Written assessment IAW Federal HRAI Certification	75%	Remedial Training / Re-test / TRB
EC 001.03	EO 001.03	Perform Basic mathematics	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 001.04	EO 001.04	Interpret Drawings	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
PC 001.05	EO 001.05	Maintain DRMIS Portal as a Maintainer	IAW DRMIS Module 1		
EC 002.01	EO 002.01	Apply Foundational Science	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 002.02	EO 002.02	Inspect Fluid System Components	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 002.03	EO 002.03	Inspect Components of Pressurized Systems	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 002.04	EO 002.04	Inspect the Liquid Damage Control Systems	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 002.05	EO 002.05	Inspect Fuel, Oil and Lubricant Systems	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
PC 002.06	EO 002.06	Repair Metal Components Without Heat	Practical; install metal shapes and components using a schematic	SAT all checklist items	Remedial Training / Re-test / TRB
EC 002.07	EO 002.07	Maintain Pumps and Valves	Practical; maintenance of random basic valve for invigilator inspection and pressurization	SAT all checklist items	Remedial Training / Re-test

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 002.08	EO 002.08	Repair Pipes, Hoses and Insulation	Practical; produce a section of pipe with two flanges and threaded Tee with reducer; installation of short length of tube and finishing cap; pressurized to 50 PSI	SAT all checklist items	Remedial Training / Re-test
EC 002.09	EO 002.09	Maintain Miscellaneous Fluid Components	Practical; in pairs, perform maintenance on randomly selected fluid component	Sat on all checklist items	Remedial Training / Re-test
EC 002.10	EO 002.10	Maintain Air Systems	Practical; perform maintenance on randomly selected air system	Sat on all checklist items	Remedial Training / Re-test
PC 002.11	EO 002.11	Maintain Refrigeration Systems	Practical; perform refrigeration leak testing routine	Sat on all checklist items	Remedial Training / Re-test / TRB
EC 003.01	EO 003.01	Inspect DC Circuits	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 003.02	EO 003.02	Inspect AC Circuits	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 003.03	EO 003.03	Inspect AC Machines	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 003.04	EO 003.04	Inspect DC Machines	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 003.05	EO 003.05	Troubleshoot Basic Control Devices	Practical Assessment	SAT all checklist items	Remedial Training / Re-test
EC 003.06	EO 003.06	Troubleshoot Ship's Service Lighting	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
PC 003	PO 003	Apply LOTO and Troubleshoot Ship's Cabling and Connectors	Practical Assessment	SAT all checklist items	Remedial Training / Re-test / TRB
EC 003.08	EO 003.08	Report Status of Shipboard PG&D	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 004.01	EO 004.01	Inspect Hull Structures	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
PC 004	PO 004	Maintain Shipboard Structures	Practical; In pairs, perform maintenance on watertight door or hatch	SAT on all checklist items	Remedial Training / Re-test / TRB
EC 005.01	EO 005.01	Inspect small engine	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 005.02	EO 005.02	Maintain Small Engines	Practical Assessment	SAT all checklist items	Remedial Training / Re-test
EC 006.01	EO 006.01	Inspect Gearing and Shaft Lines	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
EC 006.02	EO 006.02	Inspect Steering System	Written assessment on CBT; 30 questions from bank	60%	Remedial Training / Re-test
PC 007	PO 007	Repair Metal Using Heat	Practical Assessment	SAT all checklist items	Remedial Training / Re-test / TRB

ANNEX L – TEST DETAILS

EC 001.01: Adhere to Safety Requirements for Engineering Operations

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will complete a theoretical online assessment derived from a combination of all the content from EO 001.01. The assessment will be delivered utilizing a CBT framework. Questions to include pictures of safety equipment for identification, matching, etc.

Pass/fail criteria: 80% pass mark

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LMS account.

IMPACT OF FAILURE: Remedial Training / Re-test

PC 001.02: Apply ODS Regulations

PC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainee to complete a theoretical online assessment derived from a combination of all the content from EO 001.02. The assessment will be delivered IAW the Federal HRAI Certification. The assessment will be invigilated by an instructor or regional instructor.

Pass/fail criteria: Pass 75% (IAW Federal HRAI Certification)

ASSESSMENT RESOURCES:

- Test booklet; and
- Pencil with eraser.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 001.03: Calculate Basic Mathematics

EC Time: 50 mins (concurrent) + 10 min debrief (concurrent)

ASSESSMENT ACTIVITY: Trainee to complete a theoretical online assessment that is derived from a combination of all the content from EO 001.03. The assessment will be invigilated by an instructor or regional instructor.

Pass/fail criteria: Pass 60%

ASSESSMENT RESOURCES:

- Classroom / location to complete the assessment
- Test booklet
- Pencil with eraser

IMPACT OF FAILURE: Remedial Training / Re-test

EC 001.04: Interpret Drawings

EC Time: 50 mins (concurrent) + 10 min debrief (concurrent)

ASSESSMENT ACTIVITY: Trainee to complete a theoretical online assessment that is derived from a combination of all the content from EO 001.04. The assessment will be invigilated by an instructor or regional instructor.

Pass/fail criteria: Pass 60%

ASSESSMENT RESOURCES:

- Classroom / location to complete the assessment
- Test booklet
- Pencil with eraser

IMPACT OF FAILURE: Remedial Training / Re-test

PC 001.05: Maintain DRMIS Logs

PC Time: 60 mins

ASSESSMENT ACTIVITY:

Pass/fail criteria:

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LMS Account.

IMPACT OF FAILURE: Remedial training / Re-test

EC 002.01: Apply Foundational Science

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical online assessment derived from a combination of all the content from EO 002.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: Pass 60%.

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LMS Account.

IMPACT OF FAILURE: Remedial training / Re-test

EC 002.02: Inspect Fluid System Components

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical online assessment derived from a combination of all the content from EO 002.02. The assessment will be delivered utilising a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

- DLN LMS Account; and
- Computer with DLN LMS access.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 002.03: Inspect Components of Pressurized Systems

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical online assessment derived from a combination of all the content from EO 002.03. The assessment will be delivered utilising a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LM account.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 002.04: Inspect the Liquid Damage Control Systems

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical online assessment derived from a combination of all the content from EO 002.04. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: Pass mark 60%

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LMS Account.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 002.05: Inspect Fuel, Oil, and Lubricant Systems

EC Time: 60 mins

ASSESSMENT ACTIVITY: Trainees to complete a theoretical online assessment derived from a combination of all the content from EO 002.05. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: Pass 60%

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and

- DLN LMS Account.

IMPACT OF FAILURE: Remedial Training / Re-test

PC 002.06: Repair Metal Components (Without Heat)

CEC Time: 400 mins (concurrent) + 25 min debrief (for all trainees)

ASSESSMENT ACTIVITY: Trainee to complete a practical assessment to install metal shapes and components. The trainee will be given a schematic and will produce a project IAW the checklist. The schematic will be an item that can be achieved through performing basic tool skills listed in the checklist within the given time limit.

Pass/fail criteria: Trainees must achieve a min of 60% SAT on each of Parts 1-2 and 100% SAT on the critical components in safety.

- Part 1: 7 / 11 SAT
- Part 2: 3 / 5 SAT
- SAFETY: 2 / 2 SAT Critical

ASSESSMENT RESOURCES:

- Appropriate PPE;
- Appropriate tools;
- Materials; and
- Schematic to produce.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – PC 002.06 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a 60% SAT on each section of the checklist to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Prepare Components			
Cut/File Aluminum Block ● IAW Drawing ● Tolerance +/- 1/32	/1		
Punch Aluminum Block IAW drawing	/1		
Drill and tap Top hole Aluminum Block ● ½" 13 TPI ● IAW Drawing ● Tolerance + 1/32	/1		
Drill and tap Side hole Aluminum Block ● 1/4" 20 TPI ● IAW Drawing ● Tolerance + 1/32	/1		
Cut/file Aluminum Flat Bar ● IAW Drawing ● Tolerance +/- 1/32	/1		
Punch flat bar IAW Drawing	/1		
Drill and Countersink flat bar	/1		
Cut/file Aluminum Round Stock ● IAW Drawing	/1		

● Tolerance +/- 1/32			
Thread bolt ● IAW Drawing ● Tolerance - 1/32	/1		
Did not require extra materials	/2		
2. Install Components			
Main Block ● Flat bar filed square with block ● Post is square on each side with main block after installation	/2		
¼” 20 screw ● Countersunk and flush with flat bar	/1		
Installed project file marks (2 file marks = 1 point)	/2		
3. Safety			
Utilize appropriate PPE ● Safety goggles ● Boots ● Sleeves rolled up	/1		
Follow safety practices	/1		

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 002.07: Maintain Basic Pumps and Valves

EC Time: 30 mins (concurrent) + 20 min inspections/debrief (concurrent)

ASSESSMENT ACTIVITY: Trainees will complete a practical assessment that is derived from a combination of all the content from EO 002.07. Each trainee will be given a random basic valve to perform maintenance on. Once the trainee is satisfied that the valve has been maintained, they will present the valve to the instructor for inspection. Once the valve has been inspected, it will then be presented back to the trainee to place within a test bench to be pressurized. Training facilities may only require one test bench as trainees will complete their basic valve maintenance at different times.

The trainee may be randomly tested on one of the following basic valves:

1 Gate Valve;
2 Globe Valve;
3 Diaphragm Valve;
4 Ball Valve;
5 Butterfly Valve;
6 Check Valve;
7 Non-return Valve;
8 Steam trap;
9 Pressure Relief Valve; and
10 Pressure Regulating valve.
11 Assembling basic valves;

12 Disassembling basic valves;

13 Lubricating valve spindles;

14 Replacing packing;

15 Replacing gaskets; and

16 Testing basic valves under pressure.

Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:

- Appropriate tools;
- Appropriate PPE;
- Basic Valves;
- Workbench;
- Maintenance material (grease, gaskets and packing materials);

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 002.07 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Maintain Basic Valves			
a. Disassemble the Valve			
b. Lubricate valve spindles			
c. Replace the packing			
d. Replace the gaskets			
e. Reassemble the valve			
f. Pressurize the valve			
g. Ensure the valve functions as intended with no leaks when pressurized			
2. Perform Tasks Safely			
a. Wear Appropriate PPE			

b. Adhere to appropriate safety practices throughout			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 002.08: Repair Pipes, Hoses, and Insulation

EC Time: 240 mins (concurrent) + 10 min debrief (for all trainees)

ASSESSMENT ACTIVITY: Trainee to complete a practical assessment that is derived from a combination of content from EO 002.08. The trainee is to produce a section of pipe with two flanges on either end. One flange will be soldered into place, while the other end is threaded. In the middle of the pipe will be a threaded Tee with a reducer in which a tubing fitting can be attached. A short length of tube will be installed, finishing with a cap on the other end. Once produced, the project will be pressure tested to hold 50 PSI. Once no leaks are found, the trainee will lag the project.

Pass/fail criteria: SAT on all checklist items

ASSESSMENT RESOURCES:

- Materials (Piping, solder, fittings, etc);
- Appropriate PPE;
- Properly equipped workshop; and
- Test bench to pressurize the project.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 002.08 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Perform Basic Maintenance on pipes, hoses and insulation			
Measure the piping;			
Cut the piping;			
Solder one flange;			
Thread one flange;			
Thread a Tee reducer;			

Install a tubing fitting;			
Measure a tubing length;			
Cut a tubing length;			
Install a tubing;			
Install a plug;			
Pressurize the project to 50 PSI with no leaks			
Lag the project.			
2. Perform tasks safely			
Utilize appropriate PPE			
Utilize appropriate safety practices			

ADDITIONAL COMMENTS			
STRENGTHS:		**AREAS FOR DEVELOPMENT:**	
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:

Candidate	Name:	Date:	Signature:

EC 002.09: Maintain Miscellaneous Fluid Components

EC Time: 80 mins (concurrent) + 10 min debrief (per pair of trainees)

ASSESSMENT ACTIVITY: Trainee to complete a practical assessment on the content from EO 002.10. The trainees will work in pairs to perform basic maintenance on one of the following miscellaneous fluid components:

a. Fuel Coalescor;

b. Centrifuge; and

c. Marine vacuum toilet.

The component the pair will receive shall be chosen at random. The basic maintenance shall be IAW the EC checklist.

To limit the need for multiple invigilators, the trainee's will be given milestones based on the EC checklist. Once they have achieved a milestone, they will signal to the invigilator who will verify the milestone and instruct the trainees to continue with the assessment or redo the milestone in question.

Pass/fail criteria: Sat on all checklist items.

ASSESSMENT RESOURCES:

- Appropriate Tools;
- Appropriate PPE; and
- Combination of the following miscellaneous fluid components:
    - Fuel Coalescor;
    - Centrifuge; and
    - Marine vacuum toilet.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 002.09 Assessment Checklist

Candidate Name:	Assessor Name:	Date:	Session:

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Perform Basic Maintenance on a Marine vacuum toilet (In Pairs)			
1 Disassemble a marine vacuum toilets common component, to include: a. Vacuum Interface valve; b. Discharge valve housing; c. VT Control mechanism; d. Discharge valve assembly; and e. EVAC toilet assembly.			
2 Inspect the marine vacuum toilet components for the following: a. Corrosion; f. Tears and rips; and g. Cracks and holes.			Have known faults to see if the trainees correctly inspect and inform the instructor of these faults.
3 Repair by replace any damaged parts; and			
4 Assemble a marine vacuum toilets common component, to include: a. Vacuum Interface valve;			

h. Discharge valve housing; i. VT Control mechanism; j. Discharge valve assembly; and k. EVAC toilet assembly.			
2. Perform Tasks Safely			
Utilize appropriate PPE			
Utilize safe practices			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 002.10: Maintain Air Systems

EC Time: 140 mins (concurrent) + 10 min debrief (per pair of trainees)

ASSESSMENT ACTIVITY: Trainee to complete a practical assessment on the content from EO 002.11. The trainees will work in pairs to perform basic maintenance on one of the following air systems:

1 Air compressor;

2 Booster Cooler (KU); and

3 Fan Coil Unit (FCU).

The air system the pair of trainees will receive shall be chosen at random. The basic maintenance shall be IAW the EC checklist.

To limit the need for multiple invigilators, the trainee's will be given milestones based on the EC checklist. Once they have achieved a milestone, they will signal to the invigilator who will verify the milestone and instruct the trainees to continue with the assessment, or re-due the milestone in question.

Pass/fail criteria: Sat on all checklist items

ASSESSMENT RESOURCES:

- Appropriate Tools;
- Appropriate Personal Protective Equipment (PPE);
- Multiple Air Compressors;
- Multiple KU mock-ups; and
- Multiple FCU mock-ups.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 002.10 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Perform Basic Maintenance on an Air compressor (In Pairs)			
Inspecting the following components: 1 Hoses; 2 Piping; 3 Discharge valves; and 4 Pressure gauge panel.			
Remove the oil filter;			
Replace the oil filter;			
Remove the air filter;			
Replace the air filter;			
Drain the Oil; and			
Refill the Oil.			
2. Perform Basic Maintenance on a Booster Cooler (KU) (In Pairs)			
Change filter media on booster cooler units;			
Lubricate pillow block bearings;			
Remove the drive pulley;			
Describe proper alignment of drive pulley;			

Replace the drive pulley;			
Adjust motor alignment; and			
Adjust belt tension.			
3. Perform Basic Maintenance on a Fan Coil Unit (FCU) (In Pairs)			
Replace air filter on fan coil unit;			
Remove three-way valve;			
Replace three-way valve;			
Remove the motor/fan combo; and			
Replace the motor/fan combo.			
4. Perform Tasks Safely			
Utilize appropriate PPE			
e proper LOTO practices			
e safe practices			

ADDITIONAL COMMENTS		
STRENGTHS:	**AREAS FOR DEVELOPMENT:**	
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT		
OVERALL ASSESSMENT (circle one)	**FAIL**	**PASS**

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002: Maintain Refrigeration Systems

PC Time: 15 mins (per trainee) + 20 min debrief (for all trainees)

ASSESSMENT ACTIVITY: Trainee to complete a practical assessment that is derived from a combination of the content within EO 002.11. The trainee will perform a refrigeration leak testing routine, as per the EC Checklist.

As this assessment requires to be closely invigilated, the addition of extra invigilators for PC 002.11 will greatly reduce the overall PC assessment timings.

Pass/fail criteria: Sat on all checklist items

ASSESSMENT RESOURCES:

- Appropriate leak testing equipment;
- Appropriate PPE; and
- Properly equipped workshop to assess refrigeration leak testing.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 002.11 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Perform basic maintenance on a refrigeration system			
Explain the nature of the refrigerant gas: • Characteristics of the gas; and • Potential safety hazards.			Refrigerant gas is heavier than air and will collect close to the ground, and large refrigerant leaks will displace oxygen, creating a big safety hazard. Skin exposed to the refrigerant gas may result in severe frostbites.
Demonstrate the operational knowledge of the respiratory protection in case of a major refrigerant gas leak.			
Demonstrate the procedures in case of skin/eye.			
Demonstrate the working knowledge of leak detection systems.			The trainee to be able to use the Bachrach Leak Detector, Automatic Halogen Leak detector, and Para sense.
Perform a refrigeration leak test.			
2. Perform Tasks Safely			
Utilize appropriate PPE			
Utilize safe practices			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 003.01: Inspect DC Circuits

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all the content from EO 002.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

- Computer with access to DLN LMS

IMPACT OF FAILURE: Remedial Training / Re-test

EC 003.02: Inspect AC Circuits

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 002.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

Computer with DLN LMS access

IMPACT OF FAILURE: Remedial Training / Re-test

EC 003.03: Inspect DC Machines

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 002.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

Computer with DLN LMS access

IMPACT OF FAILURE: Remedial Training / Re-test

EC 003.04: Inspect AC Machines

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 002.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

Computer with DLN LMS access

IMPACT OF FAILURE: Remedial Training / Re-test

EC 003.05: Troubleshoot Basic Control Devices

EC Time: 50 mins (concurrent/per trainee)

ASSESSMENT ACTIVITY: EC 003.05 will be a written assessment based on content from EO 003.05. Schematic diagram is to be provided and trainee are to label schematic symbols

Pass/fail criteria: Pass mark will be 60%.

ASSESSMENT RESOURCES:

The trainee will be marked IAW master drawing. To consist of:

a. Start/Stop;
b. Forward contractor;
c. Circuit Breaker;
d. Control transformer;
e. Run indication (green lamp);
f. Overload protection;
g. Overload indication (red lamp); and
h. Motor.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 003.05 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS	
Personal Safety				
Use of PPE ● Safety glasses				
Lock out Tag out				
Observed all safety considerations			Failure to observe safety constitutes an automatic failure	
Student picks a random fault from the envelope (without looking) and leaves the room while the instructor inserts fault without changing wiring. Student re-enters the room to troubleshoot the circuit unassisted.				
Fault chosen:	Fault details (exact location of fault in circuit):			
Circuit Understanding				
Explain circuit operation using supplied drawing				
Confirm circuit operation				
Determining the Fault				
Test circuit				
Test meter				

Use meter correctly				
Explain logic behind chosen troubleshooting technique				
Correctly identify fault			Failure to identify the fault constitutes an automatic failure	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 003.06: Troubleshoot Ship's Service Lighting

EC Time: 100 mins (concurrent/per trainee) + 20 min debrief

ASSESSMENT ACTIVITY: Trainees will be required to build and test a lighting circuit unassisted IAW the EC checklist. Their circuit will be tested on both qualities of build and functionality.

Pass/fail criteria: The trainee must receive 9 SAT to pass this EC.

ASSESSMENT RESOURCES:

- Cabling and connectors
- Schematics
- Electrical tool kit
- Multimeters for each trainee and instructor
- Soldering equipment shared by the group
- heat guns shared by the group

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 003.06 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

4. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Personal Safety			
Use of PPE • Safety glasses			
Lock out Tag out			
Observed all safety considerations			Failure to observe safety constitutes an automatic failure
2. Confirm Operation of Lighting Circuit			
Verify power supply			

Operate Switch			
3. Inspect Lighting			
Inspect Cabling			
Inspect Fixture			
Inspect Bulbs			
Check Voltage			
Check Ballast			
4. Use DVM			
Verify power ● at input ● at switch ● at light fixture			
5. Repair Fixture			
Locate and repair the fault			
6. Test Lighting			
Verify Operation ● Turn on power ● Check that light works			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 003: Troubleshoot Ship's Cabling and Connectors

PC Time: 100 mins (concurrent/per trainee) + 20 min debrief

ASSESSMENT ACTIVITY: Trainees will be required to complete a LOTO form and apply a lock to a device prior to building and testing a lighting circuit unassisted IAW the EC checklist. Their circuit will be tested on both qualities of build and functionality.

Pass/fail criteria: The trainee must receive 9 SAT to pass this EC.

ASSESSMENT RESOURCES:

- Cabling and connectors
- schematics
- electrical tool kit
- multimeters for each trainee and instructor
- Soldering equipment shared by the group
- heat guns shared by the group

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – PC 003 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
Personal Safety			
Use of PPE • Safety glasses			
Lock out Tag out			
Maintain proper tool use			
Observed all safety considerations			Failure to observe safety constitutes an automatic failure
Stage 1: Assemble a Lighting Circuit			
Project Board			
Plug is attached to the junction box IAW drawing			
Polarity of connections is IAW drawing			
Connections are tight - can rotate easily (too loose) - no deformities (too tight)			
Terminals are secured			
Stak-ons are secured			
Glands are secured			
Cables are an appropriate length between components			

- no excess cabling - not under tension			
Wires are of an appropriate length inside components (rotary switch, junction box, plug, receptacle). - no excess wiring - adequate length to allow for connector replacement if necessary			
Wire insulation is intact - smooth with no nicks - wire is not visible			
Wires are intact - not pinched or kinked			
Heat shrink glands are smooth - no air gaps (under-heated) - no stress marks or deformities (overheated)			
O-Rings are installed correctly IAW drawing			
Grounding strap is connected between junction box and switch			
Correct components installed IAW drawing			
Functionality of Circuit Board			
Correct use of a multimeter to conduct a continuity test			
Light turns on when switch is turned to the ON position			
INVIGILATOR CHECK	**SAT**	**UNSAT**	

Stage 2: Repair Lighting Circuit			
Splice and Solder			
Correct # of turns - Min 4 - Max 6	1		
Turns are tight and well defined	1		
Spacing is narrow enough to allow insulation overlap of splice joints	1		
Overall splice length is appropriate	1		
INVIGILATOR CHECK	**SAT**	**UNSAT**	**/4**
Distance from edge of solder joint to conductor insulation is appropriate	1		
Even tension between wires	1		
Wire strands are intact - no nicks	1		
Solder joints correct (not cold soldered) on at least 2 of 3 colours	1		
Solder joints are straight (no bends) on at least 2 of 3 colours	1		
No wire protrusions	1		
Solder is clean (no excess) on at least 2 of 3 colours	1		
Solder concentration on joints is sufficient (wires do not move) for at least 2 of 3 colours	1		
Solder connections and surrounding areas are cleaned of excess solder, flux residue, etc on at least 2 of 3 colours	1		

INVIGILATOR CHECK	SAT	UNSAT	/9
Taping and Heat Shrink			
Order and amount of tape is appropriate (rubber, plastic, friction)	1		
Start and end of tape is in the center	1		
Taping covers all copper	1		
Taping is neat and tight	1		
INVIGILATOR CHECK	**SAT**	**UNSAT**	**/4**
Heat shrinks selected matches job requirements - Not too loose - No splitting or discoloration from overheating	1		
INVIGILATOR CHECK	**SAT**	**UNSAT**	**TOTAL /18**
Stage 3: Test a Lighting Circuit			
Correct use of a multimeter to conduct a continuity test			
Light turns on when switch is turned to the ON position			
Repairs proven correct			
INVIGILATOR CHECK	**SAT**	**UNSAT**	**TOTAL /18**

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 003.08: Report Status Shipboard PG&D

EC Time: 50 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 002.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

- DLN LMS Account; and
- Computer with DLN LMS access.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 004.01: Inspect Hull Structures

EC Time: 60 mins (concurrent)

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 004.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: Pass 60%

ASSESSMENT RESOURCES:

- DLN LMS Account; and
- Computer with DLN LMS access.

IMPACT OF FAILURE: Remedial Training / Re-test

PC 004: Maintain Shipboard Structures

PC Time: 90 mins (concurrent)

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 004.01. Trainees will work in pairs to complete a practical assessment in which they will be required to perform maintenance on a watertight hatch or door. Prior to working on the Door/Hatch, Students will be required to properly fill out a Confined Space Sentries and Workers Instruction (Entry/Exit Log). The choice of a watertight door or hatch will be randomly chosen for each pair of trainees. On completion of the PC, to pass this test the door or hatch must also function correctly.

To limit the need for multiple invigilators, the trainee's will be given milestones based on the PC checklist. Once they have achieved a milestone, they will signal to the invigilator whom will verify the milestone and instruct the trainees to continue with the assessment, or redo the milestone in question.

Pass/fail criteria: Pass 60% SAT on all checklist items

ASSESSMENT RESOURCES:

- DLN LMS Account;
- Workshop;
- Door and hatch; and
- Computer with DLN LMS access.

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – PC 004 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items for the closure they were given and safety to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Maintain Watertight Hatches			
Disassemble a quick acting hatch.			
Reassemble a quick acting hatch.			
Prove full working order of hatch.			
Ensure spring tension is set corrected *critical* safety.			
Perform a chalk test.			
2. Maintain Watertight Doors			
Disassemble a quick acting door.			
Reassemble a quick acting door.			
Prove full working order of the door.			
Ensure door hold back is correctly holding door handle *critical* safety.			
Perform a chalk test.			

3. Perform tasks safely			
Utilize appropriate PPE			
Follow safety practices			

ADDITIONAL COMMENTS			
STRENGTHS:		**AREAS FOR DEVELOPMENT:**	
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 005.01: Inspect small engine

EC Time: 50 mins (concurrent)

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 005.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: Pass 60%

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet);

- DLN LMS account.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 005.02: Maintain Small Engines

EC Time: 10 mins (per trainee; two trainees at a time) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will be given a small gas engine and will be required to operate it including doing pre-start checks, starting it, and shutting it off.

Pass/fail criteria: Pass/Fail

ASSESSMENT RESOURCES:

- Engine (min of one diesel and one gas)

IMPACT OF FAILURE: Remedial Training / Re-test

Marine Technician RQ-S3 – EC 005.02 Assessment Checklist	
Candidate Name:	**Assessor Name:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a 60% SAT on each section of the checklist to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

OPERATE A SMALL GAS ENGINE

ITEM	SAT	UNSAT	COMMENTS
1. Personal Safety			
Utilized PPE for duration of assessment: ● Safety glasses or face shield ● No jewelry or loose clothing ● Ear protection ● Gloves			
Maintained personal and team safety throughout with implementation of appropriate safety practices			
2. Component Identification			
Identify the following components ● Start/Stop switch ● Throttle lever ● Overspeed trip ● Air intake ● Exhaust muffler ● Air filter ● Fuel line ● Spark plug ● Choke lever ● Pump ● Oil type/mixture			
3. Operate a Small Gas Engine			
Conduct Pre-start checks ● Check fluid level and fill with recommended fuel/oil mixture if necessary			

• Ensure fuel line is connected to engine • Ensure air intake and exhaust are free of obstruction			
Start the engine • Attach quick handle of the fuel supply to the WAJAX • Open air vent • Connect water discharge hose • Connect water suction hose • Fill pump with water • Close choke shutter • Check overspeed trip position • Toggle to “START” • Move throttle lever to start and warm up position • Allow engine to warm-up prior to using full throttle to adjust speed			
Stop the engine • Move throttle to “STOP” position • Run the engine in this position for 1 min • Toggle to “STOP” position			
4. Safety			
Proper PPE always worn			
No open flame, sparks, or naked light in the area			
Properly ventilated			

Fuel handling/spill response			
An Unsat in ANY of the critical safety will result in an automatic failure of EC 5.02			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EC 006.01: Inspect Gearing and Shaft lines

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 006.01. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LMS account.

IMPACT OF FAILURE: Remedial Training / Re-test

EC 006.02: Inspect the Steering System

EC Time: 60 mins via DLN

ASSESSMENT ACTIVITY: Trainees to complete a theoretical assessment derived from a combination of all of the content from EO 006.02. The assessment will be delivered utilizing a CBT framework.

Pass/fail criteria: 60%

ASSESSMENT RESOURCES:

- Access to a computer (DWAN or internet); and
- DLN LMS account.

IMPACT OF FAILURE: Remedial Training / Re-test

PC 007: Repair Metal (Using Heat)

PC Time: 400 mins

ASSESSMENT ACTIVITY: Trainees to complete a practical assessment derived from a combination of all the content from PO 007. Trainees will be required to safely operate an oxy acetylene torch to include both cutting and welding.

Pass/fail criteria: (Passing % or taken from invigilator notes of PC checklist)

ASSESSMENT RESOURCES:

- Workshop;
- PPE;
- Oxy Acetylene equipment;
- Metal;
- Access to a computer (DWAN or internet);
- DLN LMS account;

IMPACT OF FAILURE: Remedial Training / TRB after second unsuccessful attempt

Marine Technician RQ-S3 – PC 007 Assessment Checklist	
Candidate Name:	**Assessor Name:**

INVIGILATOR NOTES:

1 PASS / FAIL CRITERIA: Trainees must achieve a 60% SAT on each section of the checklist to be successful with this assessment.

2 IMPACT OF FAILURE: Remedial training, re-test.

Repair Metal - OXY ACETYLENE TORCH			
ITEM	**SAT**	**UNSAT**	**COMMENTS**
Personal Safety			
Use of proper PPE ● Eye protection ● Ear protection ● Welding safety equipment ● Respirator			
Observed all safety considerations			Failure to observe safety constitutes an automatic failure
Equipment Safety			
Visual inspection ● damage, oil or dirt ● damaged hoses or threads			

• Safe handling of bottles			
Acetylene safety			
Equipment Set up			
Purge both oxygen and acetylene cylinder valves			
Inspect regulators, threads and fittings for damage, oil or grease			
Connect regulator to cylinder			
Back out adjusting screw			
Open then shut cylinder valves and check for a pressure drop on first stage regulator			
Inspect and connect the flashback arrestor			
Inspect and connect hoses, blow hose regulators before connecting to the torch body			
When using a flash back arrestor for cutting inspect screen and threads prior to connection to torch body			
Inspect inlet threads and overall body of the torch before connecting it to the assembly hose			
Inspect and connect the tip			
Check all valves are closed and regulator screws backed off			

Slowly open the oxygen cylinder valve fully to allow double seat valve to seat			
Open acetylene cylinder valve ¼ turn max			
Adjust regulator to working pressure **Never use acetylene more than 15 PSI**			
Check for leaks at all fitting, threads, finger and cylinder valves			
Purge system			
Check both cylinders to ensure they are secure			
Cut Metal			
Cut a metal shape • Cut 7x4x3/8" mild steel plate along 4" dimension			
Repair Metal			
a) Weld flat butt joint IAW following standard: • FACE width min 5/16" and max 7/16" and max of ten			
Trainee Instruction: Once weld is complete, bring completed and cooled (do not quench) test piece to the instructor with EC Checklist for evaluation.			
b) Welding defects:			
bead wandering: any portion of a bead that wanders from a straight line.			

bead uniformity: a bead should maintain equal width and height along the entire distance.			
inclusion: foreign matter introduced and remaining in the weld.			
porosity: gas pockets trapped inside the weld or surface of the weld.			
undercut: depression at the toe of the weld which is below the surface of the base metal.			
lack of face reinforcement: protrusion of the weld over the surface of the base metal on the side the welding was done.			
lack of root reinforcement: protrusion of the weld over the surface of the base metal on the opposite side the welding was done.			
Instructor Actions: Measure welded flat butt joint to be no more than 2” above vice Root bend butt joint approx. 90 degrees IAW destructive hammer test			

ADDITIONAL COMMENTS

STRENGTHS:	AREAS FOR DEVELOPMENT:

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

ANNEX M - TEL PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta

ANNEX N - TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS

The following information was compiled during the QSPWB (date) to give context for developers. It can be removed from the QSP if desired after initial course development.

Key Target Population Details - (Course name) candidates

"The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" -CFITES Vol 4

Characteristic	**Target population details**
General data	
Potential group size Expected annual throughput Location(s) of the intended learners Access to computers (job) Gender Age Trades	
Learning skills and preferences	
Competencies (general)	
Reading and writing level	
Preferences, attitudes towards specific type of learning material and equipment (i.e. TEL, aversion to lectures)	
Educational level	

Characteristic	Target population details
Previous military training experience	
Current subject-matter competence	
Experience with target performance	
Ability to perform any part of the POs	
Perception of the impact of achieving the POs on self, work or career	
Job/career stage	
Social Characteristics	
Relationship to peers	
Tendencies toward cooperation or competition	

Describing the learning environment - NFS (division)

The “learning system” is composed of all factors that affect and are affected by the learning that takes place: learners (Target Population), instructional materials, the instructors, instructional equipment (such as computers, projectors, LMS, etc), the instructional facilities, and the community or organization.

What are the characteristics of the teachers/trainers who will be using these materials?
What are the interests and preferences of the instructors? How do instructors feel about having instruction delivered via media or other non-traditional methods? Do instructors like to use technology-based instruction? What experience have instructors had with technology-based instruction?
Are there existing courses/programs into which this piece of instruction must fit?
If so, what is the strategy used in these materials?
What hardware is commonly available in the potential learning environments?
Are computer workstations available? How many? DWAN, standalone? What software or other materials are available?
What are the characteristics of the classes and facilities that will use the new instruction?

ANNEX O - MODULARIZATION DETAILS

This Annex is to be used when scheduling and planning for a session to be run using method 1 (as described in Chapter 1 para 2. b.). For PRes personnel, completing the DL portions from their home NRD, the preferred method is to have a regional instructor within each applicable NRD or region for easier access by the candidates. The NFS instructor will still be overall responsible for the course with the regional instructor simply to help as necessary

Module A:

- This module shall be the first module and is a prerequisite for all other modules.
- It is to be run using DL (from home).
- There will be a designated DL Instructor at an NFS (recommendation is using staff at NFS(Q)) with a 60:1 instructor/trainee maximum ratio having the following quals:
    - Instructional Techniques (or equivalent civilian qual);
    - Distributed Learning Instructor (DLI);

EO ##.###	EO title	Time
001.03	Perform basic mathematics	DL: 1080 + EC 60 = 1140 mins
	ATT Sub-total	3 days DL

Module B:

- This module shall be the second module and is a prerequisite for all modules except Module A.
- It will be run using DL (from home).
- There will be a designated DL Instructor at an NFS (recommendation is using staff at NFS(Q)) with a 60:1 instructor/trainee maximum ratio having the following quals:
    - USQ-S1 Mech Spec or above or MAR TECH Res S1 or above with RQ-S3 and Roundsperson qualification or legacy Cert B or above;
    - Instructional Techniques (or equivalent civilian qual); and
    - DLI.

EO ##. ###	EO title	Time
Mod 0	Intro to the RCN Intro to MARTECH Training	DL 180 mins
001.01	Maintain safety requirements for engineering operations	DL 410 + EC 60 = 470 mins
001.04	Interpret drawings	DL: 150 + EC 60 = 210 mins
004.01	Inspect hull structures	DL 440 + EC 60 = 500 mins
002.01	Apply foundational science	DL 400 + EC 60 = 460 mins

002.02	Inspect fluid system components	DL 550 + EC 60 = 610 mins
002.03	Inspect components of pressurized systems	DL 635 + EC 60 = 695 mins
002.04	Maintain the liquid damage control systems	DL 175 + EC 60 = 235 mins
002.05	Maintain POL systems	DL 325 + EC 60 = 385 mins
002.06.01	Repair metal without heat	DL 150 mins Total: 3895
	ATT Sub-total	10 days DL

Module C:

- This module shall be the third module and is a prerequisite for all modules except Modules A and B.
- It will be run as a combination of DL (from home) on the DLN, CBT, and face to face (F2F) training in-house at an NFS.
- EO 001.05, DRMIS will be run at a Training Establishment IAW the DRMIS Module 1 course MITE 119096
- For the DL, there will be a designated DL Instructor at an NFS (recommendation is using staff at NFS(Q)) with a 60:1 instructor/trainee maximum ratio having the following quals:
    - USQ-S1 Mech Spec or above or MAR TECH Res S1 or above with RQ-S3 and Roundsperson qualification or legacy Cert B or above MESO for PRes);
    - Instructional Techniques (or equivalent civilian qual);
    - DLI;
- For the F2F at an NFS the instructor must have following quals:
    - USQ-S1 Mech or above; and
    - Instructional Techniques course (or equivalent civilian qual)
- After completion of 001.05, trainees awarded MITE 119096

EO ##.###	**EO title**	**Time**
001.05	Maintain DRMIS portal as a maintainer	CBT at the TE: 700 + EC 60 = 760 mins
002.06	Repair metal without heat	002.06.02/3 F2F: 650 mins F2F CEC 425
002.07	Maintain pumps and valves	F2F: 1200 + EC 50 = 1250 Total: 3085 mins
	ATT Sub Total	8 days at TE

Module D:

- This module shall be completed after Modules A, B, and C. It is preferred, though not mandatory, for it to precede Modules E and F.
- It will be run as a combination of DL (from home) on the DLN, DL not on DLN and F2F training in-house at NFS (A) or (P).

    - 001.02 ODS regulations will be completed DL (from home) through HRAI Canada
- For the DL, there will be a designated DL Instructor at an NFS (recommendation is using staff at NFS(Q)) with a 60:1 instructor/trainee maximum ratio having the following quals:
    - USQ-S1 Mech Spec or above or MAR TECH Res S1 or above with RQ-S3 and Roundsperson qualification or legacy Cert B or above;
    - Instructional Techniques (or equivalent civilian qual);
    - DLI.
- For the CBT and F2F at the TE the instructor must have:
    - USQ-S1 Mech or above;
    - Instructional Techniques (or equivalent civilian qual).
- 001.02 will be completed via DL (at home) through HRAI of Canada and not on DLN

EO ##.###	**EO title**	**Time**
002.08	Maintain pipes, hoses, insulation	F2F: 1370 + EC 210 =1580 min
002.09	Maintain miscellaneous fluid components	F2F: 740 + EC 40 = 780 mins
002.10	Perform basic maintenance on Air Systems	F2F: 500 mins + EC 100 = 600 mins
001.02	Apply ODS regulations	DL: 525 + EC 60 = 585 mins
002.11	Maintain refrigeration systems	F2F: 150 + DL 200 + EC 200 = 550 mins F2F PC 002 = 340 mins Total: 890 mins
004.02	Maintain basic closures	F2F: 1150 + EC 100 = 1250 mins
007.01	Assemble Oxy-acetylene torch	007.01.01 DL: 100 mins & F2F 150 mins = 250 mins 007.01.02/03 F2F: 1300 mins + EC 250 = 1550 mins Total = 1800 mins
007.02	Cut metal with oxy-acetylene torch	F2F: 2300 + EC 400 = 2700 mins
007.03	Weld metal with oxy-acetylene torch	F2F: 1050 + EC 300 = 1350 mins Total: 12,345
007.04	Cut metal with exothermic torch	F2F: 1050 + EC 300 = 1350 mins Total: 12,345

Total training days	31 days (2 days DL and 29 days at TE)

Module E:

- This module shall be completed after Modules A, B, and the DL portion of Module C. It is preferred, though not mandatory, for it to come after the DL portion of Module E.
- It will be run of a combination of DL (from home) and in-house at NFS(P)/(A)
- For the DL, there will be a designated DL Instructor at an NFS (recommendation is using staff at NFS(Q) with a 60:1 instructor/trainee maximum ratio having the following quals:
    - USQ-S1 Mech Spec or above or MAR TECH Res S1 or above with RQ-S3 and Roundsperson qualification or legacy Cert B or above;
    - Instructional Techniques (or equivalent civilian qual);
    - DLI.
- For F2F training the instructors must have the following quals:
    - USQ-S1 Mech or above;
    - Instructional Techniques (or equivalent civilian qual).

EO ##.###	EO title	Time
005.01	Describe internal combustion engine	DL 525 mins + EC 60 = 585 mins
005.02	Perform basic maintenance on internal combustion engine	F2F 930 mins + EC 20 = 950 mins PC 005 100 mins Total: 1050 mins
006.01	Describe gearing and shaft lines	DL 405 mins + EC 60 mins = 465 mins
006.02	Describe steering systems	DL 225 mins + EC 60 = 285 mins Total: 2385
	Total training days	6 days (3 days DL and 3 days at TE)

Module F:

- This module shall be completed after Modules A, B, and the DL portion of Module C. It is preferred, though not mandatory, for it to come after the DL portion of Module D.
- It will be run on a combination of DL (from home) and in-house at NFS(P)/(A).

- For the DL, there will be a designated DL Instructor at an NFS (recommendation is using staff at NFS(Q)) with a 60:1 instructor/trainee maximum ratio having the following quals:
    - USQ-S1 Elect or above or MAR TECH Res S1 or above RQ-S3 and Roundsperson qualification or legacy Cert B. MAR TECH Res will require support from an Elec Instructor from a NFS (A) or (P) as there is no Elect Specialist at NFS(Q).
    - Instructional Techniques (or equivalent civilian qual);
    - DLI.
- For F2F training the instructors must have the following quals:
    - USQ-S1 Elec or above;
    - Instructional Techniques (or equivalent civilian qual).

EO ##.###	**EO title**	**Time**
003.01	Analyze DC Circuits	DL: 720 mins + EC 60 = 780 mins
003.02	Analyze AC circuits	DL 330 mins + EC 60 = 390 mins
003.03	Describe AC Machines	F2F: 500 mins + DL 500 mins + EC 60 = 1060 mins
003.04	Describe DC Machines	F2F 700 mins + DL 150 mins + EC 60 = 910 mins
003.05	Maintain Control Devices	F2F: 1900 mins + DL 120 mins + EC 90 = 2110 mins
003.06	Maintain Ship's Service Lighting	F2F 200 mins + DL: 115 + EC 60 = 375 mins
003.07	Maintain Ship's Cabling and Connectors	F2F 955 mins + DL: 125 mins + EC 100 = 1180 mins
003.08	Describe shipboard PG&D	DL 250 mins + EC 60 = 310 mins PC 003 300 mins Total: 7415 mins
	Total training days	19 days (7 days DL and 12 days at TE)